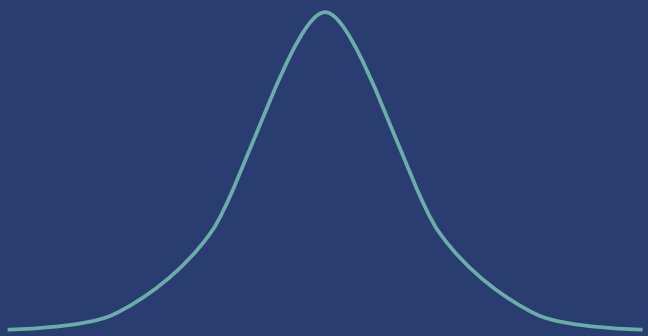
AN INTRODUCTION TO PROBABILITY AND STATISTICS

Chance, Data, and Decisions

A visual, inquiry-driven course with exercises





One-semester course packet

events → random variables → samples → inference

How to use this course

Big idea

Probability starts with a model of what might happen; statistics works backward from what did happen. The course repeatedly follows one cycle:

$$\text{model} \longrightarrow \text{simulate} \longrightarrow \text{observe} \longrightarrow \text{infer} \longrightarrow \text{check}.$$

The point is not merely to calculate a probability or a confidence interval, but to understand which assumptions made that calculation meaningful.

Audience and prerequisites

This packet is designed for a first undergraduate course or a mathematically ambitious secondary course. Algebra, functions, and summation notation suffice for the finite probability models and their expectations and variances. Differentiation, integration, and series enter later with continuous models and approximations. Multivariable calculus is not required. Proofs are included, with an emphasis on modeling, interpretation, and computation.

Appendix A rebuilds the calculus as it is needed: limits, derivatives, optimization, definite and improper integrals, series, and Taylor estimates. It includes worked applications to birthdays and derangements, followed by 18 exercises with solutions. Chapters 1–2 develop the exact finite counting arguments before using those approximations. The probability-space and random-variable introductions use concrete outcomes, events, weights, and inequalities.

For the current discrete route, work through Chapter 4's weighted averages, linearity, indicators, variance, and sums, then its 18 core discrete exercises. Additional indicator examples count records, inversions, spaghetti loops, and crossing edges; five further exercises develop these applications. Chapter 5 next develops uniform, Bernoulli, binomial, and hypergeometric distributions from their experiments. Geometric waits and Poisson counts then meet in the coupon collector. Countably infinite models and continuous densities are marked for later reading. The coupon section develops both uniform and unequal probabilities. Further sections derive negative-binomial, gamma, and beta laws from their mechanisms, then use Poissonization and exponential clocks for collection and occupancy problems. Explicit de-Poissonization bounds justify returning to fixed draw counts; the completion-window limit is optional asymptotic reading.

Learning goals

By the end, students should be able to:

1. translate verbal questions into sets, events, and probability models;
2. use counting, conditioning, independence, and Bayes' rule correctly;
3. distinguish permutations from combinations, prove basic binomial identities, and apply inclusion–exclusion to both counts and probabilities;
4. derive the birthday-paradox formula from an explicit probability space and distinguish any-pair matches from matches with one designated person;
5. derive the derangement formula and bound its approximation by $1/e$;
6. translate a condition such as $X \geq a$ into an event in the original sample space;
7. move fluently among CDFs, probability masses, and densities;
8. compute and interpret expectation, variance, covariance, and correlation;
9. recognize standard distributions and justify an appropriate model;
10. obtain marginal and conditional distributions from a joint model;
11. explain the law of large numbers and central limit theorem through both mathematics and simulation;
12. evaluate estimators by bias, standard error, and mean squared error;
13. construct and interpret basic confidence intervals and hypothesis tests;
14. perform a simple Bayesian update and compare credible and confidence intervals;
15. distinguish exchangeability from independence and explain how a shared random parameter produces dependence and a random limiting frequency.

A 14-week route

The design assumes two 75-minute meetings per week. "Lab" means that students make a prediction, run an experiment or simulation, and explain a discrepancy.

Week	Core mathematics	Suggested lab or assessment
1	Sets, events, Venn diagrams, counting	Inclusion–exclusion gallery
2	Probability spaces and axioms	Build and critique sample spaces
3	Conditional probability and trees	Medical-test simulation
4	Independence, total probability, Bayes	Monty Hall and base-rate lab
5	Discrete random variables, PMFs, CDFs	Outcomes, values, and probability tables
6	Expectation, variance, indicators, sums	Dice, payoffs, hats, and birthday pairs
7	Discrete families and waiting times	Sampling and coupon collection
8	Densities; uniform, exponential, normal, beta	Integrals, memorylessness, and normal areas
9	Joint, marginal, conditional laws	Heat maps and independence
10	Covariance, correlation, conditional expectation	Correlation counterexamples
11	Laws of large numbers; Monte Carlo	Running-average experiment
12	Central limit theorem; sampling distributions	Visual CLT; Problem Set 5
13	Estimation, confidence intervals, tests	Coverage and power simulation
14	Bayesian inference and synthesis	Prior-to-posterior capstone

Assessment design

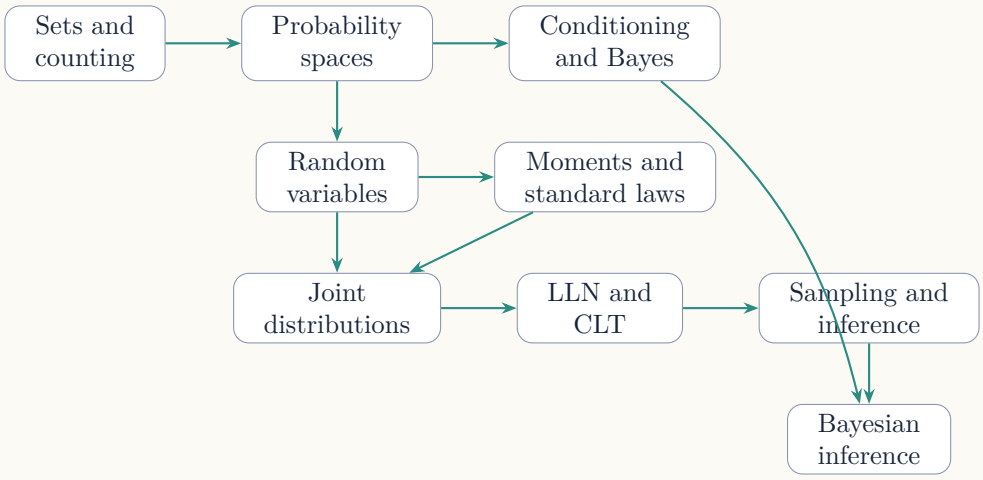
Optional synthesis. Chapter 10 follows Bayesian inference as an exchangeability capstone. Its coin and urn examples can anchor a week-14 synthesis session; the optional proof is extension reading, not an added prerequisite for the first weeks. A short preview in Chapter 3 plants the distinction between symmetry and independence.

Early-course companion. Chapters 1 and 2 now provide a fuller development of permutations, combinations, binomial coefficients, both forms of inclusion–exclusion, finite probability spaces, and the birthday paradox. Problem Sets 2 and 3 revisit these foundations and connect counting to birthdays and derangements. Chapter 4 follows the same examples into probability tables, expected counts, and variances. Chapter 5 gives the next sequence of distribution examples. The calculus refresher can be assigned in short sections as its tools become needed. Assignment solutions are supplied separately through the instructor library.

One workable balance is 35% problem sets, 20% short quizzes, 20% computational labs, and 25% a modeling capstone. Exercises use three labels:

- **Fluency**: calculations and translations that make later reasoning automatic.
- **Reasoning**: derivations, short proofs, and careful interpretation.
- **Exploration**: simulation, visualization, model criticism, or open-ended comparison.

Dependency map



Checkpoint

A good opening prompt. A family has two children and at least one is a girl. What is the probability that both are girls? Ask students to defend not just an answer, but a sample space. Then change how the information was obtained and ask whether the answer changes.

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Chapter 1

The grammar of uncertainty

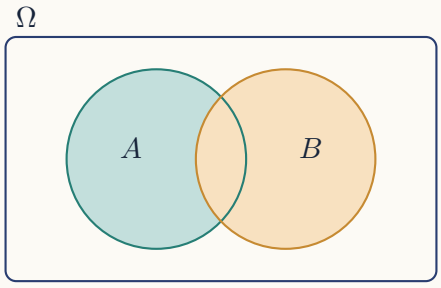
Sets, permutations, combinations, and inclusion–exclusion

Guiding question 1.1. Before assigning probabilities, how do we describe precisely what could happen?

A *sample space* is built from possible outcomes; an *event* is a collection of outcomes. Set notation is therefore the grammar of probability.

1.1 Set operations

Let Ω be a universe of outcomes and $A, B \subseteq \Omega$. The union $A \cup B$ means "A or B (or both)," the intersection $A \cap B$ means "both," and the complement A^c means "not A." The difference $A \setminus B$ contains outcomes in A but not B.



Overlap represents $A \cap B$

De Morgan's laws translate negated statements:

$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$

Common trap

In everyday speech, "or" is sometimes exclusive. In probability, $A \cup B$ is inclusive unless explicitly stated otherwise.

1.2 Products and sequences

The Cartesian product $A \times B$ contains ordered pairs (a, b). Repeated trials naturally live in product spaces. Three coin flips have sample space $\{H, T\}^3$, with eight ordered strings. Order matters: HTT and THT are different outcomes.

1.3 The multiplication principle

Start by deciding what one outcome records. An ordered list, an unordered group, and a list with repetitions are different objects. Before using a formula, describe when two proposed outcomes count as the same.

The *addition principle* says that counts add over disjoint cases. For example, a club offers either one of three morning sessions or one of five afternoon sessions. Choosing exactly one session gives $3 + 5 = 8$ possibilities. Choosing one session in *each* period gives $3 \cdot 5 = 15$. The words "either" and "each" describe different experiments.

If a first choice has m possibilities and, after *each* first choice, a second choice has n possibilities, there are mn ordered outcomes. More generally, if the successive numbers of choices are $n_1, \ldots, n_k$, the number of outcomes is $n_1 \cdots n_k$. The available choices may depend on earlier choices; their *number* at each stage must be the stated one. If different first choices leave different numbers of continuations, count each branch and add its contributions instead.

Example 1.2 (Codes and officers)**.** A four-digit code may begin with zero. If digits can repeat, there are 10^4 codes. If digits must be distinct, the count is $10 \cdot 9 \cdot 8 \cdot 7 = 5040$. Assigning president, secretary, and treasurer from ten students similarly gives $10 \cdot 9 \cdot 8 = 720$ outcomes when no student holds two offices.

1.4 Permutations and combinations

There are $n! = n(n-1)\cdots 1$ orders of n distinct objects, with $0! = 1$. For $0 \le k \le n$, an ordered selection of k distinct objects from n has

$$(n)_k = n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}$$

possibilities. This is sometimes written $P(n, k)$. An unordered selection of k objects is a *combination*. Every such selection has exactly $k!$ orderings, so

$$\binom{n}{k} = \frac{(n)_k}{k!} = \frac{n!}{k!(n-k)!}.$$

In particular, $\binom{n}{0} = \binom{n}{n} = 1$: there is one empty selection and one selection containing everything. We use $\binom{n}{k} = 0$ when $k < 0$ or $k > n$ and n is a nonnegative integer.

Experiment	What distinguishes outcomes?	Count
k draws with replacement	Order matters; repetition allowed	n^k
k draws without replacement	Order matters; objects distinct	$(n)_k$
A k-element subset	Order ignored; no repetition	$\binom{n}{k}$
An unordered selection with repetition	Only the number of each type matters	$\binom{n+k-1}{k}$

Example 1.3 (A committee is not a list of officers)**.** From ten students there are $\binom{10}{3} = 120$ three-person committees, but 720 assignments to three distinct offices. A committee with one designated chair has $3\binom{10}{3} = 360$ possibilities: the two other members have no distinct roles. The question about order must be answered before choosing a formula.

Proposition 1.4 (Arrangements with repeated objects)**.** *If n objects consist of r types with multiplicities $n_1, \dots, n_r$, where $n_1 + \cdots + n_r = n$, the number of distinct arrangements is $n!/(n_1! \cdots n_r!)$.*

Proof. Temporarily distinguish objects of the same type with labels. There are $n!$ arrangements of the labeled objects. Each arrangement after erasing the labels comes from exactly $n_1! \cdots n_r!$ labeled arrangements. Dividing by that constant multiplicity gives the result. □

For example, `BANANA` has $6!/(3!2!) = 60$ distinct arrangements.

Unordered selections with repetition

Suppose k scoops are chosen from $n \geq 1$ flavors, with repetition allowed and order ignored. An outcome specifies nonnegative counts $x_1 + \cdots + x_n = k$. Arrange k stars and $n - 1$ bars in a row: stars before the first bar count flavor 1, those between the first two bars count flavor 2, and so on. Empty groups are allowed. Choosing the k star positions among $n + k - 1$ positions gives $\binom{n+k-1}{k}$ outcomes. For two scoops from three flavors A, B, C, these are AA, AB, AC, BB, BC, CC, a total of $\binom{4}{2} = 6$.

Common trap

Divide a count only when every desired outcome has the same number of representations. Ordered pairs AB, BA both become AB, but AA has only one ordering. Thus dividing 3^2 by 2! does not count the six two-scoop selections. The equal-multiplicity argument for combinations applies to distinct objects.

1.5 Binomial coefficients count in two directions

Proposition 1.5 (Symmetry and Pascal's identity)**.** *For $0 \leq k \leq n$,*

$$\binom{n}{k} = \binom{n}{n-k},$$

and, for $n \geq 1$,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Proof. A k-element subset determines its $(n-k)$-element complement, proving symmetry. For Pascal's identity, single out one object. A k-element subset either contains it, leaving $k-1$ choices from the other $n-1$ objects, or omits it, leaving k choices from those objects. These two cases are disjoint and exhaustive. □

Theorem 1.6 (Binomial theorem)**.** *For a nonnegative integer n,*

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

Proof. Expand the product of n copies of $(x+y)$. To produce $x^{n-k}y^k$, choose the k factors that supply y; all the others supply x. There are $\binom{n}{k}$ choices. □

Example 1.7 (Coefficients carry both counts and weights)**.** In $(2-x)^4$, choosing k copies of $-x$ and $4-k$ copies of 2 gives

$$(2-x)^4 = 16 - 32x + 24x^2 - 8x^3 + x^4.$$

For instance, the x^2 coefficient is $\binom{4}{2}2^2(-1)^2 = 24$. The binomial coefficient counts the positions; the remaining factors give the contribution from each choice.

The same distinction explains a probability formula. In n independent trials with success probability p, a *specified* string with k successes has probability $p^k(1-p)^{n-k}$. There are $\binom{n}{k}$ such strings, so the probability of exactly k successes is $\binom{n}{k}p^k(1-p)^{n-k}$. Their sum is $(p+(1-p))^n = 1$. Chapter 5 develops this binomial distribution; for $p = 0$ or 1, every trial has its certain outcome.

Taking $x = y = 1$ gives $\sum_{k=0}^{n}\binom{n}{k} = 2^n$. This also counts all subsets in two ways: by their sizes, or by making an include/exclude choice for each object. Taking $x = 1, y = -1$ gives, for $n \geq 1$, $\sum_{k=0}^{n}(-1)^k\binom{n}{k} = 0$. This alternating sum is the engine behind the general inclusion–exclusion formula.

1.6 Inclusion–exclusion for finite sets

Proposition 1.8 (Two and three sets)**.** *For finite sets A, B, C,*

$$\begin{aligned} |A \cup B| &= |A| + |B| - |A \cap B|, \\ |A \cup B \cup C| &= |A| + |B| + |C| \\ &\quad - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|. \end{aligned}$$

For two sets, subtraction repairs the double count of the overlap. With three sets, an element in all three is added three times and subtracted three times, so it must be added back once. An element in exactly two sets contributes $2 - 1 = 1$; an element in exactly one contributes 1. An element outside the union contributes zero. This proves the formulas.

Common trap

$A \cap B$ includes the triple intersection $A \cap B \cap C$. It does not mean "in A and B but outside C." Confusing these two regions produces incorrect inclusion–exclusion calculations.

Example 1.9 (Three activities)**.** Among 100 students, 39 attend activity A, 29 attend B, and 25 attend C. The pairwise intersection counts are 13, 11, and 9, respectively, and 5 students attend all three. Thus the union has $39 + 29 + 25 - 13 - 11 - 9 + 5 = 65$ students; 35 attend none. The exactly-two count is $(13-5) + (11-5) + (9-5) = 18$. The exactly-one count is $65 - 18 - 5 = 42$.

Theorem 1.10 (Finite inclusion–exclusion)**.** *For finite sets $A_1, \dots, A_r$,*

$$\left|\bigcup_{i=1}^{r} A_i\right| = \sum_{\varnothing \neq J \subseteq \{1,\dots,r\}} (-1)^{|J|+1} \left|\bigcap_{j \in J} A_j\right|.$$

Equivalently: add the single-set sizes, subtract the pairwise intersection sizes, add the triple-intersection sizes, and continue with alternating signs through the intersection of all r sets.

Proof. An element belonging to exactly $t \geq 1$ sets occurs in $\binom{t}{k}$ of the k-fold intersections. Its total contribution is

$$\sum_{k=1}^{t} (-1)^{k+1} \binom{t}{k} = 1 - (1-1)^t = 1.$$

An element in no set contributes zero. Thus each element of the union is counted exactly once. □

Example 1.11 (Using all three symbols)**.** There are 3^4 strings of length four over $\{A, B, C\}$. To count those using every symbol, subtract strings missing at least one symbol. Each missing-symbol event has 2^4 strings; each pair of missing-symbol events has one string; no length-four string misses all three symbols. Therefore the required count is

$$3^4 - 3 \cdot 2^4 + 3 \cdot 1^4 = 36.$$

These are counts; turning them into probabilities requires a specified probability model.

Example 1.12. A five-card poker hand is an unordered choice from 52 cards, so there are $\binom{52}{5}$ hands. Exactly two aces means choosing two of four aces and three of the remaining 48 cards:

$$\binom{4}{2}\binom{48}{3}.$$

Explore

Use the website's counting lab to compare committees, ordered selections, and repeated draws. Then use the inclusion–exclusion lab to vary the eight disjoint regions of three sets, including the region outside the union. Recover the marginal and intersection totals from those regions. With all seven marginal/intersection totals specified, the seven interior region counts are determined; knowing the universe size also determines the outside region. If totals are omitted, the remaining data need not determine a unique diagram.

Exercises

1. Write $(A \cap B^c) \cup (A^c \cap B)$ in words. [Fluency]

2. Verify both De Morgan laws using membership tables. [Reasoning]

3. How many eight-character strings contain exactly three letters and five digits? Use 26 letters and 10 digits, allow repetition, and allow a leading zero. [Fluency]

4. Count five-card hands containing at least one ace. [Reasoning]

5. Prove $\binom{n}{k} = \binom{n}{n-k}$ by a counting argument. [Reasoning]

6. Among integers from 1 to 1,000, how many are divisible by 2 or 3 or 5? [Reasoning]

7. Design two different sampling procedures that produce the same set of possible outcomes but not the same probabilities. [Exploration]

8. From nine students, count a four-person committee, a list of four distinct officers, and a four-person committee with one chair. [Fluency]

9. Count the distinct arrangements of the letters in `LEVEL`. Explain the divisor in your formula. [Fluency]

10. Prove Pascal's identity by deciding whether a distinguished object is selected. [Reasoning]

11. Find the coefficient of x^2 in $(1+3x)^5$, and explain its binomial coefficient. [Fluency]

12. Count length-five strings over $\{A, B, C\}$ that use all three symbols. [Reasoning]

13. In five independent trials with success probability $1/3$, find the probability of exactly three successes. Distinguish one specified string from all strings with three successes. [Fluency]

14. Choose four scoops from three flavors, allowing repetition. Count ordered lists and unordered selections. Explain why dividing the first count by 4! is invalid. [Reasoning]

15. Count the distinct arrangements of `MISSISSIPPI`. Identify every repeated-letter factor. [Fluency]

16. Prove $\sum_{k=0}^{n} k\binom{n}{k} = n2^{n-1}$ for $n \geq 1$ by counting subsets with one distinguished member in two ways. Appendix A also obtains this identity by differentiation. [Reasoning]

Chapter 2

A measure of possibility

Outcomes, events, birthdays, and derangements

Guiding question 2.1. What rules must any coherent numerical description of uncertainty obey?

2.1 A probability space is a model

Definition 2.2. A *probability space* describes three things: the possible outcomes Ω, the events we can ask about, and a rule $\mathbb{P}$ assigning their probabilities. An event is a set of outcomes; it occurs when the observed outcome belongs to that set. For a finite space, every subset is an event. The probability rule satisfies

1. probabilities are nonnegative;

2. $\mathbb{P}(\Omega) = 1$;

3. probabilities add for disjoint events: if $A \cap B = \varnothing$, then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.

The last rule also applies to a finite or countably infinite list of pairwise disjoint events, by adding all their probabilities.

The basic examples in this course use finite sets, lists of integers, or points in intervals and planar regions. In continuous models we will describe events using intervals, inequalities, and their unions, intersections, and complements. These are enough for the calculations here.

Example 2.3 (One outcome, several events)**.** Roll a die. An outcome is a number such as 4. The event "even" is $\{2, 4, 6\}$; the event "at least five" is $\{5, 6\}$. The single-outcome event $\{4\}$ is a set too. For a fair die, the two larger events have probabilities $1/2$ and $1/3$. If the die is weighted, the outcomes and events can stay the same while their probabilities change.

Build the model before taking a ratio

Write down what one outcome records, list the possible outcomes, assign their probabilities, and then identify the event in the question. Use "favorable divided by total" only after explaining why the individual outcomes are equally likely.

2.2 Building a finite probability space

Choose nonnegative numbers w_ω for the individual outcomes such that $\sum_{\omega\in\Omega} w_\omega = 1$, and define

$$\mathbb{P}(A) = \sum_{\omega\in A} w_\omega.$$

The total mass is one; for disjoint events, the sums use disjoint outcomes, which gives additivity. Since Ω is finite, only finitely many events in a disjoint family can be nonempty, so countable additivity also follows. Conversely, the probabilities of individual outcomes determine every event probability by addition. The uniform model is the special choice $w_\omega = 1/|\Omega|$ for each outcome.

Example 2.4 (Unequal weights)**.** Let $\Omega = \{a, b, c, d\}$ have masses $1/10, 2/10, 3/10, 4/10$. For $A = \{a, b\}$ and $B = \{b, d\}$,

$$\mathbb{P}(A) = \frac{3}{10}, \quad \mathbb{P}(B) = \frac{6}{10}, \quad \mathbb{P}(A\cap B) = \frac{2}{10}, \quad \mathbb{P}(A\cup B) = \frac{7}{10}.$$

Although A contains half the outcomes, it does not have probability $1/2$. Counting outcomes and adding their probability masses are different operations unless all outcomes have equal mass.

Example 2.5 (A coin need not be fair)**.** Toss a coin independently twice, with heads probability $1/3$ each time. The ordered outcomes and their probabilities are

ω	TT	TH	HT	HH
$\mathbb{P}(\{\omega\})$	$4/9$	$2/9$	$2/9$	$1/9$

Exactly one head is the event $\{TH, HT\}$, with probability $4/9$. At least one head is its union with $\{HH\}$, with probability $5/9$; the complement calculation is $1 - \mathbb{P}(\{TT\}) = 1 - 4/9$. Independence justifies multiplying the two single-toss probabilities. It does not say that the four strings are equally likely.

Proposition 2.6. *For events A, B,*

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \qquad \mathbb{P}(A\cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A\cap B).$$

If $A \subseteq B$, then $\mathbb{P}(A) \le \mathbb{P}(B)$.

Each statement follows by splitting a set into disjoint pieces. This is the basic proof move in probability.

The two-event formula. Write $A \cup B = A \sqcup (B \setminus A)$, where $\sqcup$ denotes disjoint union. Also $B = (A\cap B) \sqcup (B \setminus A)$. Additivity gives $\mathbb{P}(A\cup B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A)$ and $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A\cap B)$; substitute the second identity into the first. □

2.3 Inclusion–exclusion for probabilities

For any three events, the same overlap correction gives

$$\begin{aligned}\mathbb{P}(A\cup B\cup C) = {}& \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C)\\ & - \mathbb{P}(A\cap B) - \mathbb{P}(A\cap C) - \mathbb{P}(B\cap C) + \mathbb{P}(A\cap B\cap C).\end{aligned}$$

No independence or equal-likelihood assumption is needed. One proof applies the two-event formula to $(A\cup B)\cup C$, then to $(A\cap C)\cup(B\cap C) = (A\cup B)\cap C$.

Theorem 2.7 (Finite probabilistic inclusion–exclusion)**.** *For events $A_1, \dots, A_r$ in any probability space,*

$$\mathbb{P}\left(\bigcup_{i=1}^{r} A_i\right) = \sum_{\varnothing \neq J \subseteq \{1,\dots,r\}} (-1)^{|J|+1} \mathbb{P}\left(\bigcap_{j \in J} A_j\right).$$

Proof. Partition the sample space by the exact list of events that occur. There are at most 2^r such disjoint regions. A region in exactly $t \geq 1$ events contributes its probability mass with coefficient $\sum_{k=1}^{t} (-1)^{k+1} \binom{t}{k} = 1$ on the right. The region in no event contributes zero. Finite additivity therefore gives the probability of the union, even when the original sample space is infinite. □

Example 2.8 (Counts become probabilities under a uniform model)**.** Choose a student uniformly from the 100 students in the activity example in Chapter 1. Dividing all its counts by 100 gives

$$\mathbb{P}(A \cup B \cup C) = .39 + .29 + .25 - .13 - .11 - .09 + .05 = .65.$$

The eight disjoint regions have probabilities $.20, .12, .10, .08, .06, .04, .05, .35$ (the three single-only regions, three pair-only regions, triple region, and outside region). They are nonnegative and sum to one, so they also specify a valid eight-outcome probability space with *unequal* outcome weights.

Common trap

Inclusion–exclusion always uses the *actual* intersection probabilities. Replacing $\mathbb{P}(A \cap B)$ with $\mathbb{P}(A)\mathbb{P}(B)$ requires an additional independence assumption. The formula itself makes no such assumption. Likewise, counts divided by a total give probabilities only for a justified uniform model.

2.4 Equally likely is an assumption

If a finite sample space has N equally likely outcomes and A contains m, then $\mathbb{P}(A) = m/N$. But equality of probabilities is part of the model, not a consequence of counting.

Example 2.9. For two independent fair dice, the ordered-pair space has 36 equally likely outcomes. The possible sums $2, 3, \dots, 12$ are not equally likely: sum 7 has six representations, while sum 2 has one.

Recording a coarser outcome carries its probabilities along: for a recorded value s, add the masses of all original outcomes producing s. Replacing the resulting weights by equal weights changes the experiment.

2.5 The birthday paradox: count the complement

Guiding question 2.10. How large must a class be before a shared birthday is more likely than not?

State the model first. There are n labeled people and $d \geq 1$ possible days. Their birthdays are independent and uniform on those days. Thus

$$\Omega = \{1, \dots, d\}^n, \qquad \mathbb{P}(\{\omega\}) = d^{-n}.$$

For the usual classroom model set $d = 365$: ignore leap days and seasonal variation. Independence and equal daily probabilities are modeling assumptions, not claims about every real class.

Proposition 2.11 (Exact collision probability)**.** *For $0 \le n \le d$, the probability $q_d(n)$ of at least one shared birthday is*

$$q_d(n) = 1 - \frac{d(d-1)\cdots(d-n+1)}{d^n} = 1 - \prod_{j=0}^{n-1}\left(1 - \frac{j}{d}\right).$$

The empty product is one, so $q_d(0) = q_d(1) = 0$. For $n > d$, $q_d(n) = 1$ by the pigeonhole principle.

Proof. Count the complement: all birthdays are distinct. There are d choices for the first birthday, $d-1$ for the second, and so on, yielding $(d)_n$ ordered birthday lists. All d^n lists are equally likely, so divide and subtract from one. Counting *at least one* matching pair directly is harder because different matching-pair events overlap. □

Class size n	10	20	22	23	30
$q_{365}(n)$	.11695	.41144	.47570	.50730	.70632

Adding a person cannot destroy an existing match, so the probability is nondecreasing. Since $q_{365}(22) < 1/2 < q_{365}(23)$, the first class size above 50% is 23.

The familiar approximation $1 - e^{-n(n-1)/(2d)}$ comes from taking logarithms of the product. Section A.10 derives it, explains its error, and checks the threshold against the exact formula. The counting argument itself requires no calculus.

Any pair versus one particular person

In a class of $n \ge 1$ people, the probability that someone else shares one *designated* person's birthday is

$$1 - \left(1 - \frac{1}{d}\right)^{n-1}.$$

For $d = 365, n = 23$, this is about .05857, much smaller than .50730. The designated-person question compares that person with 22 others; the any-pair question considers $\binom{23}{2} = 253$ possible pairs.

2.6 Birthday matches meet inclusion–exclusion

For three people, let E_{12}, E_{13}, E_{23} be the events that the indicated people have the same birthday. Each event has probability $1/d$. Any two of these events force all three birthdays to agree, so each pairwise intersection has probability $1/d^2$. The triple intersection is the *same event*, with probability $1/d^2$, not $1/d^3$. Therefore inclusion–exclusion gives

$$q_d(3) = \frac{3}{d} - \frac{3}{d^2} + \frac{1}{d^2} = \frac{3}{d} - \frac{2}{d^2},$$

agreeing with $1 - (1 - 1/d)(1 - 2/d)$. This illustrates why treating all matching-pair events as independent gives a wrong answer.

More generally there are $\binom{n}{2}$ matching-pair events, each with probability $1/d$. The union bound gives the useful check

$$q_d(n) \le \min\left(1, \frac{\binom{n}{2}}{d}\right).$$

The sum $\binom{n}{2}/d$ counts overlapping pair events; it is not itself the collision probability. Later, linearity of expectation will identify it as the expected number of matching pairs.

The birthday experiment

Use the website to compare the exact product, the designated-person probability, and the union bound. Predict the 50% threshold, then simulate classes. Change the calendar from 365 days to 30 or 10 days. Separate sampling fluctuations from changes in the underlying model.

2.7 Derangements: nobody gets their own hat

Return n labeled hats to their n owners, giving each person one hat. Model the return by a uniformly chosen permutation π, where $\pi(i)$ is the hat given to person i. There are $n!$ equally likely outcomes. A *fixed point* is an i with $\pi(i) = i$; a *derangement* has no fixed points. Let D_n count derangements, and set $D_0 = 1$ for the one empty arrangement.

For $n = 3$, write the three received hats in recipient order. The six permutations are

$$123, \quad 132, \quad 213, \quad 231, \quad 312, \quad 321.$$

Only 231 and 312 have no fixed point, so $D_3 = 2$ and the probability is $1/3$.

Theorem 2.12 (The derangement formula)**.** *For $n \ge 0$,*

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}, \qquad \mathbb{P}(\textit{no fixed point}) = \frac{D_n}{n!}.$$

Proof. Let A_i be the event that person i gets their own hat. If a specified set of k people gets its own hats, the remaining hats can be assigned in $(n-k)!$ ways. There are $\binom{n}{k}$ choices of that specified set. Inclusion–exclusion for avoiding all the A_i gives

$$D_n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)! = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

The term $k = 0$ counts all permutations. Divide by $n!$ to obtain the probability under the stated uniform model. □

n	0	1	2	3	4	5	6
D_n	1	0	1	2	9	44	265
$D_n/n!$	1	0	.5	.33333	.375	.36667	.36806

Subtracting the formulas for D_n and nD_{n-1} leaves just the last term, giving the convenient exact recurrence

$$D_n = nD_{n-1} + (-1)^n \quad (n \ge 1), \qquad D_0 = 1.$$

For example, $D_5 = 5 \cdot 9 - 1 = 44$ and $D_6 = 6 \cdot 44 + 1 = 265$.

Common trap

For $n \geq 2$, $\mathbb{P}(A_i) = 1/n$, but for $i \neq j$, $\mathbb{P}(A_i \cap A_j) = 1/[n(n-1)] \neq 1/n^2$. Getting one person's hat right changes the remaining assignments. The probability that nobody gets their own hat is therefore not $(1 - 1/n)^n$.

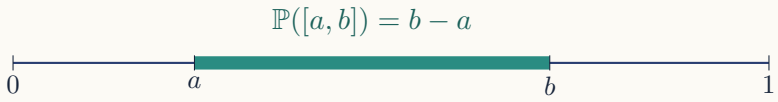
Example 2.13 (Exactly one correct hat)**.** For four people, choose the one fixed point and derange the other three: $\binom{4}{1} D_3 = 8$ permutations, giving probability $8/24 = 1/3$. More generally, exactly r fixed points can be chosen and arranged in $\binom{n}{r} D_{n-r}$ ways. Requiring a *derangement* on the rest is essential: an arbitrary permutation could introduce more fixed points.

Why do the probabilities settle near .367879? The sum is an initial segment of the series for e^{-1}. Section A.11 derives that series and proves the explicit error bound $|D_n/n! - e^{-1}| < 1/(n+1)!$. Thus the approximation is accompanied by a quantitative guarantee. Chapter 4 returns to the number of correct hats as a random variable.

2.8 Continuity and geometry

Choose a point uniformly from $[0, 1]$. Every individual point has probability zero, yet the entire interval has probability one. Probabilities attach to regions: $\mathbb{P}([a, b]) = b - a$. Zero probability does not mean impossible; it means negligible under the model.

Here $0 \leq a \leq b \leq 1$. For example, the event "the point is at least $3/4$" is the interval $[3/4, 1]$ and has probability $1/4$. There is no positive equal weight to put on each individual point: probability is assigned through lengths. In a uniform unit square the corresponding rule uses areas. The event $x + y < 1$ is a triangle of area $1/2$.



An infinite list can also carry probabilities

On $\Omega = \{1, 2, 3, \ldots\}$, assign outcome k probability 2^{-k}. The geometric series $1/2 + 1/4 + 1/8 + \cdots = 1$ makes this a valid model. The event $\{4, 5, 6, \ldots\}$ has probability $1/8$ by the same series. An infinite sum means the limit of its finite partial sums; Section A.8 reviews exactly this calculation.

2.9 Simulation and the model–world gap

A probability model is a deliberate simplification. Simulation samples from the model; data come from the world. Agreement between simulation and data supports a model, but never proves it uniquely correct.

Theorem 2.14 (Union bound)**.** *For any events $A_1, \ldots, A_n$,*

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} \mathbb{P}(A_i).$$

The events need not be disjoint. Double counting makes the right side larger, which is harmless for an upper bound.

Exercises

1. Derive $\mathbb{P}(\varnothing) = 0$ from the axioms. [Reasoning]

2. Two independent fair dice are rolled. Find the probability that the maximum is 4. [Fluency]

3. Prove the union bound for three events. [Reasoning]

4. A point is chosen uniformly from the unit square. Find the probability that $x + y < 1$. [Fluency]

5. Give an event with probability zero that is not logically impossible. [Reasoning]

6. Estimate by simulation the probability that two random points in $[0, 1]$ are within 0.1 of each other; then derive it geometrically. [Exploration]

7. On $\Omega = \{1, 2, 3, 4\}$, set $\mathbb{P}(\{i\}) = i/10$. Compute the probability of an even outcome and compare it with the fraction of outcomes that are even. [Fluency]

8. Prove three-event probabilistic inclusion–exclusion using only the two-event formula. [Reasoning]

9. With independent uniform birthdays on 30 days, find the first class size whose collision probability exceeds $1/2$. [Fluency]

10. For four people and a five-day calendar, count the no-collision lists and calculate the collision probability. [Fluency]

11. Explain why $\binom{n}{2}/d$ can exceed one without contradicting the probability axioms. [Reasoning]

12. An urn contains three labeled red balls and two labeled blue balls. Draw twice without replacement. Describe a uniform ordered sample space and find the probability of two reds. Check using unordered pairs. [Fluency]

13. In a uniform permutation of five objects, find the probability of exactly two fixed points. [Reasoning]

14. For four hats, compute $\mathbb{P}(A_1)$ and $\mathbb{P}(A_1 \cap A_2)$. Compare with $\mathbb{P}(A_1)\mathbb{P}(A_2)$. [Fluency]

15. Deduce $D_n = (n-1)(D_{n-1} + D_{n-2})$ for $n \geq 2$ from the alternating-sign recurrence, and calculate D_7. [Reasoning]

16. In the model $\mathbb{P}(\{k\}) = 2^{-k}$ on the positive integers, find the probability of an even outcome by summing a geometric series. [Fluency]

Chapter 3

Information changes the space

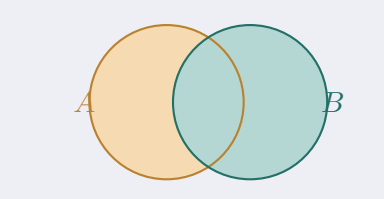
Conditional probability, independence, and Bayes' rule

Guiding question 3.1. When evidence arrives, which probabilities should change—and by how much?

Definition 3.2. If $\mathbb{P}(B) > 0$, the conditional probability of A given B is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Conditioning restricts attention to B and renormalizes so that B becomes the new whole.



Within B, measure the part also in A

3.1 Multiplication and trees

Rearranging the definition gives

$$\mathbb{P}(A \cap B) = \mathbb{P}(B)\mathbb{P}(A \mid B) = \mathbb{P}(A)\mathbb{P}(B \mid A).$$

Along a probability tree, multiply down branches and add across disjoint terminal paths.

3.2 Independence

Events A and B are independent when

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

When probabilities are positive, this is equivalent to $\mathbb{P}(A \mid B) = \mathbb{P}(A)$. Disjointness is almost the opposite: two nontrivial disjoint events cannot be independent.

Common trap

Pairwise independence does not imply mutual independence. Three events can be independent in every pair while the third is determined by the first two.

3.3 Total probability and Bayes

If $B_1, \dots, B_k$ partition Ω, then

$$\mathbb{P}(A) = \sum_{i=1}^{k} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

Bayes' rule reverses the condition:

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.$$

Example 3.3 (A diagnostic test)**.** Suppose prevalence is 1%, sensitivity is 95%, and specificity is 90%. Among 10,000 people, expect 100 cases, of which 95 test positive. Among 9,900 non-cases, 990 test positive. Thus

$$\mathbb{P}(\text{case} \mid +) = \frac{95}{95 + 990} \approx 0.0876.$$

A highly sensitive test can have a modest positive predictive value when the base rate is low.

Explore

Represent the test example with 10,000 dots rather than formulas. Change prevalence while holding sensitivity and specificity fixed. Plot the positive predictive value; locate the prevalence at which a positive result becomes more likely correct than incorrect.

3.4 A preview: symmetry is not independence

Choose once between coins with heads probabilities 1/4 and 3/4, giving each coin probability 1/2, and then repeatedly toss the chosen coin. Conditional on the choice, the tosses are independent. Without that information, if H_i means heads on toss i, then

$$\mathbb{P}(H_1) = \mathbb{P}(H_2) = \frac{1}{2}, \qquad \mathbb{P}(H_1 \cap H_2) = \frac{1}{2}\left(\frac{1}{4}\right)^2 + \frac{1}{2}\left(\frac{3}{4}\right)^2 = \frac{5}{16} \neq \frac{1}{4}.$$

The tosses are dependent, yet any reordering of a finite string has the same probability as the original string. This symmetry is called *exchangeability*. Chapter 10 returns to the example and proves de Finetti's surprising converse for infinite zero–one sequences: exchangeability gives a representation by a shared random bias and conditionally independent trials.

Exercises

1. Draw a tree for two draws without replacement from an urn with three red and two blue balls. [Fluency]

2. Prove that if A and B are independent, then A^c and B are independent. [Reasoning]

3. Construct three pairwise independent events that are not mutually independent. [Reasoning]

4. Solve the diagnostic-test example with prevalence 5%. [Fluency]

5. A randomly selected coin is fair with probability .8 and double-headed with probability .2. It lands heads three times. Find the posterior probability it is double-headed. [Reasoning]

6. Explain precisely why changing the information protocol can change the "two children" answer. [Exploration]

7. Simulate Monty Hall and explain the result using conditioning rather than frequency alone. [Exploration]

Chapter 4

Numbers attached to outcomes

Discrete random variables, expectation, and variance

Guiding question 4.1. How can a numerical record of an experiment answer a probability question?

The next step: averages on discrete spaces

Begin with the finite experiments already studied: coins, dice, hats, and birthdays. Probability tables lead to expectation, linearity, and variance using sums and algebra. Work through Sections 4.5–4.10 and the 18 discrete exercises before moving to the named distributions in Chapter 5. The two sections marked "For later" extend the ideas to infinite lists and continuous models.

4.1 A rule for recording a number

Definition 4.2. A *random variable* X assigns a real number $X(\omega)$ to each outcome ω of an experiment. The rule is fixed; the outcome to which we apply it is uncertain. Conditions such as $X \geq a$ must describe events in the probability model, so that their probabilities are defined.

On a finite sample space this last requirement is automatic: every set of outcomes is an event. For the continuous examples in this course, we use familiar functions and events described by intervals and inequalities. Think of X as the answer to a numerical question about the outcome: the number of heads, a waiting time, a sum of dice, or the number of people receiving the correct hat.

Example 4.3 (Three tosses, one count)**.** Toss a fair coin independently three times, and let X be the number of heads. The eight equally likely outcomes are grouped by their recorded value:

x	outcomes with $X = x$	$\mathbb{P}(X = x)$
0	TTT	1/8
1	HTT, THT, TTH	3/8
2	HHT, HTH, THH	3/8
3	HHH	1/8

The outcome HTH is a string; the value $X(HTH) = 2$ is a number. Different outcomes can give the same number. Consequently, the four possible values $0, 1, 2, 3$ are not equally likely.

4.2 An inequality defines an event

The notation $\{X \geq a\}$ abbreviates

$$\{\omega \in \Omega : X(\omega) \geq a\}.$$

It is a subset of the *original* sample space. To find it, keep the outcomes whose recorded numbers satisfy the inequality. For the three-toss example,

$$\{X \geq 2\} = \{HHT, HTH, THH, HHH\}, \qquad \mathbb{P}(X \geq 2) = \frac{4}{8}.$$

Likewise, $\{X > 2\} = \{HHH\}$ has probability $1/8$. We usually omit the braces when writing probabilities.

Event operations follow the same logic as before:

$$\{a < X \leq b\} = \{X > a\} \cap \{X \leq b\}, \qquad \{X \geq a\}^c = \{X < a\}.$$

This is all the event structure needed for most calculations with a random variable.

Example 4.4 (Several records of the same dice)**.** For two dice, an outcome is an ordered pair (i, j). We can record $S(i, j) = i + j$, $M(i, j) = \max(i, j)$, or $I(i, j) = 1$ if $i = j$ and 0 otherwise. The event $S = 4$ consists of $(1, 3), (2, 2), (3, 1)$, while $M = 4$ contains seven pairs. For independent fair dice their probabilities are $3/36$ and $7/36$. The functions answer different questions about the same experiment.

4.3 The distribution collects the weights

For a variable with finitely or countably many possible values, its *probability mass function* (PMF) is

$$p_X(x) = \mathbb{P}(X = x).$$

It is nonnegative and its sum over all possible values is one. To find the distribution, group the original outcomes that have the same value and add their probabilities. For any set B of possible recorded values,

$$\mathbb{P}(X \in B) = \sum_{x \in B} p_X(x).$$

Thus we can answer questions about X without retaining every detail of the original outcome.

Outcome, value, event, probability

For the three-toss example, HHT is an outcome, 2 is its recorded value, $\{X \geq 2\}$ is an event, and $1/2$ is that event's probability. These four objects play different roles.

4.4 The cumulative distribution function

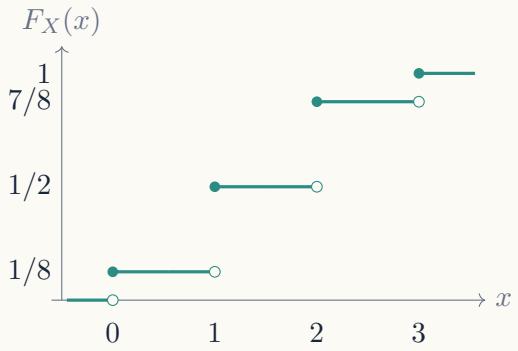
The *cumulative distribution function* (CDF) records all threshold probabilities at once:

$$F_X(x) = \mathbb{P}(X \leq x).$$

For the number of heads in three fair tosses,

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1/8, & 0 \le x < 1, \\ 1/2, & 1 \le x < 2, \\ 7/8, & 2 \le x < 3, \\ 1, & x \ge 3. \end{cases}$$

The jump at each value is its probability mass. A larger threshold includes more outcomes, so a CDF cannot decrease. It approaches zero far to the left and one far to the right, and is right-continuous: the value at a jump includes the mass at that point.



If $a < b$, subtraction of nested events gives

$$\mathbb{P}(a < X \le b) = F_X(b) - F_X(a), \qquad \mathbb{P}(X > a) = 1 - F_X(a).$$

For an integer-valued variable and an integer k, $\mathbb{P}(X = k) = F_X(k) - F_X(k-1)$ and $\mathbb{P}(X \ge k) = 1 - F_X(k-1)$.

Common trap

$1 - F_X(a)$ means $\mathbb{P}(X > a)$, not $\mathbb{P}(X \ge a)$. The distinction matters when X can equal a with positive probability. In the toss example, $\mathbb{P}(X > 2) = 1/8$ but $\mathbb{P}(X \ge 2) = 1/2$.

4.5 Expectation is a weighted average

Suppose the experiment has a finite sample space, with outcome weights $w_\omega = \mathbb{P}(\{\omega\})$. The *expectation*, or *mean*, of X is

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega) w_\omega.$$

Multiply each recorded value by the probability of its outcome, then add. If the N outcomes are equally likely, this becomes the ordinary average $\frac{1}{N}\sum_\omega X(\omega)$. Otherwise the weights matter.

We can do the same calculation from the distribution:

$$\boxed{\mathbb{E}[X] = \sum_x x\, p_X(x).}$$

Here the sum runs over the distinct values of X. To prove the equality, group outcomes with the same recorded value x. Their contribution is $x \sum_{\omega: X(\omega)=x} w_\omega = x\, p_X(x)$.

Example 4.5 (A mean need not be a possible result)**.** A fair die U has mean $(1+2+\cdots+6)/6 = 7/2$. The number of heads in three independent fair tosses has mean $0(1/8)+1(3/8)+2(3/8)+3(1/8) = 3/2$. Neither 3.5 on a die nor 1.5 heads is an outcome of one experiment. Expectation is a balance point, not a prediction of the next individual result.

Example 4.6 (A game with unequal weights)**.** Roll a fair die. The net payoff X is -2 on faces $1, 2, 3$, 1 on faces $4, 5$, and 4 on face 6. Its distribution and two useful weighted columns are

x	-2	1	4	sum
$p_X(x)$	1/2	1/3	1/6	1
$x\, p_X(x)$	-1	1/3	2/3	0
$x^2 p_X(x)$	2	1/3	8/3	5

Thus $\mathbb{E}[X] = 0$. Averaging the three possible payoffs equally would give 1, which is wrong: the three values do not have equal probabilities. The game is *fair in expectation.* It still loses money half the time. The last row will measure its variability.

With many independent repetitions, averages tend to settle near the mean; Chapter 7 will state the hypotheses and prove a law of large numbers. The definition of expectation itself is the weighted sum above. It does not require running a simulation or assuming that a short run will resemble the average.

4.6 Functions and sums of random variables

If $Y = g(X)$, apply g to the value actually observed. The event $\{Y \geq a\}$ is $\{\omega : g(X(\omega)) \geq a\}$. Several values of X may give the same value of Y; their probabilities must be combined when finding the distribution of Y.

Theorem 4.7 (Expectation of a function, often called LOTUS)**.** *On a finite probability space,*

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x).$$

Proof. Group the original outcome sum by $X(\omega) = x$. Within each group, $g(X(\omega))$ has the constant value $g(x)$. □

In particular, $\mathbb{E}[X^2] = \sum_x x^2 p_X(x)$: square the values, leaving their probabilities unchanged. There is no need to compute the distribution of X^2 first.

Theorem 4.8 (Linearity)**.** *For random variables on the same finite probability space and constants a, b,*

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

More generally, the expectation of a finite sum is the sum of its expectations. Independence is not required.

Proof. Expand the defining sum:

$$\sum_{\omega\in\Omega}(aX(\omega)+bY(\omega))w_\omega = a\sum_{\omega}X(\omega)w_\omega + b\sum_{\omega}Y(\omega)w_\omega.$$

□

Example 4.9 (Two dice, and one die recorded twice)**.** For two fair dice U, V, $\mathbb{E}[U+V]=7$ and $\mathbb{E}[U-V]=0$. These statements hold whenever each die is marginally fair, even if the dice are dependent. If the second record is the first die again, $V=U$, linearity still gives $\mathbb{E}[U+V]=7$. Independence will matter when we ask about the spread of that sum.

Common trap

Linearity applies to sums and constant multiples. In general, $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$. In the payoff example, $\mathbb{E}[X^2]=5$ but $(\mathbb{E}[X])^2=0$. Likewise $\mathbb{E}[XY]=\mathbb{E}[X]\mathbb{E}[Y]$ requires an additional justification; putting $Y=X$ already supplies a counterexample.

4.7 Variance measures spread around the mean

Write $m=\mathbb{E}[X]$. The signed deviations cancel: $\mathbb{E}[X-m]=0$. To measure spread, square before averaging.

Definition 4.10. The *variance* and *standard deviation* are

$$\mathrm{Var}(X)=\mathbb{E}[(X-m)^2]=\sum_x (x-m)^2 p_X(x), \qquad \sigma_X=\sqrt{\mathrm{Var}(X)}.$$

Variance is nonnegative. It is zero exactly when $X=m$ on every outcome with positive probability. The standard deviation has the same units as X; variance has squared units. In the game above, variance is 5 and standard deviation is $\sqrt{5}$, in payoff units.

Proposition 4.11 (A useful computation formula)**.**

$$\mathrm{Var}(X)=\mathbb{E}[X^2]-(\mathbb{E}[X])^2.$$

Proof. Expand $(X-m)^2=X^2-2mX+m^2$ and use linearity: $\mathbb{E}[(X-m)^2]=\mathbb{E}[X^2]-2m^2+m^2$. □

The square of an average and the average of squares are different. The formula also proves $(\mathbb{E}[X])^2 \le \mathbb{E}[X^2]$.

Example 4.12 (Compute both moments before subtracting)**.** For a fair die,

$$\mathbb{E}[U]=\frac{7}{2}, \qquad \mathbb{E}[U^2]=\frac{1+4+9+16+25+36}{6}=\frac{91}{6}, \qquad \mathrm{Var}(U)=\frac{91}{6}-\frac{49}{4}=\frac{35}{12}.$$

For the three-toss head count, $\mathbb{E}[X]=3/2$ and $\mathbb{E}[X^2]=(3+12+9)/8=3$, so $\mathrm{Var}(X)=3/4$.

A reliable calculation

List distinct values and their probabilities; check the probabilities sum to one. Compute $\sum_x xp_X(x)$ and $\sum_x x^2 p_X(x)$. Subtract the *square* of the first total from the second total. A negative result signals a calculation error. These are properties of a specified probability model; no sample-variance denominator such as $n-1$ appears here.

Proposition 4.13 (Shifts and changes of scale)**.** *For constants a, b,*

$$\mathbb{E}[aX+b] = a\mathbb{E}[X]+b, \qquad \operatorname{Var}(aX+b) = a^2 \operatorname{Var}(X), \qquad \sigma_{aX+b} = |a|\sigma_X.$$

The new mean is $am+b$, so the deviation from it is $a(X-m)$. Adding a constant shifts all values and their mean together. For example, charging a fixed fee changes a game's expected net return but does not change its variance.

Mean alone does not determine variability. A constant payoff of zero and the game with payoffs $-2, 1, 4$ both have mean zero, but their variances are zero and five. Even mean and variance together do not determine the whole distribution; Exercise 7 asks for an example.

4.8 Indicators connect counting to averages

For an event A, its *indicator* $\mathbf{1}_A$ is 1 when A occurs and 0 otherwise. If $p = \mathbb{P}(A)$, then

$$\mathbb{E}[\mathbf{1}_A] = p, \qquad \mathbb{E}[\mathbf{1}_A^2] = p, \qquad \operatorname{Var}(\mathbf{1}_A) = p(1-p).$$

The second equality follows because squaring zero or one changes nothing. Chapter 5 gives this two-value distribution the name *Bernoulli*.

If $N = \sum_{i=1}^{r} \mathbf{1}_{A_i}$ counts how many events occur, linearity gives

$$\boxed{\mathbb{E}[N] = \sum_{i=1}^{r} \mathbb{P}(A_i).}$$

The events can overlap and can be dependent. A useful strategy is to find the mean of a count before attempting its full distribution.

Example 4.14 (The expected number of correct hats)**.** For a uniform permutation of $n \geq 1$ hats, let I_i indicate that person i gets the right hat. Then $N = \sum_{i=1}^{n} I_i$ and $\mathbb{E}[N] = n(1/n) = 1$. No derangement sum is needed. This does not mean every assignment has one fixed point: $\mathbb{P}(N=0) = D_n/n!$ approaches $1/e$.

Example 4.15 (Matching birthday pairs)**.** Let I_{ij} indicate a match between people i and j. The number of matching pairs is $C = \sum_{i<j} I_{ij}$, so $\mathbb{E}[C] = \binom{n}{2}/d$ in the independent uniform d-day model. With $n = 23, d = 365$, the mean is $253/365 \approx .69315$, while $\mathbb{P}(C \geq 1) \approx .50730$. Several matching pairs may occur together; an expected count and the probability of at least one occurrence answer different questions.

4.8.1 Small indicators, large answers

The difficult step is often choosing what to count. The following examples use the same identity, $\mathbb{E}[\sum I_j] = \sum \mathbb{P}(I_j = 1)$, on very different objects. Each proof uses a finite probability space.

Example 4.16 (Records in a random order)**.** Shuffle n distinct scores uniformly and reveal them one at a time. The jth score is a *record* if it exceeds every previous score; the first score is always a record. Among the first j scores, symmetry makes each position equally likely to hold the largest. Thus its record indicator I_j has mean $1/j$. The number R_n of records satisfies

$$\mathbb{E}[R_n] = \sum_{j=1}^{n} \frac{1}{j} = H_n,$$

where $H_0 = 0$ and H_n is the nth harmonic number. For example, four distinct scores give mean $H_4 = 25/12$, although an individual ordering has an integer number of records. We did not need to count orderings with each possible record total.

Example 4.17 (Inversions in a shuffle)**.** In a uniform permutation π of $1, \ldots, n$, an *inversion* is a pair $i < j$ with $\pi(i) > \pi(j)$. For each of the $\binom{n}{2}$ pairs, swapping the two values pairs the greater-first and smaller-first cases, so the inversion probability is $1/2$. Therefore

$$\mathbb{E}[\text{number of inversions}] = \frac{1}{2}\binom{n}{2} = \frac{n(n-1)}{4}.$$

Indicators sharing a position need not be independent; linearity still applies. For $n = 1$ the empty sum is zero.

4.8.2 Spaghetti loops: count the closures

Start with $n \geq 1$ separate strands, each with two distinguishable ends. Pair all $2n$ ends uniformly at random and tie each pair together. There are $(2n-1)(2n-3)\cdots 1$ perfect matchings: choose the partner of a specified end, remove those two ends, and repeat. Equivalently, repeatedly choose any specified free end and choose its partner uniformly among the other free ends. Every full matching has the same product probability. Closed loops stay closed; only *free* ends can be selected.

At any intermediate stage, every unfinished component is an open chain with exactly two free ends. If there are m open chains, there are $2m$ free ends. Tying two ends has just two possibilities:



Same chain:
one loop closes

Different chains:
one longer chain remains

Either operation reduces the number of open chains by one. Given any completed history with m chains remaining, the selected end has $2m-1$ possible partners, exactly one of which is the other end of its own chain. The chance that this tie closes a loop is therefore $1/(2m-1)$, regardless of the history.

Let J_m indicate closure at the tie made with m chains remaining. Since the conditional probability has the same value for every possible history, total probability gives $\mathbb{E}[J_m] = 1/(2m-1)$. Every final loop was closed by exactly one tie, so the loop count L_n is $\sum_{m=1}^{n} J_m$. Hence

$$\boxed{\mathbb{E}[L_n] = \sum_{m=1}^{n} \frac{1}{2m-1} = H_{2n} - \frac{1}{2}H_n.}$$

For $n = 1, 2, 3$ the means are $1, 4/3, 23/15$, respectively. With no strands there are no loops, consistent with the empty sum at $n = 0$. The indicators refer to *closure steps*, not to original strands: one loop may contain several original strands. No independence assertion is needed for the expectation calculation. This classical example also appears in the Stanford EE 178/278A expectation notes listed in the references.

4.8.3 An average proves that a good cut exists

A finite graph consists of vertices and edges joining distinct vertices. Suppose it has e edges. Color each vertex red or blue independently with equal probabilities. An edge crosses the resulting *cut* if its ends have different colors. For each edge this probability is $2(1/2)(1/2) = 1/2$, so

$$\mathbb{E}[\text{number of crossing edges}] = e/2.$$

At least one coloring has at least $e/2$ crossing edges: otherwise every value, and hence their average, would be smaller than $e/2$. Thus every such graph has a cut containing at least half its edges. On a triangle the edge indicators are not mutually independent, since the number of crossing edges can only be zero or two. Their dependence does not obstruct the average argument.

Choose what contributes one

For records, count successful positions. For inversions, count pairs. For loops, count closing ties. For a cut, count crossing edges. The right decomposition can make a complicated distribution unnecessary.

4.9 When variances of sums add

For finite-valued variables, independence means $\mathbb{P}(X = x, Y = y) = p_X(x)p_Y(y)$ for all possible x, y. Under that hypothesis,

$$\mathbb{E}[XY] = \sum_{x,y} xy\, p_X(x)p_Y(y) = \left(\sum_x x p_X(x)\right)\left(\sum_y y p_Y(y)\right).$$

Thus $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$. This is a consequence of independence, unlike linearity.

Put $m_X = \mathbb{E}[X]$ and $m_Y = \mathbb{E}[Y]$. Expanding the centered square gives

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\mathbb{E}[(X - m_X)(Y - m_Y)].$$

The last expectation is called the *covariance*; Chapter 6 studies it systematically. For independent variables it vanishes. The same expansion works for a finite sum:

$$\operatorname{Var}\left(\sum_{i=1}^{r} X_i\right) = \sum_{i=1}^{r} \operatorname{Var}(X_i) \quad \text{when the variables are independent.}$$

Pairwise independence already suffices, because only products of pairs appear in the expanded square.

Example 4.18 (Independent dice versus repeating the same die)**.** If U, V are independent fair dice, $\mathrm{Var}(U+V) = 2(35/12) = 35/6$. If instead $V = U$, the sum is $2U$ and its variance is $35/3$. Both sums have mean seven, but repeating the same random result produces twice the variance. If $V = 7-U$, each individual die record is still fair, while $U+V = 7$ is constant and has variance zero.

Example 4.19 (Averages of independent observations)**.** If $X_1, \ldots, X_r$ are independent copies of a finite-valued variable with mean m and variance v, then

$$\mathbb{E}\left[\frac{X_1+\cdots+X_r}{r}\right] = m, \qquad \mathrm{Var}\left(\frac{X_1+\cdots+X_r}{r}\right) = \frac{v}{r}.$$

The standard deviation of the average is $\sqrt{v}/\sqrt{r}$. Quadrupling the number of independent observations halves this spread.

Return to the hats: variance without the full distribution

For $n \geq 2$, two specified people both get their hats back with probability $1/[n(n-1)]$. Since $N(N-1) = \sum_{i \neq j} I_i I_j$ counts *ordered* pairs of fixed points,

$$\mathbb{E}[N(N-1)] = n(n-1)\frac{1}{n(n-1)} = 1.$$

Therefore $\mathbb{E}[N^2] = \mathbb{E}[N(N-1)] + \mathbb{E}[N] = 2$ and $\mathrm{Var}(N) = 2-1 = 1$. For $n = 1$ the single hat always returns correctly, so the variance is zero. The indicators are dependent; their variances cannot simply be added.

4.10 Two bounds from averages

Theorem 4.20 (Markov and Chebyshev)**.** *If $X \geq 0$, $\mathbb{E}[X] < \infty$, and $a > 0$, then $\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$. If X has mean m and finite variance v, then, for $t > 0$,*

$$\mathbb{P}(|X-m| \geq t) \leq \frac{v}{t^2}.$$

In particular, if $\sigma = \sqrt{v} > 0$ and $k > 0$, then $\mathbb{P}(|X-m| \geq k\sigma) \leq 1/k^2$.

Proof. The pointwise inequality $X \geq a\mathbf{1}_{\{X \geq a\}}$ gives $\mathbb{E}[X] \geq a\mathbb{P}(X \geq a)$. Apply this result to $(X-m)^2$ at threshold t^2 to obtain Chebyshev. □

For the birthday pair count, Markov at $a = 1$ gives $\mathbb{P}(C \geq 1) \leq \mathbb{E}[C] = \binom{n}{2}/d$, recovering the union bound. These estimates use averages alone; a full distribution can give a much sharper answer.

4.11 Practice with discrete variables

Exercises

1. Find the CDF of the number of heads in two fair independent flips. [Fluency]

2. On $\{1, 2, 3, 4\}$, let $\mathbb{P}(X = k) = k/10$. Find $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, $\mathrm{Var}(X)$, and $\mathbb{P}(X \geq 3)$. Explain why averaging the four values equally gives the wrong mean. [Fluency]

3. Prove $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$. [Reasoning]

4. Find the expected number of fixed points in a uniform permutation of $n \geq 1$ objects using indicators. [Reasoning]

5. Give a net-payoff distribution with mean zero and large variance. Compute both quantities and explain the risk of one play. [Exploration]

6. Use Chebyshev to bound the chance of lying at least three standard deviations from the mean. For a fair die, find that probability exactly. [Reasoning]

7. Construct two different discrete distributions with the same mean and variance. [Exploration]

8. For two independent fair dice, list the ordered outcomes in $\{U + V \geq 10\}$ and calculate its probability. [Fluency]

9. Toss independently twice with heads probability $1/3$. Find the PMF and CDF of the head count, its mean and variance, and $\mathbb{P}(X \geq 1)$. [Fluency]

10. For the three-fair-toss count X, let $Y = (X - 1)^2$. Find its distribution and mean, combining repeated values correctly. [Reasoning]

11. A fair die gives net payoff -2 on $1, 2, 3$, 1 on $4, 5$, and 4 on 6. Find the mean, variance, and chance of a positive payoff. What changes if a fee of one unit is charged? [Fluency]

12. For four uniformly assigned hats, use $\binom{4}{r} D_{4-r}/4!$ to find the fixed-point distribution, its mean and variance. Explain why three fixed points are impossible. [Reasoning]

13. For 23 independent uniform birthdays on 365 days, find the mean pair count and bound the collision probability. Why do they differ? [Reasoning]

14. For a fair die U, find the mean, variance, and standard deviation of $2U - 3$. [Fluency]

15. Let I indicate heads in one fair toss and put $J = 1 - I$. Compare $\mathrm{Var}(I + J)$ with $\mathrm{Var}(I) + \mathrm{Var}(J)$. Repeat with J the head indicator of an independent fair toss. [Reasoning]

16. For two independent fair dice U, V, find the means and variances of $U + V$, $U - V$, and $(U + V)/2$. [Fluency]

17. Use $N(N - 1)$ to prove that the number N of correct hats has variance one for $n \geq 2$. Explain the exceptional case $n = 1$. [Reasoning]

18. Prove $\mathbb{E}[(X - a)^2] = \mathrm{Var}(X) + (\mathbb{E}[X] - a)^2$ for a constant a. Which constant minimizes this expected squared error? [Reasoning]

4.12 For later: a countably infinite list of values

The finite calculations above require only sums and algebra. For countably many values, use the same formulas as limits of partial sums. We call the expectation finite when $\sum_x |x| p_X(x) < \infty$. This condition permits regrouping terms and preserves linearity. A finite second moment similarly makes the variance formulas valid.

A nonnegative variable may instead have expectation $+\infty$. For example, assign the positive integer k probability

$$p(k) = \frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}.$$

The first m probabilities telescope to $1 - 1/(m+1)$, so the total is one. But $\mathbb{E}[X] = \sum_{k=1}^{\infty} 1/(k+1) = \infty$. A valid probability distribution need not have a finite mean. Appendix A reviews the series involved; Chapter 5 introduces geometric and Poisson distributions with finite moments.

4.13 For later: densities and continuous averages

A *probability density* f_X is a nonnegative function with total area one. For a variable described by this density,

$$\mathbb{P}(a < X \le b) = \int_a^b f_X(t)\,dt, \qquad F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt.$$

An interval of zero width has zero area, so $\mathbb{P}(X = a) = 0$. Endpoint choices do not affect interval probabilities for these variables. The height $f_X(a)$ is not a point probability and may exceed one. Section A.5 reviews the calculus needed here.

The corresponding average of a function is

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\,dx$$

when the integral with $|g(x)|$ is finite. Taking $g(x) = x$ or x^2 gives the first two moments; the variance identity remains the same.

Example 4.21 (An increasing density)**.** Let $f_X(x) = 2x$ on $[0, 1]$ and zero elsewhere. Its total area is one, and

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x^2, & 0 \le x \le 1, \\ 1, & x > 1. \end{cases}$$

Thus $\mathbb{P}(X \ge 1/2) = 3/4$, $\mathbb{E}[X] = \int_0^1 2x^2\,dx = 2/3$, and $\mathbb{E}[X^2] = \int_0^1 2x^3\,dx = 1/2$, giving variance $1/18$. Although $f_X(3/4) = 3/2$, the probability of the single value $3/4$ is zero. Where the density is continuous, $F_X' = f_X$.

Later practice: continuous models

19. A density is proportional to x on $[0, 2]$ and zero elsewhere. Normalize it and find its mean and variance. [Fluency]

20. A CDF is 0 below zero, x^3 on $[0, 1]$, and 1 above one. Find its density, $\mathbb{P}(X \geq 1/2)$, mean, and variance. [Fluency]

Further moments

The kth raw moment is $\mathbb{E}[X^k]$, and the kth central moment is $\mathbb{E}[(X-\mathbb{E}[X])^k]$, when these expectations are finite. One later tool packages moments in $M_X(t) = \mathbb{E}[e^{tX}]$. If this moment-generating function is finite on an interval around zero, $M_X^{(k)}(0) = \mathbb{E}[X^k]$. For finite-valued variables this follows by differentiating a finite sum. Neither this tool nor continuous densities are needed for the discrete expectation and variance calculations in this chapter.

Further practice: linearity in action

21. Eight distinct scores arrive in a uniformly random order. Find the expected total number of records and the expected number of records in positions 5 through 8. [Fluency]

22. In a uniform permutation of five labels, find the expected number of inversions and the expected number of adjacent descents $\pi(i) > \pi(i+1)$. Why are these different counts? [Reasoning]

23. Enumerate the three matchings of the four ends of two strands. Find the distribution and mean of the loop count. For general $n \geq 1$, find the expected number of loops containing exactly one original strand. [Reasoning]

24. In $n \geq 1$ independent tosses with heads probability p, a head run is a maximal consecutive block of heads. Prove its expected count is $p + (n-1)p(1-p)$. Check $p = 0, 1$ and ten fair tosses. [Reasoning]

25. Color a triangle's vertices independently red or blue with equal probabilities. Find the crossing-edge count's distribution and mean; show its three edge indicators are not mutually independent. Extend the half-the-edges existence argument to nonnegative edge weights. [Synthesis]

Chapter 5

A library of chance mechanisms

Discrete distributions first; continuous models later

Guiding question 5.1. Which probability laws recur because the same mechanisms recur?

For a discrete variable, its distribution lists the possible values and their probabilities. The coin counts, dice, and hat counts of Chapter 4 already have distributions. A *named family* collects experiments whose probability tables have the same form. Learn the experiment before memorizing the name.

A route through the next topics

Start with finite distributions: a uniform choice, one success indicator, a count of independent successes, and sampling without replacement. For each, identify the possible values, calculate their probabilities, then find the mean and variance. Geometric waiting times and Poisson counts come next as countably infinite examples. The coupon collector then brings waiting times, expectation, variance, and inclusion–exclusion together, first for uniform types and then for different probabilities. The continuous section is for later, after the calculus refresher. Additional examples derive negative-binomial, gamma, and beta laws, followed by a full section on Poissonization and its connection to independent exponential clocks.

5.1 A uniform choice from a finite list

Choose one of $1, \dots, m$ uniformly, where $m \geq 1$. Then $\mathbb{P}(U = k) = 1/m$ for $k = 1, \dots, m$, and zero for other values. The usual sum and sum-of-squares identities give

$$\mathbb{E}[U] = \frac{m+1}{2}, \qquad \mathbb{E}[U^2] = \frac{(m+1)(2m+1)}{6}, \qquad \operatorname{Var}(U) = \frac{m^2-1}{12}.$$

For example, choosing a label uniformly from $1, \dots, 8$ gives mean $9/2$ and variance $21/4$. The case $m = 1$ is constant, with variance zero.

A uniform distribution on any finite set $\{u_1, \dots, u_m\}$ averages those values with weights $1/m$. The formulas above are specifically for the consecutive integers $1, \dots, m$. Uniform outcomes also need not produce uniform recorded values: the sum of two fair dice is not uniform on $2, \dots, 12$.

5.2 Bernoulli: one event, recorded as zero or one

If an event has probability p, its indicator I has distribution

$$\mathbb{P}(I = 1) = p, \qquad \mathbb{P}(I = 0) = 1 - p, \qquad 0 \le p \le 1.$$

We call this the *Bernoulli distribution* with parameter p. There are only two values, so directly

$$\mathbb{E}[I] = p, \qquad \mathbb{E}[I^2] = p, \qquad \mathrm{Var}(I) = p(1-p).$$

The parameter p is a probability, the value of I is zero or one, and its expectation can lie between these values. At $p = 0$ or 1 the result is certain and the variance vanishes.

Example 5.2. Let I record whether a fair die shows a six. Then $p = 1/6$, $\mathbb{E}[I] = 1/6$, and $\mathrm{Var}(I) = 5/36$. A success may mean a correct answer, a defective item, or a birthday match for a specified pair; a *success* need not mean something desirable.

5.3 Binomial: count successes in a fixed number of trials

Perform n independent trials, each with success probability p, and let X count the successes. For $0 < p < 1$ and $k = 0, \dots, n$,

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

A specified success/failure string has probability $p^k(1-p)^{n-k}$; choose its k success positions in $\binom{n}{k}$ ways. The binomial theorem verifies that the masses sum to one. We write $X \sim \mathrm{Bin}(n, p)$. If $p = 0$, $X = 0$; if $p = 1$, $X = n$; if $n = 0$, $X = 0$.

Write $X = I_1 + \cdots + I_n$. Linearity and the independent-variance rule give

$$\mathbb{E}[X] = np, \qquad \mathrm{Var}(X) = np(1-p).$$

This calculation avoids summing k^2 against the binomial probabilities.

Example 5.3 (Five trials)**.** For $X \sim \mathrm{Bin}(5, 1/3)$,

$$\mathbb{P}(X = 3) = \binom{5}{3}(1/3)^3(2/3)^2 = \frac{40}{243}, \qquad \mathbb{E}[X] = \frac{5}{3}, \qquad \mathrm{Var}(X) = \frac{10}{9}.$$

For a threshold it can be easier to use a complement: $\mathbb{P}(X \ge 2) = 1 - \mathbb{P}(X = 0) - \mathbb{P}(X = 1)$.

Common trap

A count is binomial only when its experiment satisfies the assumptions: a fixed number of trials, two outcomes per trial, a common success probability, and independence. Recording the same toss ten times gives a count of either zero or ten, not a binomial count of ten independent tosses. Sampling from a finite population without replacement also requires a different model.

5.4 Hypergeometric: sample without replacement

A population contains K marked objects among N distinct objects. Choose a subset of m objects uniformly, and let X count marked objects in the sample. There are $\binom{N}{m}$ equally likely samples. To obtain exactly k marked objects, choose those k from the marked group and the other $m-k$ from the unmarked group:

$$\mathbb{P}(X=k)=\frac{\binom{K}{k}\binom{N-K}{m-k}}{\binom{N}{m}}.$$

Here $N\geq 1$, $0\leq K,m\leq N$, and the possible values satisfy $\max(0,m-(N-K))\leq k\leq \min(m,K)$. The formula is normalized because these disjoint cases partition all m-element samples. This is the *hypergeometric distribution*.

Example 5.4 (Two draws from five balls)**.** An urn contains three red and two blue balls. For a uniform sample of two balls without replacement, the number X of red balls has table

k	0	1	2
$\mathbb{P}(X=k)$	1/10	6/10	3/10

Thus $\mathbb{E}[X]=6/5$, $\mathbb{E}[X^2]=9/5$, and $\mathrm{Var}(X)=9/25$. With replacement the count would instead be $\mathrm{Bin}(2,3/5)$, with the same mean but variance $12/25$.

For the general moments, regard the sample as the first m positions in a uniform random ordering of the population. Put $p=K/N$ and let I_j indicate a marked object in position j. Every position has marked probability p, so $\mathbb{E}[X]=mp$. For $N\geq 2$ and $i\neq j$,

$$\mathbb{E}[I_iI_j]=\frac{K(K-1)}{N(N-1)},\qquad \mathbb{E}[I_iI_j]-p^2=-\frac{p(1-p)}{N-1}.$$

The variance-of-a-sum expansion from Chapter 4 gives

$$\mathrm{Var}(X)=mp(1-p)-m(m-1)\frac{p(1-p)}{N-1}=mp(1-p)\frac{N-m}{N-1}.$$

For $1\leq m\leq N$, the last factor is at most one: sampling without replacement has no more variance than the corresponding binomial count. When $m=N$, the count is the fixed number K, so its variance is zero. For a one-object population the count is constant as well.

5.5 For later: geometric waiting times

Now allow a countably infinite list of possible values. In independent trials with fixed success probability $0<p\leq 1$, let T count trials up to and including the first success. Then

$$\mathbb{P}(T=k)=(1-p)^{k-1}p,\qquad k=1,2,\ldots.$$

For $0<p<1$, the first $k-1$ trials must fail and the kth succeed. The geometric series sums the masses to one. For $p=1$, $T=1$ with certainty. If $p=0$, there is no finite waiting time to a success, so it is excluded from this distribution.

The event $T > r$ means r initial failures, hence $\mathbb{P}(T > r) = (1-p)^r$ for nonnegative integers r. For $r = 0$ the probability is one, including at $p = 1$. For $0 < p < 1$ and integers $s, t \geq 0$,

$$\mathbb{P}(T > s + t \mid T > s) = \frac{(1-p)^{s+t}}{(1-p)^s} = (1-p)^t = \mathbb{P}(T > t).$$

This is *memorylessness*: after any number of failures, the distribution of the additional wait is unchanged. For $p = 1$, only conditioning events of positive probability can be used.

Mean and variance from tail sums

A positive integer T satisfies $T = \sum_{j \geq 1} \mathbf{1}_{\{T \geq j\}}$ and $T^2 = \sum_{j \geq 1} (2j-1)\mathbf{1}_{\{T \geq j\}}$. First truncate these nonnegative sums; letting the truncation grow gives

$$\mathbb{E}[T] = \sum_{j \geq 1} \mathbb{P}(T \geq j) = \sum_{j \geq 1} (1-p)^{j-1} = \frac{1}{p}.$$

The geometric-series calculation $\sum_{j \geq 1} j r^{j-1} = 1/(1-r)^2$, reviewed in Section A.8, similarly gives

$$\mathbb{E}[T^2] = \frac{2}{p^2} - \frac{1}{p} = \frac{2-p}{p^2}, \qquad \mathrm{Var}(T) = \frac{1-p}{p^2}.$$

These sums converge for $p > 0$; at $p = 1$ the values follow directly. Waiting for a six on independent fair rolls gives mean 6, variance 30, and $\mathbb{P}(T > 5) = (5/6)^5$.

Common trap

Some books count *failures before* the first success. That variable is $T - 1$, supported on $0, 1, \ldots$, with mean $(1-p)/p$ and the same variance. State which convention you use.

5.6 For later: Poisson counts

For $\lambda > 0$, the *Poisson distribution* assigns

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

The exponential series makes the total mass $e^{-\lambda} \sum_{k \geq 0} \lambda^k / k! = 1$. The limiting case $\lambda = 0$ places all mass at zero. Shifting indices in absolutely convergent sums gives

$$\mathbb{E}[X] = e^{-\lambda} \lambda \sum_{j \geq 0} \frac{\lambda^j}{j!} = \lambda, \qquad \mathbb{E}[X(X-1)] = \lambda^2.$$

Consequently $\mathbb{E}[X^2] = \lambda^2 + \lambda$ and $\mathrm{Var}(X) = \lambda$. Factorial moments make this calculation short, just as they did for hats.

The law arises from many independent rare opportunities. For fixed $\lambda > 0$ and fixed integer $k \geq 0$,

$$\binom{n}{k} (\lambda/n)^k (1 - \lambda/n)^{n-k} \longrightarrow e^{-\lambda} \frac{\lambda^k}{k!}.$$

Indeed, $(n)_k/n^k \to 1$ and $(1-\lambda/n)^n \to e^{-\lambda}$. This explains a Poisson approximation to $\text{Bin}(n,p)$ when p is small and np is near λ; it does not make every count Poisson. In particular, the model's equality of mean and variance is a consequence to check, not a rule for all counts.

For X Poisson with mean 3, $\mathbb{P}(X \geq 2) = 1 - 4e^{-3}$. The number of fixed points of a large uniform permutation approaches Poisson with mean one, as derived in Section A.11. For a finite permutation this is an approximation; the exact law is $\mathbb{P}(N=k) = \binom{n}{k} D_{n-k}/n!$.

5.7 Coupon collection: waiting to see every type

An album has $n \geq 1$ types of coupon. Each purchase gives one coupon, independently and uniformly chosen from the n types; duplicates are possible. Let T_n be the number of purchases needed to see every type, starting with an empty album. Equivalently, how many independent rolls of a fair die are needed to see all six faces?

This reverses the birthday question. There we stopped at the first repeat; here we keep going until nothing is missing. With sampling without replacement, completion would take exactly n draws. Repeated types are what make this problem interesting.

Break the wait into stages

After k distinct types have appeared, let W_k count additional draws up to and including the next new type, for $k = 0, \ldots, n-1$. There are $n-k$ unseen types. Until one appears, the chance of a new type on each draw stays at

$$p_k = \frac{n-k}{n}, \qquad W_k \sim \text{Geometric}(p_k), \qquad \mathbb{E}[W_k] = \frac{n}{n-k}.$$

In particular $W_0 = 1$: the first draw is always new. The collection time is the sum of these stages,

$$T_n = W_0 + W_1 + \cdots + W_{n-1}.$$

Linearity of expectation now does the work. Define the *harmonic numbers* by $H_0 = 0$ and $H_n = \sum_{j=1}^{n} 1/j$. Then

$$\boxed{\mathbb{E}[T_n] = \frac{n}{n} + \frac{n}{n-1} + \cdots + \frac{n}{1} = nH_n.} \tag{5.1}$$

Every stage has finite expectation, so completion occurs with probability one, although there is no fixed upper limit on the wait when $n \geq 2$. The last missing type alone takes n further draws on average.

Example 5.5 (Every face of a die)**.** For six equally likely types, the successive expected waits are

$$1, \quad \frac{6}{5}, \quad \frac{6}{4}, \quad \frac{6}{3}, \quad \frac{6}{2}, \quad 6.$$

Their sum is $6H_6 = 14.7$ rolls. Seeing the first five types takes 8.7 rolls on average, and the last one takes another six. The mean is an average over repetitions of the entire experiment, not a deadline by which every collection will finish.

For a sense of scale, comparing rectangles with the area under $1/x$ gives, for $n \geq 1$,

$$\log(n+1) = \int_1^{n+1} \frac{dx}{x} \leq H_n \leq 1 + \int_1^n \frac{dx}{x} = 1 + \log n.$$

Thus $\mathbb{E}[T_n]$ grows like $n \log n$, rather than a fixed multiple of n. Here and below, log is the natural logarithm. The integral comparison can wait until Appendix A; the exact mean above uses only geometric expectations and a finite sum.

Variance: justify independence before adding

The expectation calculation did not require independence of the stages. For the variance, we do need it. After any specified completed history, future draws are still independent and uniform. The conditional law of W_k is always geometric with parameter $(n-k)/n$, regardless of which types have appeared or how long earlier stages took. Applying the multiplication rule to these conditional probabilities shows that $W_0, \ldots, W_{n-1}$ are mutually independent.

The geometric variance formula therefore gives

$$\begin{aligned} \operatorname{Var}(T_n) &= \sum_{k=0}^{n-1} \frac{1-p_k}{p_k^2} = \sum_{j=1}^{n} \frac{1-j/n}{(j/n)^2} \\ &= \boxed{n^2 \sum_{j=1}^{n} \frac{1}{j^2} - nH_n.} \end{aligned}$$

For a die, this is 38.99 rolls squared, giving standard deviation about 6.24 rolls. For $n = 1$, $T_1 = 1$ and the variance is zero. For $n \geq 2$, the entire collection time is not itself geometric: the success probability changes each time a new type is found.

A deadline: count the types still missing

Fix an integer $m \geq 1$. Let M_m be the number of types missing after m draws, and let A_i be the event that type i is missing. Then

$$M_m = \sum_{i=1}^{n} \mathbf{1}_{A_i}, \qquad \mathbb{P}(A_i) = \left(1 - \frac{1}{n}\right)^m, \qquad \mathbb{E}[M_m] = n\left(1 - \frac{1}{n}\right)^m.$$

These missing-type indicators are generally dependent. Linearity still applies, and the union bound (equivalently, Markov at one) gives

$$\mathbb{P}(T_n > m) = \mathbb{P}(M_m \geq 1) \leq \min\left\{1, n\left(1 - \frac{1}{n}\right)^m\right\} \leq \min\{1, ne^{-m/n}\}. \tag{5.2}$$

We used $1 - u \leq e^{-u}$. Consequently, for $0 < \delta < 1$, any integer $m \geq n(\log n + \log(1/\delta))$ guarantees completion probability at least $1 - \delta$. This is a sufficient bound, not the exact cutoff.

Inclusion–exclusion gives an exact answer: a specified set of j types is missing with probability $(1 - j/n)^m$, so

$$\mathbb{P}(T_n \leq m) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} \left(1 - \frac{j}{n}\right)^m, \qquad m \geq 1. \tag{5.3}$$

At $m = 0$ the completion probability is zero and $M_0 = n$. For a die, the formula gives $\mathbb{P}(T_6 \leq 15) \approx .6442$: a deadline close to the mean is far from a guarantee.

5.7.1 Different probabilities: the rare type matters

Now the n types have probabilities $p_1, \dots, p_n > 0$, summing to one; successive purchases are still independent. Let T be the number of purchases to collect all types, and write $p_A = \sum_{i \in A} p_i$. If some required type has probability zero, completion is impossible and $T = +\infty$; the finite formulas below assume positive probabilities.

The uniform stage argument needs revision. After collecting the set S, the next-new-type probability is $1 - \sum_{i \in S} p_i$. Its value depends on the identities of the collected types, not just their number. For probabilities $1/2, 1/3, 1/6$, a first coupon of type 1 leaves new-type probability $1/2$, whereas a first coupon of type 3 leaves $5/6$. There is no single geometric parameter determined only by the stage index.

Proposition 5.6 (Exact unequal-probability collection formulas)**.** *For integers $m \geq 0$,*

$$\mathbb{P}(T > m) = \sum_{\varnothing \neq A \subseteq \{1,\dots,n\}} (-1)^{|A|+1} (1 - p_A)^m,$$

$$\mathbb{E}[T] = \sum_{\varnothing \neq A \subseteq \{1,\dots,n\}} \frac{(-1)^{|A|+1}}{p_A}.$$

In the tail formula at $m = 0$, every $(1 - p_A)^0$ means 1, including the full-set term: before any draw every specified set is missing.

Proof. For each type i, let A_i be the event that it is missing after m draws. A specified nonempty set A is entirely missing with probability $(1 - p_A)^m$. The event $T > m$ is the union of the A_i; inclusion–exclusion gives the tail formula. At zero draws the alternating coefficients sum to one, as required.

Use $\mathbb{E}[T] = \sum_{m \geq 0} \mathbb{P}(T > m)$. There are finitely many subsets, and each geometric series converges because $p_A > 0$. Summing each series gives $\sum_{m \geq 0} (1 - p_A)^m = 1/p_A$ and the stated mean. This also proves that the mean is finite and completion is almost sure. □

Example 5.7 (Three rarities)**.** For $(p_1, p_2, p_3) = (1/2, 1/3, 1/6)$,

$$\mathbb{E}[T] = 2 + 3 + 6 - \left(\frac{6}{5} + \frac{3}{2} + 2\right) + 1 = \frac{73}{10}.$$

For three equally likely types the mean is $3H_3 = 11/2$. With only two types of probabilities $p, 1 - p$,

$$\mathbb{E}[T] = \frac{1}{p} + \frac{1}{1-p} - 1 = \frac{1}{p(1-p)} - 1 \geq 3,$$

with equality exactly at $p = 1/2$. As p tends to zero the mean diverges. A very common first type does not compensate for a very rare last type.

The same calculation gives a second moment. Since $T^2 = \sum_{m=0}^{T-1} (2m + 1)$,

$$\mathbb{E}[T^2] = \sum_{\varnothing \neq A} (-1)^{|A|+1} \frac{2 - p_A}{p_A^2}, \qquad \mathrm{Var}(T) = \mathbb{E}[T^2] - (\mathbb{E}[T])^2.$$

Indeed $\sum_{m \geq 0} (2m + 1)(1 - p)^m = (2 - p)/p^2$. The sums involve finitely many convergent geometric series, so the calculation is justified without assuming independent first-appearance times.

5.7.2 Remaining waits and practical bounds

Let $R(S)$ be the expected additional draws when precisely the types in S are still missing. Its boundary value is $R(\varnothing) = 0$. For nonempty S, first-step conditioning gives

$$R(S) = 1 + (1 - p_S)R(S) + \sum_{i \in S} p_i R(S \setminus \{i\}),$$

and hence the positive recursion

$$\boxed{R(S) = \frac{1 + \sum_{i \in S} p_i R(S \setminus \{i\})}{p_S}.}$$

For the three-rarity example, if types 2 and 3 are missing, the remaining mean is $3 + 6 - 2 = 7$ draws. Knowing only that two types are missing is insufficient. The subset formula for the mean applies to a missing set as well, with the original, unnormalized probabilities: already collected types still use up draws. The recursion provides another exact way to compute these means, using only smaller missing sets.

Let $p_{\min} = \min_i p_i$. Waiting for the rarest type gives a lower bound, while the maximum of the individual first-appearance times is at most their sum. Thus

$$\frac{1}{p_{\min}} \le \mathbb{E}[T] \le \sum_{i=1}^{n} \frac{1}{p_i}.$$

For deadlines, linearity of missing-type indicators and the union bound give

$$\mathbb{E}[M_m] = \sum_i (1 - p_i)^m, \qquad \mathbb{P}(T > m) \le \min\left\{1, \sum_i (1 - p_i)^m\right\} \le \min\{1, n e^{-m p_{\min}}\}.$$

When $p_i = 1/n$, the exact formulas reduce to the uniform formulas above; group the subsets by cardinality. Section 5.12 gives a second derivation and proves that the uniform choice minimizes the expected completion time among all positive probability vectors.

A Poisson total points forward

If the number of uniform draws is an independent $K \sim \text{Pois}(t)$, then the type counts become independent Poisson variables of mean t/n. Consequently $M_K \sim \text{Bin}(n, e^{-t/n})$ and $\mathbb{P}(T_n \le K) = (1 - e^{-t/n})^n$. This is a different experiment from exactly m draws. Section 5.12 proves and generalizes the identity; its splitting argument uses only discrete probabilities, before its later continuous-clock interpretation.

5.8 Practice: choose and use a discrete model

Exercises

1. For $X \sim \text{Bin}(5, .3)$, find $\mathbb{P}(X \ge 2)$, its mean, and variance. [Fluency]

2. Prove the geometric memoryless property, stating the allowed values and the condition

that the conditioning event has positive probability. [Reasoning]

3. Approximate Bin$(1000, .002)$ by a Poisson law and compare the exact and approximate $\mathbb{P}(X = 0)$. [Exploration]
4. Choose a label uniformly from $1, \ldots, 8$. Find its mean, variance, and probability of exceeding its mean. [Fluency]
5. An event has probability .2. Find the mean and variance of its indicator. Compare counting successes in ten independent repetitions with recording one indicator ten times. [Reasoning]
6. Draw two balls without replacement from three red and two blue balls. Derive the complete red-count distribution, mean, and variance. Repeat with replacement. [Fluency]
7. For each discrete family, describe one experiment that fits and one that violates a stated modeling assumption. [Exploration]
8. Roll a fair die independently until the first six. Find the mean and variance of the number of rolls and the probability it exceeds five. What is the mean number of failures? [Fluency]
9. For a Poisson variable of mean three, find $\mathbb{P}(X \geq 2)$ and $\mathbb{E}[X(X-1)]$. Recover its variance. [Fluency]
10. A population has six marked objects among twenty. Take four draws. Compare the count's mean and variance with and without replacement. State why the probability models differ. [Reasoning]
11. Roll a fair die until all faces have appeared. Find the mean, variance, and expected additional wait when exactly two faces are still missing. [Fluency]
12. Let $T_{n,r}$ count draws until r distinct types have appeared in the uniform coupon model. Show that $\mathbb{E}[T_{n,r}] = n(H_n - H_{n-r})$ for $1 \leq r \leq n$. What is the expected remaining wait for a complete collection? [Reasoning]
13. For 100 equally likely coupon types, use the exponential bound in (5.2) to choose a whole number of draws guaranteeing at least 95% completion probability. Explain why $100H_{100}$ is not such a guarantee. [Reasoning]
14. Independently choose $K \sim \text{Pois}(12)$, then roll a fair die K times. Find the distribution of the number of unseen faces and the probability of seeing every face. Compare with exactly 12 rolls. [Exploration]

5.9 For later: continuous families

This section uses densities and integrals from Section 4.13 and Appendix A. It can be left until after the discrete expectation and variance material.

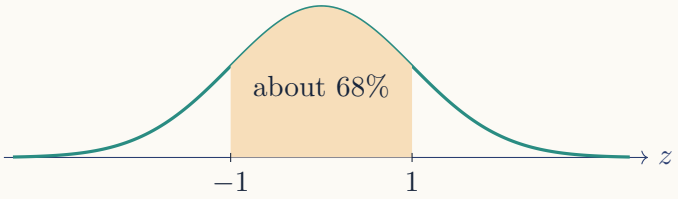
For $a < b$, Uniform(a, b) assigns constant density $1/(b-a)$ to an interval. For $\lambda > 0$, Exponential(λ) has density $\lambda e^{-\lambda x}$ on $x \geq 0$ and is the continuous memoryless law.

The normal density with mean μ and variance $\sigma^2 > 0$ is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Standardization sends $X \sim \mathcal{N}(\mu, \sigma^2)$ to

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1).$$



For $\alpha, \beta > 0$, the beta family on $[0, 1]$ has density proportional to $x^{\alpha-1}(1-x)^{\beta-1}$. It is useful for proportions and becomes a natural prior for a Bernoulli probability.

5.10 Choose by mechanism, not by silhouette

A bell-shaped histogram does not prove normality. Ask what generated the observations, what values are possible, whether trials are independent, whether rates are stable, and what approximation error is acceptable. A named distribution is useful because its assumptions describe an experiment and its formulas then answer questions about that experiment.

Later practice: continuous families

15. Find the median of an exponential distribution. [Fluency]

16. If $X \sim \mathcal{N}(10, 9)$, express $\mathbb{P}(8 < X < 13)$ using the standard normal CDF. [Fluency]

17. Derive the mean and variance of Uniform(a, b) for $a < b$. [Reasoning]

5.11 Further examples: where the standard laws come from

The distributions introduced earlier are descriptions of experiments, not just lists of formulas. The table makes their origins explicit. The negative-binomial, gamma, and order-statistic beta examples that follow are enrichment; the continuous parts use the calculus refresher.

Law	Generating experiment and assumptions
Discrete uniform	One uniformly chosen label in a finite list.
Bernoulli	Record any event of probability p as zero or one.
Binomial	Count successes in a fixed number of independent trials with common p.
Hypergeometric	Count marked objects in a uniform sample without replacement.
Geometric	Count independent trials through the first success, with fixed $p > 0$.
Poisson	Count arrivals in a fixed interval of a constant-rate Poisson model; also a limit of many rare independent trials.
Continuous uniform	Choose a point with constant density on an interval.
Exponential	Wait for the first arrival in a constant-rate Poisson model.
Normal	An exact Gaussian model when assumed; also an approximation to standardized sums under the central limit theorem.

5.11.1 Negative binomial: wait for the rth success

Use independent Bernoulli trials with success probability $0 < p \leq 1$. For an integer $r \geq 1$, let V_r count trials through the rth success. For $0 < p < 1$ and $k = r, r+1, \ldots$,

$$\boxed{\mathbb{P}(V_r = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}.}$$

The last trial must succeed, and the first $k-1$ trials must contain exactly $r-1$ successes. This proves the mass formula from its experiment. Alternatively, V_r is the sum of r independent geometric waits: after each success the unused trials are a fresh independent sequence. Each wait is finite almost surely, so the displayed masses exhaust all outcomes and sum to one. The geometric moments give

$$\mathbb{E}[V_r] = r/p, \qquad \operatorname{Var}(V_r) = r(1-p)/p^2.$$

At $p = 1$, $V_r = r$ with certainty. For $r = 1$ this is the geometric law. Some authors instead call the *failure count* $V_r - r$ negative binomial; its support starts at zero and its mean is $r(1-p)/p$.

Example 5.8 (Three successful attempts)**.** With independent success probability $1/2$, the number of attempts through success number three has mean 6, variance 6, and $\mathbb{P}(V_3 = 5) = \binom{4}{2}(1/2)^5 = 3/16$. The experiment fixes the number of successes, whereas a binomial experiment fixes the number of trials.

5.11.2 Uniform and exponential waits come from different clocks

If a bus departs every b minutes and one's arrival phase is uniform on $[0, b]$, the wait is Uniform$(0, b)$, with mean $b/2$ and variance $b^2/12$. Uniform phase is a modeling assumption; a timetable alone does not make a passenger's arrival phase uniform.

By contrast, suppose counts in an interval of length t are Poisson(λt), with independent counts in disjoint intervals. The wait E for the next arrival exceeds t exactly when there are zero arrivals, so

$$\mathbb{P}(E > t) = e^{-\lambda t}, \quad f_E(t) = \lambda e^{-\lambda t}, \quad t \geq 0.$$

This is the exponential law with *rate* λ and mean $1/\lambda$. Division of survival probabilities proves memorylessness: $\mathbb{P}(E > s + t \mid E > s) = e^{-\lambda t}$. For calls modeled at rate 3 per hour, 20 minutes has mean count one, probability $1 - 2e^{-1}$ of at least two calls, and probability e^{-1} of waiting more than 20 minutes for the next call.

There is also a precise discrete-to-continuous connection. If $G_n \sim$ Geometric(λ/n), then for fixed $t \geq 0$ and integers $n > \lambda$,

$$\mathbb{P}(G_n/n > t) = (1 - \lambda/n)^{\lfloor nt \rfloor} \longrightarrow e^{-\lambda t}.$$

Thus an exponential wait is a limit of rescaled rare-success waits. The equality before the limit concerns an integer-trial experiment; the exponential survival function is its limiting continuous model.

5.11.3 Gamma: add exponential waiting times

Let $E_1, E_2, \ldots$ be independent exponential variables of rate $\lambda > 0$, and put $S_r = E_1 + \cdots + E_r$. Then, for integer $r \geq 1$,

$$\boxed{f_{S_r}(t) = \frac{\lambda^r t^{r-1} e^{-\lambda t}}{(r-1)!}, \qquad t > 0.}$$

This is Gamma$(r, \text{rate } \lambda)$, also called an Erlang law. To derive it, start with $r = 1$. Conditioning on the previous subtotal and integrating the independent next-wait density gives

$$\begin{aligned} f_{S_{r+1}}(t) &= \int_0^t f_{S_r}(s) \lambda e^{-\lambda(t-s)}\, ds \\ &= \frac{\lambda^{r+1} e^{-\lambda t}}{(r-1)!} \int_0^t s^{r-1}\, ds = \frac{\lambda^{r+1} t^r e^{-\lambda t}}{r!}. \end{aligned}$$

This single-integral formula is called *convolution*; it sums the possibilities for the first subtotal. Integration by parts gives $\int_0^\infty t^j e^{-\lambda t}\, dt = j!/\lambda^{j+1}$ for integers $j \geq 0$, verifying normalization and the moments

$$\mathbb{E}[S_r] = r/\lambda, \qquad \text{Var}(S_r) = r/\lambda^2.$$

They also follow by adding the independent exponential moments. General positive noninteger gamma shapes exist too; the arrival-count interpretation here uses integer r.

Integrating the density, or differentiating the expression below and checking its value at zero, gives

$$\mathbb{P}(S_r > t) = e^{-\lambda t} \sum_{j=0}^{r-1} \frac{(\lambda t)^j}{j!}.$$

If arrivals occur at times $S_1, S_2, \ldots$, their count $N(t)$ obeys $\{N(t) \geq r\} = \{S_r \leq t\}$. Taking differences of these probabilities gives $\mathbb{P}(N(t) = k) = e^{-\lambda t}(\lambda t)^k/k!$ for $k \geq 0$. Thus exponential gaps construct the Poisson count model. The gamma tail tends to one as $r \to \infty$ at fixed t, so only finitely many arrivals occur in a bounded interval, almost surely.

For rate 2 per minute, the wait for the third arrival has mean $3/2$ minutes, variance $3/4$ minutes squared, and $\mathbb{P}(S_3 > 1) = e^{-2}(1 + 2 + 2^2/2) = 5e^{-2}$.

5.11.4 Beta: the location of an ordered uniform sample

Choose m independent Uniform$(0, 1)$ points and let $U_{(k)}$ be the kth smallest, where $1 \leq k \leq m$. For $0 \leq x \leq 1$, at least k points lie at or below x exactly when $U_{(k)} \leq x$. Hence

$$\mathbb{P}(U_{(k)} \leq x) = \sum_{j=k}^{m} \binom{m}{j} x^j (1-x)^{m-j}.$$

Differentiate this finite sum. The identities $j\binom{m}{j} = m\binom{m-1}{j-1}$ and $(m-j)\binom{m}{j} = m\binom{m-1}{j}$ cancel adjacent terms, leaving

$$\boxed{f_{U_{(k)}}(x) = \frac{m!}{(k-1)!(m-k)!} x^{k-1}(1-x)^{m-k}, \qquad 0 < x < 1.}$$

This is Beta$(k, m+1-k)$. Its normalization follows from the CDF; the integer beta integral in Appendix A gives

$$\mathbb{E}[U_{(k)}] = \frac{k}{m+1}, \qquad \operatorname{Var}(U_{(k)}) = \frac{k(m+1-k)}{(m+1)^2(m+2)}.$$

For example, the middle of three uniform points is Beta$(2, 2)$, with density $6x(1-x)$, mean $1/2$, and variance $1/20$. Its chance of lying between $1/4$ and $3/4$ is $11/16$, using the CDF $3x^2 - 2x^3$. This sampling origin of beta laws is different from their use as priors in Chapter 9; neither interpretation is forced by the name alone.

5.11.5 Normal: a model and a limiting shape

For independent copies X_i of a variable with finite mean μ and finite nonzero variance σ^2, the central limit theorem in Chapter 7 states that

$$\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \quad \text{converges in distribution to } \mathcal{N}(0, 1).$$

This explains the normal approximation to many accumulated small contributions. It requires assumptions; a common random shock or an infinite variance can invalidate this particular theorem.

For 100 independent fair tosses, the head count is exactly $\mathrm{Bin}(100, 1/2)$, with mean 50 and standard deviation 5. The exact probability of 45 through 55 heads is $2^{-100}\sum_{j=45}^{55}\binom{100}{j}$. A continuity-corrected normal approximation is $\Phi(1.1) - \Phi(-1.1)$: integrate from 44.5 to 55.5 before standardizing. The binomial law is exact under the toss assumptions; the normal answer is an approximation. For continuous measurement errors, normality can instead be an explicit model assumption to assess from the mechanism and data, rather than a conclusion from a bell-shaped histogram.

5.12 Poissonization: randomize a total to reveal independence

With exactly $m \geq 1$ draws from $n \geq 2$ positive-probability coupon types, the counts add to m and are dependent. Poissonization replaces that fixed total by an independent Poisson total. The resulting independence is exact for the new experiment; returning to a fixed total requires conditioning or a separate comparison. The argument also applies to balls in boxes, requests sent to servers, and other independent categorical choices.

5.12.1 Split a Poisson count

Let $K \sim \mathrm{Pois}(t)$, $t > 0$, independently of an i.i.d. sequence of type labels with probabilities $p_1, \ldots, p_n > 0$. Among the first K labels, let C_i count type i. For nonnegative integers c_i with sum s, conditioning on the total gives

$$\begin{aligned}\mathbb{P}(C_1 = c_1, \ldots, C_n = c_n) &= \frac{e^{-t}t^s}{s!}\frac{s!}{\prod_i c_i!}\prod_i p_i^{c_i} \\ &= \prod_i e^{-tp_i}\frac{(tp_i)^{c_i}}{c_i!}.\end{aligned}$$

Thus all the C_i are independent Poisson variables with means tp_i. This is the *Poisson splitting* or *thinning* identity. Conversely, conditioning these independent counts on $\sum_i C_i = m$ recovers the multinomial probabilities $m!\prod_i p_i^{c_i}/\prod_i c_i!$. For $n \geq 2$ and $m \geq 1$, conditioning reintroduces dependence. With one type or zero draws, the fixed-total count vector is constant instead.

Let M_K count the missing types. The missing indicators are now independent Bernoulli variables with probabilities e^{-tp_i}, so

$$\mathbb{E}[M_K] = \sum_i e^{-tp_i}, \qquad \mathrm{Var}(M_K) = \sum_i e^{-tp_i}(1 - e^{-tp_i}), \qquad \boxed{\mathbb{P}(T \leq K) = \prod_i (1 - e^{-tp_i}).}$$

The missing count is binomial for uniform probabilities, $M_K \sim \mathrm{Bin}(n, e^{-t/n})$; with unequal probabilities it is generally a sum of Bernoulli variables with different parameters.

Example 5.9 (A load limit for every box)**.** Send a Poisson(t) number of independent requests to boxes with routing probabilities p_i. For integer $b \geq 0$, independence gives

$$\mathbb{P}(\text{every box has at most } b \text{ requests}) = \prod_i \left[e^{-tp_i}\sum_{j=0}^{b}\frac{(tp_i)^j}{j!}\right].$$

The expected occupied-box count is $\sum_i(1 - e^{-tp_i})$. With a fixed total m its expectation instead is $\sum_i[1 - (1 - p_i)^m]$, by linearity; the corresponding occupancy indicators generally are dependent.

5.12.2 Exponential clocks for a collection

Give the original label sequence independent time gaps $E_1, E_2, \ldots \sim$ Exponential(1), also independent of all labels. The arrival times $S_m = \sum_{j=1}^{m} E_j$ construct a rate-one Poisson stream, by Section 5.11.3. It also has independent counts in disjoint intervals. Indeed, conditional on the completed arrival history up to a fixed time, memorylessness leaves a fresh rate-one exponential residual gap, and subsequent gaps are still independent. Repeating this argument at successive fixed interval boundaries proves the independent-increment assertion.

Mark each arrival with its coupon label. Splitting in each interval makes the type-i stream have rate p_i. To check that their first arrival times Y_i really are independent, fix thresholds $t_i \geq 0$ and subdivide $[0, \max_i t_i]$ at those thresholds. In each subinterval, the event $Y_i > t_i$ forbids labels whose threshold has not yet passed. The probability of zero forbidden arrivals is the exponential of minus the subinterval length times the sum of those labels' probabilities. Independence between intervals therefore gives

$$\mathbb{P}(Y_i > t_i \text{ for all } i) = \exp\left(-\sum_i p_i t_i\right) = \prod_i e^{-p_i t_i}.$$

This joint survival factorization proves that $Y_1, \ldots, Y_n$ are independent exponentials of rates $p_1, \ldots, p_n$. The time τ when the collection is complete is their maximum, hence

$$\mathbb{P}(\tau \leq t) = \prod_i (1 - e^{-tp_i}), \qquad t \geq 0.$$

There is an exact bridge back to the *number* of draws. Since T depends only on labels and all the gaps are independent of those labels,

$$\tau = \sum_{j \geq 1} E_j \mathbf{1}_{\{T \geq j\}}, \qquad \mathbb{E}[\tau] = \sum_{j \geq 1} \mathbb{E}[E_j]\mathbb{P}(T \geq j) = \mathbb{E}[T].$$

The interchange follows by truncating the nonnegative sum and taking an increasing limit. This proves the equality of means directly; no optional stopping theorem is being assumed. With arrival rate λ, the same argument gives $\mathbb{E}[\tau] = \mathbb{E}[T]/\lambda$.

For a nonnegative variable, integrating its tail gives its expectation: write its value as $\int_0^\infty \mathbf{1}_{\{\tau > t\}}\, dt$ and average nonnegative areas. Thus

$$\begin{aligned}\mathbb{E}[T] = \mathbb{E}[\tau] &= \int_0^\infty \left[1 - \prod_i (1 - e^{-tp_i})\right] dt \\ &= \sum_{\varnothing \neq A} (-1)^{|A|+1} \int_0^\infty e^{-tp_A}\, dt = \sum_{\varnothing \neq A} \frac{(-1)^{|A|+1}}{p_A}.\end{aligned}$$

This recovers the unequal-coupon formula. For uniform probabilities, the substitution $u = e^{-t/n}$ instead gives

$$\mathbb{E}[T_n] = n \int_0^1 \frac{1 - (1-u)^n}{u}\, du = n \sum_{j=0}^{n-1} \int_0^1 (1-u)^j\, du = nH_n.$$

Common trap

Equal means do not make the clock time and the draw count identically distributed. Conditional on $T = r$, the clock time is Gamma$(r, 1)$, with second moment $r(r + 1)$. Therefore $\mathrm{Var}(\tau) = \mathrm{Var}(T) + \mathbb{E}[T]$. For one coupon, $T = 1$ deterministically, whereas τ is exponential. Treating their variances as equal would lose the randomness added by the clock.

5.12.3 Why uniform probabilities minimize the mean

For fixed $t > 0$, the function $g(x) = \log(1 - e^{-tx})$ on $x > 0$ has

$$g''(x) = -\frac{t^2 e^{tx}}{(e^{tx} - 1)^2} < 0.$$

It is strictly concave. The concavity inequality for a finite average gives $n^{-1} \sum_i g(p_i) \le g(1/n)$, so

$$\prod_i (1 - e^{-tp_i}) \le (1 - e^{-t/n})^n.$$

For completeness, that finite-average inequality follows by repeated application of the two-point chord inequality for a concave function; negative second derivative proves the chord inequality by elementary calculus. If $n \ge 2$, equality holds exactly when all $p_i = 1/n$. Comparing the integrated survival functions and using the equality of clock and draw means proves

$$\boxed{\mathbb{E}[T] \ge nH_n,}$$

strictly for a nonuniform vector. The comparison just proved is a stochastic comparison of *clock times*, together with a comparison of draw-count means. It does not, by itself, establish a comparison of fixed-draw completion probabilities.

5.12.4 Return to a fixed total: de-Poissonization

Write $F(m) = \mathbb{P}(T \le m)$ and $\widetilde{F}(t) = \mathbb{P}(T \le K_t)$ for an independent $K_t \sim \mathrm{Pois}(t)$. Conditioning gives the exact transform

$$\widetilde{F}(t) = e^{-t} \sum_{m \ge 0} F(m) \frac{t^m}{m!}.$$

For two equally likely types, $F(2) = 1/2$, whereas $\widetilde{F}(2) = (1 - e^{-1})^2$. Replacing K_t by its mean is not an exact operation.

Monotonicity provides a rigorous comparison. For an integer $m \ge 1$ and $0 < t_- < m < t_+$, couple both experiments to the same label sequence. The pointwise inequalities $F(K_{t_-}) \le F(m) + \mathbf{1}_{\{K_{t_-} > m\}}$ and $F(m) \le F(K_{t_+}) + \mathbf{1}_{\{K_{t_+} < m\}}$ give

$$\boxed{\widetilde{F}(t_-) - \mathbb{P}(K_{t_-} > m) \le F(m) \le \widetilde{F}(t_+) + \mathbb{P}(K_{t_+} < m).}$$

Intersect these bounds with $[0, 1]$. Exact Poisson tails can be used, or Chebyshev bounds them by $t_-/(m - t_-)^2$ and $t_+/(t_+ - m)^2$, respectively. The comparison states the error rather than silently equating two sampling models.

Example 5.10 (The completion window, as optional asymptotics)**.** For uniform coupons, at clock time $t = n(\log n + c)$, fixed real c,

$$\widetilde{F}(t) = \left(1 - \frac{e^{-c}}{n}\right)^n \longrightarrow \exp(-e^{-c}).$$

This holds for all sufficiently large n, when $t > 0$. It also describes the fixed-draw experiment. Set $m = \lfloor n \log n + cn \rfloor$ and $t_\pm = m \pm n^{3/4}$. Both Poisson-tail errors above tend to zero, since they are at most $(m + n^{3/4})/n^{3/2}$, of order $\log n/\sqrt{n}$. Also $t_\pm/n - \log n \to c$, so both product bounds tend to the same $\exp(-e^{-c})$. Squeezing gives $\mathbb{P}(T_n \le \lfloor n \log n + cn \rfloor) \to \exp(-e^{-c})$. Poissonization made the coordinates independent; de-Poissonization justified transferring the limit back to a fixed number of draws.

Further practice: mechanisms and Poissonization

18. Coupon types have probabilities $1/2, 1/4, 1/4$. Find the mean and variance of the complete collection time and the expected remaining draws if only types 2 and 3 are missing. [Reasoning]

19. Two coupon types have probabilities $1/10, 9/10$. Find the exact mean and compare with the bounds $1/p_{\min}$ and $\sum_i 1/p_i$. What changes if a required type has probability zero? [Fluency]

20. Draw an independent $K \sim \text{Pois}(12)$ coupons with probabilities $1/2, 1/4, 1/4$. Give the joint law of the three type counts, the mean missing-type count, and the complete-collection probability. Is the missing count binomial? What is the first type's count conditional on $K = 12$? [Reasoning]

21. Independent attempts succeed with probability $1/3$. Find the probability that success number two occurs on attempt four, the mean and variance of its attempt number, and the mean failure count. [Fluency]

22. With independent exponential gaps of rate 3 per minute, find the density, mean, variance, and probability of exceeding one minute for the wait to arrival number two. [Fluency]

23. Find the density, mean, variance, and probability of exceeding $1/2$ for the second smallest of four independent Uniform$(0, 1)$ points. Derive the last probability by counting points below $1/2$. [Reasoning]

24. Send a Poisson(4) number of requests independently to two equally likely boxes. Find the probability that neither box receives more than one request. Compare with exactly four requests. [Reasoning]

25. Attach independent rate-one exponential gaps to the coupon sequence in Exercise 18. Find the mean and variance of its continuous completion time. Repeat with arrival rate 2. [Reasoning]

26. For two equally likely coupon types, use the de-Poissonization bounds with $m = 2, t_- = 1, t_+ = 4$ and exact Poisson tails. Write explicit lower and upper bounds and check them against $F(2) = 1/2$. [Synthesis]

27. For $B \sim \text{Bin}(25, 1/2)$, give exact and continuity-corrected normal expressions for $\mathbb{P}(10 \le B \le 15)$. Identify the exact model. [Reasoning]

Chapter 6

Variables together

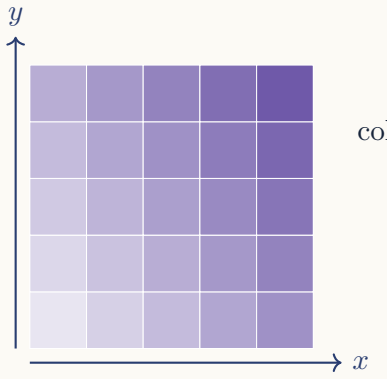
Joint, marginal, and conditional distributions

Guiding question 6.1. How do we describe dependence without confusing association, causation, and prediction?

For discrete variables, a joint mass function $p(x, y) = \mathbb{P}(X = x, Y = y)$ is a table whose entries sum to one. Marginals are obtained by summing:

$$p_X(x) = \sum_y p(x, y), \qquad p_Y(y) = \sum_x p(x, y).$$

For a continuous joint density $f(x, y)$, replace sums with integrals.



color = joint probability

6.1 Conditional laws and independence

For $p_Y(y) > 0$,

$$p_{X|Y}(x \mid y) = \frac{p(x, y)}{p_Y(y)}.$$

Variables are independent when their joint law factors:

$$p(x, y) = p_X(x)p_Y(y)$$

or $f(x, y) = f_X(x)f_Y(y)$ throughout the support.

6.2 Covariance and correlation

$$\operatorname{Cov}(X,Y) = \mathbb{E}[(X-\mathbb{E}X)(Y-\mathbb{E}Y)] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Correlation is the dimensionless quantity

$$\rho_{X,Y} = \frac{\operatorname{Cov}(X,Y)}{\sigma_X \sigma_Y}.$$

It lies between -1 and 1. Independence implies zero covariance when moments exist, but zero covariance does not generally imply independence.

Theorem 6.2.

$$\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y).$$

For independent variables, the covariance term vanishes.

6.3 Conditional expectation

$\mathbb{E}[X \mid Y]$ is itself a random variable: for each observed value of Y, it gives the mean of the corresponding conditional distribution. The tower property says

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X].$$

The conditional mean is also the best mean-square predictor of X using only Y.

Common trap

Correlation measures linear association. A perfect U-shaped relationship can have correlation zero. And neither correlation nor conditioning alone establishes causation.

Exercises

1. Given a 2×3 joint probability table, compute both marginals and one conditional distribution. [Fluency]
2. Prove that independence implies $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$. [Reasoning]
3. Let X be uniform on $\{-1, 0, 1\}$ and $Y = X^2$. Show $\operatorname{Cov}(X,Y) = 0$ although X, Y are dependent. [Reasoning]
4. Derive the variance-of-a-sum identity. [Reasoning]
5. If X, Y are independent Poisson variables, use generating functions to show $X+Y$ is Poisson. [Reasoning]
6. Create three scatterplots with correlations near -1, 0, and 1, then create a nonlinear plot with correlation near zero. [Exploration]
7. Explain the tower property using a population divided into groups. [Exploration]

Chapter 7

From repetition to regularity

Laws of large numbers, the central limit theorem, and sampling

Guiding question 7.1. Why do averages stabilize, and why do bell curves emerge from non-normal data?

Let $X_1, \dots, X_n$ be independent copies of a variable with mean μ and variance σ^2. Their average is

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad \mathbb{E}[\bar{X}_n] = \mu, \qquad \operatorname{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

Theorem 7.2 (Weak law of large numbers)**.** *For every $\varepsilon > 0$,*

$$\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \longrightarrow 0.$$

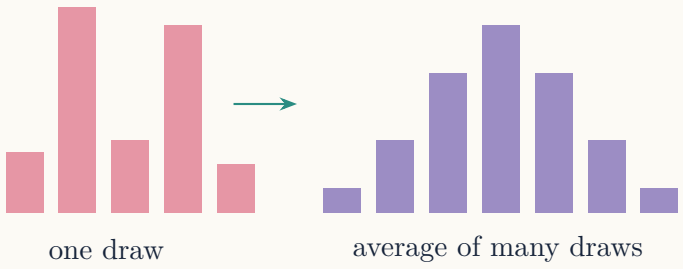
With finite variance, Chebyshev gives the proof immediately:

$$\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}.$$

Theorem 7.3 (Central limit theorem)**.** *If $0 < \sigma^2 < \infty$, then*

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1).$$

The LLN says where the average goes; the CLT describes the scale and approximate shape of its remaining fluctuations.



7.1 Sampling distributions

A *statistic* is a function of the sample. Before data are observed it is random, and its probability law is its sampling distribution. The standard deviation of that distribution is the *standard error*.

For a Bernoulli sample, the sample proportion $\hat{p}$ has

$$\mathbb{E}[\hat{p}] = p, \qquad \mathrm{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}.$$

7.2 Monte Carlo and bootstrap

Monte Carlo approximates an expectation by simulating from a known model. The bootstrap approximates a sampling distribution by repeatedly resampling, with replacement, from observed data. The bootstrap is powerful but cannot repair a badly biased or unrepresentative original sample.

Exercises

1. Compute the standard error of the mean when $\sigma = 12$ and $n = 64$. [Fluency]
2. Prove the weak law using Chebyshev's inequality. [Reasoning]
3. Approximate $\mathbb{P}(480 \le X \le 520)$ for $X \sim \mathrm{Bin}(1000, .5)$ using the CLT and a continuity correction. [Fluency]
4. Explain why quadrupling the sample size halves a standard error. [Reasoning]
5. Simulate sample means from a skewed distribution for $n = 1, 2, 5, 30$. Quantify convergence rather than only describing the pictures. [Exploration]
6. Compare Monte Carlo error at $n = 100$, 10^4, and 10^6. Is the $n^{-1/2}$ rate visible? [Exploration]
7. Give an example where bootstrap resampling cannot reveal a sampling bias. [Reasoning]

Chapter 8

Learning from a sample

Estimation, confidence intervals, and hypothesis tests

Guiding question 8.1. How can a random sample support a calibrated statement about an unknown population?

An estimator $T = T(X_1, \dots, X_n)$ targets a parameter θ. Its bias, variance, and mean squared error are

$$\mathrm{Bias}_\theta(T) = \mathbb{E}_\theta[T] - \theta, \qquad \mathrm{MSE}_\theta(T) = \mathrm{Var}_\theta(T) + \mathrm{Bias}_\theta(T)^2.$$

Unbiasedness is attractive, but MSE captures the bias–variance tradeoff.

8.1 Method of moments and likelihood

The method of moments matches sample moments to model moments. Maximum likelihood chooses the parameter under which the observed data have greatest likelihood.

For Bernoulli observations $x_1, \dots, x_n$, the likelihood is

$$L(p) = p^{\sum x_i}(1-p)^{n-\sum x_i},$$

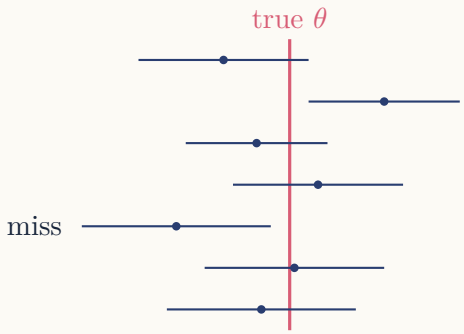
and the maximizer is $\hat{p} = \bar{x}$.

8.2 Confidence intervals

An approximate 95% interval for a mean with known or well-estimated standard error is

$$\text{estimate} \ \pm \ 1.96 \times \text{standard error}.$$

The confidence level refers to the long-run success rate of the procedure, not a posterior probability that a fixed parameter lies in one realized interval.



8.3 Tests and p-values

A hypothesis test specifies a null model H_0, a test statistic, and a rejection rule. The p-value is the probability, assuming H_0, of a result at least as incompatible with H_0 as the observed one.

It is not the probability that H_0 is true. A Type I error rejects a true null; a Type II error fails to reject a false null. Power is the probability of rejecting under a specified alternative.

Common trap

Statistical significance is not practical importance. With enough data, a tiny effect can yield a tiny p-value; with too little data, an important effect can be missed.

8.4 Model checking and design

Random sampling supports generalization; random assignment supports causal comparison. No inferential formula compensates for selection bias, poorly measured outcomes, hidden multiplicity, or a model contradicted by the data.

Exercises

1. Show that the sample mean is unbiased for μ. [Reasoning]
2. Find the Bernoulli maximum-likelihood estimator. [Fluency]
3. Compare two estimators when one has small bias but much smaller variance. [Reasoning]
4. Construct a 95% normal interval for a proportion from 600 observations with 330 successes. [Fluency]
5. Explain the repeated-sampling meaning of that interval without saying there is a 95% chance the fixed parameter is inside. [Reasoning]
6. A test reports $p = .03$. List three claims that do not follow. [Reasoning]
7. Simulate confidence-interval coverage and test power as functions of sample size. [Exploration]
8. Describe how optional stopping or multiple testing can inflate false discoveries. [Exploration]

Chapter 9

Probability for unknowns

An introduction to Bayesian inference

Chapter 10 will revisit this chapter's shared-parameter models from another direction: under infinite exchangeability, a conditional-i.i.d. representation is a theorem rather than a starting assumption.

Guiding question 9.1. What changes when an unknown parameter is itself represented by a probability distribution?

Bayesian inference combines a prior distribution $\pi(\theta)$ with a likelihood $p(y \mid \theta)$. Bayes' rule produces the posterior

$$\pi(\theta \mid y) = \frac{p(y \mid \theta)\pi(\theta)}{\int p(y \mid u)\pi(u)\,du} \propto p(y \mid \theta)\pi(\theta).$$

9.1 Beta–binomial updating

Suppose $p \sim \text{Beta}(\alpha, \beta)$ and, conditional on p, $X \sim \text{Bin}(n, p)$. After observing x successes,

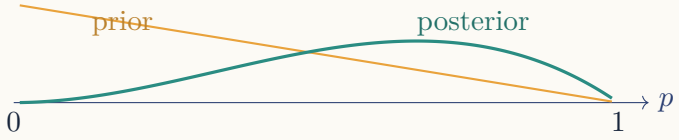
$$p \mid X = x \sim \text{Beta}(\alpha + x, \beta + n - x).$$

The prior parameters behave like pseudocounts: $\alpha - 1$ prior successes and $\beta - 1$ prior failures when both exceed one.

The posterior mean is

$$\mathbb{E}[p \mid x] = \frac{\alpha + x}{\alpha + \beta + n},$$

a weighted compromise between the prior mean and the sample proportion.



9.2 Credible intervals and prediction

A 95% posterior credible interval contains 95% of the posterior probability. This is a conditional probability statement about the parameter under the model and prior. A posterior predictive

distribution averages future outcomes over posterior uncertainty:

$$p(y_{\text{new}} \mid y) = \int p(y_{\text{new}} \mid \theta)\pi(\theta \mid y)\, d\theta.$$

9.3 Prior sensitivity and checking

Priors make assumptions visible; they do not make assumptions disappear. Compare reasonable priors, examine posterior predictive simulations, and ask whether the fitted model reproduces important features of the data.

Big idea

Frequentist and Bayesian intervals answer different probability questions. The productive comparison is not "which formula looks similar?" but "what is random, what is conditioned upon, and what operating guarantee or posterior statement is being claimed?"

Exercises

1. Update a Beta$(2, 2)$ prior after 8 successes in 10 trials. [Fluency]
2. Compute the posterior mean and compare it with the sample proportion. [Fluency]
3. Derive beta–binomial conjugacy by collecting powers of p and $1 - p$. [Reasoning]
4. Under a Beta(α, β) posterior, find the predictive probability that the next Bernoulli trial succeeds. [Reasoning]
5. Compare Beta$(1, 1)$, Beta$(2, 2)$, and Beta$(20, 20)$ priors for the same small dataset. [Exploration]
6. Explain why a credible interval and confidence interval can have the same numerical endpoints but different interpretations. [Reasoning]
7. Design a posterior predictive check for a coin model that might have time-varying bias. [Exploration]

Capstone: one question, two inferential languages

Choose a binomial, count, or measurement dataset. State a generative model, inspect the data, estimate the target, and quantify uncertainty both frequentistically and Bayesianly. End with a model check and a paragraph explaining which conclusions are robust to the modeling choices.

Chapter 10

Exchangeability: when symmetry produces a parameter

A capstone on dependence, learning, and long-run frequency

Guiding question 10.1. If the order of observations carries no information, must the observations be independent?

No. But for an *infinite* sequence of zeros and ones, a remarkably precise substitute is available: the sequence behaves as though a success probability were chosen once, and independent trials were then run with that probability. This is de Finetti's representation theorem.

This chapter joins three earlier ideas: conditioning can remove dependence, Bayes' rule updates an unknown parameter, and repeated observations reveal a long-run frequency. Read the examples first. Section 10.9 is an optional proof, using a birthday-collision bound and one explicitly stated compactness fact from analysis.

10.1 Symmetry is weaker than independence

Definition 10.2. A finite random vector $(X_1, \dots, X_n)$ is *exchangeable* if permuting its coordinates does not change its joint distribution. An infinite sequence $X_1, X_2, \dots$ is exchangeable if every finite prefix is exchangeable.

For a zero–one vector, the definition says that two particular strings with the same number of ones have the same probability. Thus 1100, 1010, and 0011 must be equally likely. It does *not* say that 1100 and 1111 are equally likely, or that the coordinates are independent.

Independent and identically distributed (i.i.d.) variables are exchangeable: permuting a product of identical marginal laws changes nothing. Exchangeability also implies identical marginal distributions, because we can swap any two positions. Neither implication reverses in general.

Example 10.3 (One coin, chosen once)**.** Choose a coin with heads probability $1/4$ or $3/4$, each with probability $1/2$. Keep the chosen coin and toss it repeatedly, independently *given the choice*. Write $X_i = 1$ for heads.

For a specified length-n string containing s heads, its probability is

$$\frac{1}{2}\left(\frac{1}{4}\right)^{s}\left(\frac{3}{4}\right)^{n-s}+\frac{1}{2}\left(\frac{3}{4}\right)^{s}\left(\frac{1}{4}\right)^{n-s}.$$

Only the count s matters, so the sequence is exchangeable. Nevertheless,

$$\mathbb{P}(X_1=1)=\frac{1}{2},\qquad \mathbb{P}(X_1=X_2=1)=\frac{1}{2}\frac{1}{16}+\frac{1}{2}\frac{9}{16}=\frac{5}{16}\neq\frac{1}{4}.$$

Seeing a head makes the high-bias coin more plausible, and therefore makes another head more likely. The tosses are dependent when the coin's identity is unknown.

Checkpoint

If we independently choose a fresh coin before *each* toss, the results are instead i.i.d. fair bits. Sharing one random choice across all trials is the source of dependence in Example 10.3.

10.2 The Bernoulli representation theorem

Theorem 10.4 (de Finetti, Bernoulli case)**.** *An infinite zero–one sequence is exchangeable if and only if there is a probability measure μ on $[0,1]$ such that, for every $n\geq 1$ and every $(x_1,\dots,x_n)\in\{0,1\}^n$ with $s=\sum_i x_i$,*

$$\mathbb{P}(X_1=x_1,\dots,X_n=x_n)=\int_{[0,1]}p^{s}(1-p)^{n-s}\,\mu(dp). \tag{10.1}$$

The measure μ is unique. Equivalently, the sequence has the law of conditionally i.i.d. Bernoulli(Θ) trials, where Θ is chosen once with distribution μ.

An exponent zero contributes a factor of one, including at $p=0$ or 1. The integral averages over a probability *measure*: μ may have a density, point masses, or both. It need not be a beta distribution. We call μ the *mixing law* or *de Finetti measure*.

Two levels of randomness

First choose $\Theta\sim\mu$. Conditional on $\Theta=p$, generate $X_1,X_2,\dots$ independently with success probability p. Conditional independence concerns the second level; exchangeability describes the joint law after averaging over the first.

The easy direction is worth proving now: every integrand in (10.1) depends on the string only through s, so every mixture is exchangeable. The difficult direction says that *every infinite exchangeable Bernoulli law* admits such a representation.

Let $S_n=X_1+\cdots+X_n$. There are $\binom{n}{s}$ strings with s ones, so

$$\mathbb{P}(S_n=s)=\binom{n}{s}\int_{[0,1]}p^{s}(1-p)^{n-s}\,\mu(dp). \tag{10.2}$$

The binomial coefficient belongs to the *count* probability, not the probability of one specified ordered string. This distinction is exactly the permutations-and-combinations distinction from Chapter 1.

Ordinary i.i.d. Bernoulli(p_0) trials correspond to $\mu = \delta_{p_0}$, a point mass at p_0. At the other extreme, choose a fair bit Z and put $X_i = Z$ for every i. The sequence is exchangeable, all trials agree, and $\mu = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$.

What the theorem does not choose

Exchangeability specifies a class of models, not a particular prior. Uniqueness means that the *entire infinite joint law* determines μ; it does not mean that finitely many observations determine it. Nor does the representation prove that nature literally selected a hidden coin. Different mechanisms can have exactly the same observable law.

10.3 Learning about the shared coin

Return to Example 10.3. After a specified string with s heads in n tosses, Bayes' rule gives

$$q_{n,s} := \mathbb{P}\left(\Theta = \frac{3}{4} \,\middle|\, X_1 = x_1, \ldots, X_n = x_n\right) = \frac{3^s}{3^s + 3^{n-s}}.$$

The probability of heads on the next toss is therefore

$$\mathbb{P}(X_{n+1} = 1 \mid X_1 = x_1, \ldots, X_n = x_n) = \frac{1}{4}(1 - q_{n,s}) + \frac{3}{4}q_{n,s}.$$

After two heads, $q_{2,2} = 9/10$ and the predictive probability is $7/10$. After a head and a tail, $q_{2,1} = 1/2$ and it is $1/2$. The order HT versus TH does not affect either answer.

For an arbitrary mixing law, the same calculation gives

$$\mathbb{P}(X_{n+1} = 1 \mid X_1 = x_1, \ldots, X_n = x_n) = \frac{\int p^{s+1}(1-p)^{n-s}\,\mu(dp)}{\int p^s(1-p)^{n-s}\,\mu(dp)}, \tag{10.3}$$

provided the observed string has positive probability. This is the posterior mean of Θ. Conditioning on the count $S_n = s$ gives the same result, since the factor $\binom{n}{s}$ cancels.

There is no conflict between exchangeability and learning. Exchangeability says that the *order* of the observed bits carries no additional information beyond their count. It does not say that the count carries no information about future bits.

10.4 Pólya's urn: reinforcement becomes a mixture

Start with one red and one blue ball. Draw a ball uniformly, replace it, and add one extra ball of the same color. Repeat. Let $X_i = 1$ mean that draw i is red. After s red draws among n draws, the next draw is red with probability

$$\frac{s+1}{n+2}. \tag{10.4}$$

This mechanism seems to remember its past: drawing red makes future reds more likely. Yet its sequence is exchangeable.

Example 10.5 (Calculate an ordered path)**.** The path $RRRB$ has probability

$$\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{1}{5} = \frac{1}{20}.$$

The path $RBRR$ has probability $\frac{1}{2}\cdot\frac{1}{3}\cdot\frac{2}{4}\cdot\frac{3}{5} = 1/20$ as well. These calculations are not the probability of getting three reds in four draws: there are four possible positions for the blue draw, giving $4/20 = 1/5$ for that event.

For any specified path with s red and $n-s$ blue draws, the red numerators, in their order of appearance, are $1, 2, \ldots, s$; the blue numerators are $1, 2, \ldots, n-s$. The denominators are $2, 3, \ldots, n+1$. Consequently

$$\mathbb{P}(X_1 = x_1, \ldots, X_n = x_n) = \frac{s!(n-s)!}{(n+1)!}. \tag{10.5}$$

The order changes where the factors appear, but not their product.

Repeated integration by parts gives the elementary beta integral

$$\int_0^1 p^s(1-p)^{n-s}\,dp = \frac{s!(n-s)!}{(n+1)!}.$$

Comparing this with (10.1) identifies the mixing law: Θ is uniform on $[0,1]$. Thus two different descriptions generate the same law of observed colors:

1. reinforce the color just drawn, using the urn rule;

2. choose one uniform random bias Θ, then generate conditionally independent Bernoulli(Θ) draws.

This is equality of joint distributions, not a claim that the physical urn contains a concealed coin.

An especially striking consequence is

$$\mathbb{P}(S_n = s) = \binom{n}{s}\frac{s!(n-s)!}{(n+1)!} = \frac{1}{n+1}, \qquad s = 0, \ldots, n.$$

The *count* is uniform, even though the individual strings are not. For four draws, all five counts $0, 1, 2, 3, 4$ have probability $1/5$. Compare independent fair tosses, whose count probabilities are $1, 4, 6, 4, 1$, divided by 16.

Other starting compositions

With a red and b blue balls initially, where a, b are positive integers, define the rising factorial $a^{\overline{r}} = a(a+1)\cdots(a+r-1)$ and $a^{\overline{0}} = 1$. It is different from the falling factorial used to count ordered selections without replacement. The path probability becomes

$$\frac{a^{\overline{s}}b^{\overline{n-s}}}{(a+b)^{\overline{n}}} = \int_0^1 p^s(1-p)^{n-s}\frac{p^{a-1}(1-p)^{b-1}}{B(a,b)}\,dp,$$

where $B(a,b) = \int_0^1 p^{a-1}(1-p)^{b-1}\,dp$. The mixing law is Beta$(a,b)$.

The Bayesian model also makes sense for arbitrary real $a, b > 0$, whether or not we picture whole balls. After s successes in n trials, the posterior is Beta$(a+s, b+n-s)$ and the predictive success probability is $(a+s)/(a+b+n)$. For the one-red, one-blue urn, this reproduces (10.4). Chapter 9's beta update now has a concrete reinforcement interpretation.

10.5 A law of large numbers with a random limit

For fixed p, the strong law for independent Bernoulli(p) trials says $S_n/n \to p$ with probability one. This means that the set of infinite sample paths where convergence fails has probability zero; it is stronger than convergence in probability in Chapter 7. A short proof of this special strong law appears in Section 10.9.

Apply the strong law conditional on the shared bias. It gives

$$\frac{S_n}{n} \longrightarrow \Theta \qquad \text{with probability one.} \tag{10.6}$$

In particular, the mixing law is the distribution of the limiting empirical frequency. The parameter is not merely an algebraic trick: it can be recovered from the infinite sequence itself.

Example 10.6 (Same marginal, different limits)**.** In each of these models $\mathbb{P}(X_i = 1) = 1/2$:

Model	Limit of S_n/n
Independent fair tosses	1/2, with probability one
One coin of bias 1/4 or 3/4	1/4 or 3/4, each with probability 1/2
One-red, one-blue Pólya urn	A uniform random value in $[0, 1]$
One fair bit repeated forever	0 or 1, each with probability 1/2

Identical one-trial probabilities tell us very little about the behavior of a long sequence.

We can see the persistent variation without invoking a limit theorem. Conditional independence and the tower property give, for $i \neq j$,

$$\mathbb{E}X_i = \mathbb{E}\Theta, \qquad \mathbb{E}[X_iX_j] = \mathbb{E}[\Theta^2], \qquad \boxed{\mathrm{Cov}(X_i, X_j) = \mathrm{Var}(\Theta) \geq 0.}$$

Also, conditional on Θ, the sample mean has variance $\Theta(1-\Theta)/n$ and mean Θ. Taking expectations in the decomposition into within-group and between-group variation gives

$$\boxed{\mathrm{Var}(S_n/n) = \frac{\mathbb{E}[\Theta(1-\Theta)]}{n} + \mathrm{Var}(\Theta).} \tag{10.7}$$

For the two-coin model this is $3/(16n) + 1/16$; for the one-red, one-blue urn it is $1/(6n) + 1/12$.

Increasing the number of observations reduces the sampling noise *within a run*. It does not remove the variation of the shared bias *between runs*. In the urn example, for instance, $\mathbb{P}(\lim S_n/n > 3/4) = 1/4$, despite the perfectly symmetric starting urn.

Explore: one long run versus many runs

Simulate 100 independent copies of the one-red, one-blue urn, each for 5,000 draws. Plot a few running red frequencies; then make a histogram of the 100 final frequencies. Compare with 100 runs of independent fair tosses and with 100 runs of the two-coin model. Predict all three histograms before simulating. A single long trajectory estimates its own shared bias, not the distribution of biases across runs.

10.6 Why infinity is essential

Example 10.7 (A symmetric pair that cannot continue)**.** Let (X_1, X_2) be $(1, 0)$ or $(0, 1)$, each with probability $1/2$. This pair is exchangeable, but

$$\mathrm{Cov}(X_1, X_2) = 0 - \frac{1}{2}\frac{1}{2} = -\frac{1}{4}.$$

It cannot be a mixture of conditionally i.i.d. Bernoulli trials, since every such mixture has nonnegative covariance. In fact, it cannot even be the first two coordinates of an exchangeable triple: all three pairs would have to disagree with probability one, which three bits cannot do.

More generally, sample without replacement from N balls, M red and $N - M$ blue. The N color indicators are exchangeable, but for $i \neq j$ and $p = M/N$,

$$\mathbb{P}(X_i = X_j = 1) = \frac{M(M-1)}{N(N-1)}, \qquad \mathrm{Cov}(X_i, X_j) = -\frac{p(1-p)}{N-1} \quad (N > 1).$$

For $0 < M < N$, exhausting red balls creates negative dependence. Finite symmetry is real, but it need not be the symmetry of an infinite sequence.

The finite bridge

Conditional on its total number of ones, every finite exchangeable zero–one vector is a uniform arrangement of those ones and zeros. It is therefore a mixture of sampling-without-replacement models. For a short sample from a large population, replacement makes little difference. This observation is the bridge to de Finetti's theorem.

10.7 Beyond coins, and the limits of symmetry

The corresponding theorem for an infinite exchangeable sequence of *real-valued* random variables replaces the random number Θ by a random probability distribution Q on $\mathbb{R}$. Conditional on Q, the variables are i.i.d. with law Q. For Borel sets $A_1, \ldots, A_n$,

$$\mathbb{P}(X_1 \in A_1, \ldots, X_n \in A_n) = \mathbb{E}\left[\prod_{i=1}^{n} Q(A_i)\right].$$

This general result is stated for perspective, not proved here. The real-valued setting matters: unrestricted measurable spaces require additional care.

For example, let M be a random group mean and let $X_i = M + \varepsilon_i$, where the ε_i are i.i.d. $\mathcal{N}(0, 1)$ and independent of M. Conditional on M, the observations are i.i.d. $\mathcal{N}(M, 1)$; unconditionally, they are exchangeable and, when $\mathrm{Var}(M) > 0$ is finite, have covariance $\mathrm{Var}(M) > 0$. Here $Q = \mathcal{N}(M, 1)$ is a random distribution.

Exchangeability is a modeling assumption to defend. Time trends, different treatment groups, and changing sampling protocols can make permutations inappropriate. Declaring observations exchangeable does not erase such structure. Its value is to make a precise connection: *a symmetry of the observable law can justify a conditional i.i.d. representation, without assuming unconditional independence.*

10.8 Exercises

Exchangeability practice

1. In the two-coin model, observe three heads and one tail. Find the posterior probability that the bias is $3/4$ and the predictive probability of another head. Would $HTHH$ give a different answer? [Fluency]
2. Prove that independently selecting a fresh bias from $\{1/4, 3/4\}$ before every toss produces i.i.d. fair bits. Identify which part of the shared-coin model has changed. [Reasoning]
3. For the one-red, one-blue urn, find the probability of the specified path $RRBRR$, and then the probability of exactly four reds in five draws. Given this path, find the probability of a red next. [Fluency]
4. An urn starts with two red and three blue balls and uses Pólya reinforcement. After four red and two blue draws, identify the mixing law before observing data, the posterior law, and the next-red probability. [Fluency]
5. Derive (10.7). Under uniform mixing, compute $\mathrm{Var}(S_{10}/10)$ and $\mathrm{Var}(S_{100}/100)$. Explain why neither should be approximated by $1/(4n)$. [Reasoning]
6. Suppose an infinite exchangeable Bernoulli sequence is pairwise independent. Prove that it is i.i.d. What hypothesis prevents this from contradicting Chapter 3's pairwise-independence warning? [Reasoning]
7. Let an exchangeable vector of length $N \geq 2$ have common finite variance v and common pairwise covariance c. Prove $c \geq -v/(N-1)$. Use this to rule out an exchangeable extension of the always-disagreeing pair to any length $N > 2$. [Reasoning]
8. Run the experiment in the Explore box. For each model, state whether final frequencies from many independent runs should cluster near one point, two points, or across an interval. Separate a simulation observation from a theorem that explains it. [Exploration]
9. Use the coupling in Section 10.9 to bound the error for *any event* concerning the first five draws from an exchangeable zero–one population of size 1,000. Explain why the same bound need not be informative for 100 draws. [Optional proof]

Sources and a next step

The representation theorem is due to Bruno de Finetti. For another elementary Bernoulli proof, using moments, see Werner Kirsch, *An elementary proof of de Finetti's Theorem* (2018). Finite exchangeability and sampling approximations are treated systematically in Persi Diaconis and David Freedman, *Finite Exchangeable Sequences* (1980).

10.9 Optional proof: finite urns and birthday collisions

Here is a proof of the Bernoulli representation, its uniqueness, and the random-frequency interpretation. The finite approximation is elementary. We isolate the one compactness input rather than conceal it in a limit.

Step 1: condition on the finite total

Fix $n \le N$ and a particular length-n string with s ones. Conditional on $S_N = k$, exchangeability makes all $\binom{N}{k}$ length-N strings with k ones equally likely, whenever $\mathbb{P}(S_N = k) > 0$. Hence

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n \mid S_N = k) = \frac{\binom{N-n}{k-s}}{\binom{N}{k}}.$$

A binomial coefficient is zero when its lower argument is outside its allowed range. This is precisely the probability of that ordered color string when sampling without replacement from k ones and $N - k$ zeros.

Step 2: allow replacement and bound the error

Draw n independent uniform labels $J_1, \dots, J_n$ from $\{1, \dots, N\}$. If no label repeats, use these same labels as the without-replacement sample. If a label repeats, generate a new uniform ordered sample of distinct labels for the without-replacement sample. This construction has the correct marginal sampling laws and makes their color strings identical whenever the original labels are distinct.

For each pair $i < j$, $\mathbb{P}(J_i = J_j) = 1/N$. The union bound from the birthday problem gives

$$\mathbb{P}(\text{a repeated label}) \le \min\left\{1, \frac{\binom{n}{2}}{N}\right\}. \tag{10.8}$$

Two coupled random strings that agree except on an event of probability ε assign probabilities differing by at most ε to *any* event: their event indicators differ only when the strings differ. With replacement, the colors are independent Bernoulli(k/N). Thus, writing μ_N for the distribution of S_N/N, and averaging over k,

$$\left|\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) - \int p^s (1-p)^{n-s}\, \mu_N(dp)\right| \le \frac{\binom{n}{2}}{N}. \tag{10.9}$$

The left-hand probability is fixed as N grows: all prefixes belong to the *same* infinite sequence.

Step 3: take a subsequence of probability laws

Lemma 10.8 (Compactness on a closed bounded interval)**.** *Every sequence of probability measures on $[0,1]$ has a subsequence ν_j and a probability measure ν on $[0,1]$ such that $\int f\, d\nu_j \to \int f\, d\nu$ for every continuous function $f : [0,1] \to \mathbb{R}$. Also, two probability measures with equal integrals against all such functions are equal.*

This is the analysis input, also called weak sequential compactness. One route to its proof selects convergent subsequences of distribution functions at each rational point, then uses monotonicity to

obtain a limiting distribution function at its continuity points. Compact support prevents probability from escaping to infinity. We use the lemma here without developing that analysis.

Apply the lemma to μ_N. Along one subsequence, μ_{N_j} converges weakly to some μ. For fixed n and s, the function $p^s(1-p)^{n-s}$ is continuous, so (10.9) yields (10.1). The same subsequence works for every continuous test function, hence for every finite string. This proves existence.

Step 4: show that the mixing law is unique

Under *any* candidate mixture, conditional on $\Theta = p$, $S_n \sim \mathrm{Bin}(n, p)$. Chebyshev gives the uniform bound

$$\mathbb{P}(|S_n/n - p| > \delta \mid \Theta = p) \le \frac{1}{4n\delta^2} \qquad (\delta > 0).$$

For a continuous f on $[0,1]$, let $K = \sup|f|$ and $\omega_f(\delta) = \sup_{|x-y|\le\delta} |f(x) - f(y)|$. Split according to whether $|S_n/n - \Theta| \le \delta$ to get

$$|\mathbb{E}f(S_n/n) - \mathbb{E}f(\Theta)| \le \omega_f(\delta) + \frac{2K}{4n\delta^2}.$$

First let $n \to \infty$, then $\delta \downarrow 0$; uniform continuity makes the right-hand side vanish. Thus the distribution of S_n/n converges weakly to the mixing law. The observable finite-dimensional laws already determine the left-hand side, so two candidate mixing laws have equal continuous-function integrals and are equal by Lemma 10.8.

Step 5: justify the almost-sure random limit

We first prove the strong law for fixed p without assuming it. For i.i.d. Bernoulli(p) trials, Chebyshev along square indices gives

$$\mathbb{P}\left(\left|\frac{S_{m^2}}{m^2} - p\right| > \varepsilon\right) \le \frac{1}{4m^2\varepsilon^2}.$$

The probability that at least one such deviation occurs for some $m \ge M$ is at most $\sum_{m\ge M}(4m^2\varepsilon^2)^{-1}$, which tends to zero. Therefore only finitely many deviations occur with probability one. Intersecting these probability-one conclusions for $\varepsilon = 1, 1/2, 1/3, \ldots$ proves $S_{m^2}/m^2 \to p$ almost surely.

Since the partial sums are nondecreasing, for $m^2 \le n \le (m+1)^2$,

$$\frac{S_{m^2}}{(m+1)^2} \le \frac{S_n}{n} \le \frac{S_{(m+1)^2}}{m^2}.$$

Both bounds tend to p, proving $S_n/n \to p$ almost surely.

Now construct the mixture by choosing $\Theta \sim \mu$ and then the conditionally independent trials. The fixed-p result has conditional probability one for every p, so averaging gives $S_n/n \to \Theta$ with probability one. The constructed sequence has the same finite-string probabilities as the original sequence. These determine the probability law on infinite zero–one sequences (including convergence events, which use countable unions and intersections). Thus the original sequence also has an almost-sure empirical limit, distributed as μ.

Taking that limit as the original sequence's Θ transfers the conditional-i.i.d. representation as well: the pair consisting of the sequence and its empirical limit has the same law in the two descriptions. This completes the promised interpretation of the theorem.

Appendix A

Calculus refresher: tools for probability

A second encounter with limits, derivatives, integrals, and series

Calculus returns in probability whenever a sum becomes an area, a product needs an approximation, or a parameter must be chosen to maximize a likelihood. This appendix rebuilds those tools through worked examples. Read with a pencil: pause before each calculation and predict its sign, scale, or limiting value. The exercises at the end have worked answers.

Choose a route through the refresher

For birthdays, review logarithms, finite sums, and Taylor approximation, then read Section A.10. For derangements, review series and the exponential function, then read Section A.11. For continuous random variables, start with definite integrals and the fundamental theorem in Section A.5. The final beta-integral example is an optional bridge to Bayesian inference.

A.1 Algebra, functions, and finite sums

Keep track of the variable

In $f(x) = \lambda e^{-\lambda x}$, x is the input and $\lambda > 0$ is a fixed parameter. Differentiating with respect to x holds λ constant. Later, maximizing a likelihood in λ holds the observed data constant instead. Before differentiating, say which symbol is allowed to vary.

The natural logarithm $\log x$ is defined for $x > 0$ and is the inverse of the exponential function. Its essential rules are

$$e^{u+v} = e^u e^v, \qquad \log(ab) = \log a + \log b, \qquad \log(a^r) = r \log a \quad (a, b > 0).$$

Also $e^0 = 1$, $\log 1 = 0$, and $e^{\log a} = a$. Taking logarithms turns a product of positive factors into a sum. It does not turn a sum into a sum of logarithms: $\log(1 + 1) = \log 2 \neq 0 = \log 1 + \log 1$.

A sum is an instruction, not a new operation

The index in $\sum_{j=0}^{n-1} j$ runs through $0, 1, \ldots, n-1$; there are n terms. The two sums needed for birthday estimates are

$$\sum_{j=0}^{n-1} j = \frac{n(n-1)}{2}, \qquad \sum_{j=0}^{n-1} j^2 = \frac{n(n-1)(2n-1)}{6}.$$

Pair the first sum with its reversed order: each of the n paired terms is $n-1$. For the second identity, induction works: increasing n to $n+1$ adds n^2 to both sides. Both sides start at zero when $n = 1$.

The product notation $\prod_{j=0}^{n-1}(1 - j/d)$ means $1(1-1/d)\cdots(1-(n-1)/d)$. A sum with no terms is zero; a product with no factors is one. These conventions make the cases $n = 0$ work without separate algebraic exceptions.

Factorials are finite products, $n! = 1 \cdot 2 \cdots n$, with $0! = 1$. Cancel common factors before using a calculator:

$$\frac{n!}{(n-2)!} = n(n-1), \qquad \frac{\binom{n}{k}(n-k)!}{n!} = \frac{1}{k!}.$$

The second cancellation explains the simple denominators in the derangement probability.

A.2 Limits and continuity

The statement $f(x) \to L$ as $x \to a$ says that $f(x)$ can be made as close as desired to L by taking x sufficiently close to a, with $x \neq a$. The value at a itself is a separate question. For example,

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = \lim_{x \to 1} (x + 1) = 2.$$

The fraction is undefined at 1, but the nearby values have a limit. A function is *continuous* at a when the limit equals $f(a)$. Polynomials are continuous everywhere; rational functions are continuous where their denominators are nonzero. Exponentials are continuous, and logarithms are continuous on their positive domain.

For a sequence a_n, the notation $a_n \to L$ means that all sufficiently late terms stay arbitrarily close to L. Thus $1/n \to 0$, while $(-1)^n$ has no limit. The expression

$$\left(1 + \frac{1}{n}\right)^n \longrightarrow e \approx 2.7182818$$

is one familiar appearance of the exponential constant. Another, $e = \sum_{k=0}^{\infty} 1/k!$, will follow from Taylor's theorem below.

Common trap

A small term can accumulate. Although $1/n \to 0$, a sum of n copies of $1/n$ is always one. Likewise, replacing every birthday factor by something "close" to it needs an estimate for the total error.

Two practical limit tools

If $0 \le a_n \le b_n$ and $b_n \to 0$, then $a_n \to 0$ by squeezing. For example, $0 \le n/(n^2+1) \le 1/n$. For quotients with the indeterminate forms $0/0$ or ∞/∞, l'Hôpital's rule can replace a quotient by the quotient of its derivatives, provided the functions are differentiable nearby, the denominator derivative is nonzero there, and the latter quotient has a limit. It is not a rule for arbitrary fractions. A useful example is

$$\lim_{x\to\infty} \frac{x}{e^x} = \lim_{x\to\infty} \frac{1}{e^x} = 0.$$

Repeated application shows $x^m e^{-\lambda x} \to 0$ for a fixed nonnegative integer m and $\lambda > 0$. This justifies the boundary terms in exponential moment calculations.

A.3 Derivatives describe local change

The derivative is the limiting slope

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}.$$

For $f(x) = x^2$, the difference quotient is $2a + h$, so $f'(a) = 2a$. For small h, the tangent-line estimate is $f(a+h) \approx f(a) + f'(a)h$. This is an approximation near a; Taylor's theorem will quantify its error.

Function or operation	Derivative
c; x^r	0; rx^{r-1} where defined
e^x; $\log x$	e^x; $1/x$ for $x > 0$
$af + bg$	$af' + bg'$
fg	$f'g + fg'$
f/g	$(f'g - fg')/g^2$ where $g \neq 0$
$f(g(x))$	$f'(g(x))g'(x)$

The last identity is the *chain rule*: differentiate the outer function and multiply by the derivative of the inner function. For example,

$$\frac{d}{dx} e^{-3x} = -3e^{-3x}, \qquad \frac{d}{dx} \log(1-x) = -\frac{1}{1-x} \quad (x < 1).$$

The product and chain rules together give $\frac{d}{dx}(x^2 e^{-3x}) = (2x - 3x^2)e^{-3x}$. The second derivative f'' describes change in the slope: $f'' > 0$ means the graph bends upward, and $f'' < 0$ means it bends downward.

Example A.1 (Differentiation remembers a counting identity)**.** Differentiate the *finite* binomial identity $(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k$. For $n \ge 1$,

$$n(1+x)^{n-1} = \sum_{k=1}^{n} k \binom{n}{k} x^{k-1}.$$

At $x = 1$ this is $\sum_{k=0}^n k\binom{n}{k} = n2^{n-1}$. The same quantity counts subsets with a distinguished member: choose the member in n ways, then decide which of the other $n-1$ objects to include.

A.4 Optimization: logarithms simplify a likelihood

A differentiable function can have an interior maximum only where its derivative is zero. That is a candidate, not a conclusion: check the sign of the derivative or the second derivative, and check boundaries.

Suppose s successes and $f = n - s$ failures have been observed in independent trials with unknown success probability p. For a specified observed string the likelihood is $L(p) = p^s(1-p)^f$. If only the count is recorded, the extra factor $\binom{n}{s}$ does not depend on p and does not change the maximizing value.

When $0 < s < n$, maximize its logarithm on $0 < p < 1$:

$$\ell(p) = s \log p + f \log(1-p), \qquad \ell'(p) = \frac{s}{p} - \frac{f}{1-p}.$$

Logarithm is increasing, so the maximizer is preserved. Setting the derivative to zero gives $s(1-p) = fp$, hence $p = s/n$. Moreover,

$$\ell''(p) = -\frac{s}{p^2} - \frac{f}{(1-p)^2} < 0,$$

so this is the unique interior maximum. The likelihood is zero at both endpoints when both counts are positive. If $s = 0 < n$, $L(p) = (1-p)^n$ is maximized at $p = 0$; if $s = n > 0$, $L(p) = p^n$ is maximized at $p = 1$. With no observations, the likelihood is constant and selects no unique parameter.

Checkpoint

With 7 successes in 10 trials the maximizing parameter is .7. Explain why this calculation treats the data as fixed and p as variable. A likelihood is not, by itself, a probability density for p.

A.5 Integrals: area and accumulated change

Divide $[a, b]$ into narrow intervals of widths Δx_i, and choose a sample point t_i in each one. The sum $\sum_i f(t_i)\Delta x_i$ adds the areas of thin rectangles. For a continuous f, these sums tend to the definite integral $\int_a^b f(x)\,dx$ as the largest width tends to zero. If $f \geq 0$, this is area; below the axis, contributions are signed. Probability densities must be nonnegative so that their areas are probabilities.

Theorem A.2 (Fundamental theorem of calculus)**.** *If f is continuous on $[a, b]$, then $A(x) = \int_a^x f(t)\,dt$ satisfies $A'(x) = f(x)$ in the interior. If $G' = f$, then*

$$\int_a^b f(x)\,dx = G(b) - G(a).$$

An antiderivative turns an area calculation into an endpoint calculation. The indefinite integral $\int f(x)\,dx = G(x) + C$ is a family of functions; the definite integral is a number. The arbitrary constant cancels between the endpoints.

The power rule gives

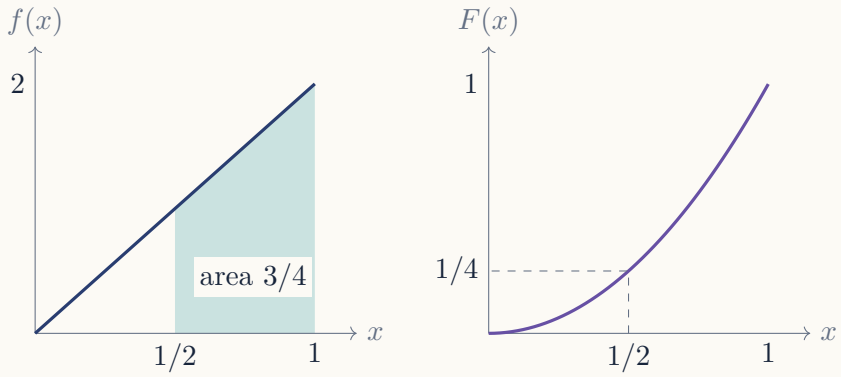
$$\int x^r\,dx = \frac{x^{r+1}}{r+1} + C \quad (r \neq -1), \qquad \int \frac{dx}{x} = \log|x| + C \quad (x \neq 0).$$

Use intervals on which the integrand is defined. Integrals distribute over finite sums and constant multiples, but not over products.

Example A.3 (Normalize, accumulate, average)**.** Suppose a proposed density is cx on $[0,1]$ and zero elsewhere. Normalization requires $1 = \int_0^1 cx\,dx = c/2$, so $c = 2$. For $0 \le t \le 1$, its CDF is $F(t) = \int_0^t 2x\,dx = t^2$. Its first two moments are

$$\mathbb{E}[X] = \int_0^1 x(2x)\,dx = \frac{2}{3}, \qquad \mathbb{E}[X^2] = \int_0^1 x^2(2x)\,dx = \frac{1}{2}.$$

Consequently $\mathrm{Var}(X) = 1/2 - 4/9 = 1/18$. The event $X \ge 1/2$ has probability $\int_{1/2}^1 2x\,dx = [x^2]_{1/2}^1 = 3/4$.



The shaded area under the density equals $1 - F(1/2)$. The height of the CDF already represents accumulated probability.

A.6 Substitution and integration by parts

Substitution reverses the chain rule

If $u = g(x)$, then $du = g'(x)\,dx$. For example,

$$\int_0^1 2xe^{-x^2}\,dx = \int_0^1 e^{-u}\,du = 1 - e^{-1}, \qquad u = x^2.$$

Both endpoints have been changed into u-values. In general,

$$\int_a^b h(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} h(u)\,du$$

under the usual continuity and differentiability assumptions. This identity also handles reversed endpoints correctly.

Common trap

After a substitution, change both the differential and the bounds. The expression $\int_a^b h(u)\,du$ still uses u as its input; old x-bounds cannot be retained unless they happen to give the same numbers.

Integration by parts reverses the product rule

Integrating $(uv)' = u'v + uv'$ gives

$$\int_a^b u(x)v'(x)\,dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x)\,dx.$$

Choose u so that differentiating it simplifies the remaining integral. For example, $u = x$, $v' = e^x$, and $v = e^x$ give

$$\int_0^1 xe^x\,dx = [xe^x]_0^1 - \int_0^1 e^x\,dx = e - (e-1) = 1.$$

This technique will also compute the mean and variance of an exponential waiting time and the beta integral used in Bayesian updating.

A.7 Improper integrals and infinite expectations

An infinite endpoint is a limit: $\int_0^\infty f(x)\,dx = \lim_{R\to\infty}\int_0^R f(x)\,dx$, when that limit exists. A singular finite endpoint is treated similarly; for example $\int_0^1 x^{-1/2}\,dx = \lim_{\varepsilon\downarrow 0}[2\sqrt{x}]_\varepsilon^1 = 2$. Do not substitute infinity as if it were an ordinary number.

Example A.4 (Exponential waiting times)**.** For $\lambda > 0$, let $f(x) = \lambda e^{-\lambda x}$ on $x \geq 0$. Its integral on $[0, R]$ is $1 - e^{-\lambda R}$, tending to one. Thus it is a density, and $\mathbb{P}(X > t) = \int_t^\infty \lambda e^{-\lambda x}\,dx = e^{-\lambda t}$ for $t \geq 0$. Integration by parts on a finite interval gives

$$\int_0^R x\lambda e^{-\lambda x}\,dx = -Re^{-\lambda R} + \frac{1 - e^{-\lambda R}}{\lambda}.$$

Since $Re^{-\lambda R} \to 0$, the mean is $1/\lambda$. Another integration by parts, using $R^2e^{-\lambda R} \to 0$, gives

$$\mathbb{E}[X^2] = 2\int_0^\infty xe^{-\lambda x}\,dx = \frac{2}{\lambda^2}, \qquad \operatorname{Var}(X) = \frac{1}{\lambda^2}.$$

The median solves $e^{-\lambda m} = 1/2$, hence $m = \log 2/\lambda$. It differs from the mean because the long right tail pulls the mean upward.

Example A.5 (A density can have no finite mean)**.** Let $f(x) = 1/(1+x)^2$ on $x \geq 0$ and zero elsewhere. Its total area is one, because $\int_0^R (1+x)^{-2}\,dx = 1 - 1/(1+R) \to 1$. But

$$\int_0^R \frac{x}{(1+x)^2}\,dx = \log(1+R) + \frac{1}{1+R} - 1 \longrightarrow \infty.$$

The distribution is valid and its expected value is $+\infty$. Normalizing a density and proving its mean finite are different calculations.

The general tail check is $\int_1^\infty x^{-r}\,dx < \infty$ exactly when $r > 1$. Integrate x^{-r} for $r \neq 1$ and use $\log x$ for $r = 1$ to verify the boundary between convergence and divergence.

A.8 Sequences, series, and geometric probabilities

An infinite series $\sum_{k=0}^{\infty} a_k$ is defined as the limit of its partial sums $S_m = \sum_{k=0}^{m} a_k$, when this limit exists. The terms a_k and their sums S_m are different sequences. Convergence requires $a_k \to 0$, since $a_k = S_k - S_{k-1}$, but that condition alone is insufficient. The harmonic series diverges:

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots .$$

Every displayed group after the first two terms contributes at least $1/2$. There are arbitrarily many such groups.

The geometric series

For $r \neq 1$, subtracting r times a finite sum from itself gives

$$1 + r + \cdots + r^m = \frac{1 - r^{m+1}}{1 - r}.$$

If $|r| < 1$, the final power tends to zero, so

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1 - r}.$$

For $0 \leq r < 1$, the tail after r^m is exactly $r^{m+1}/(1 - r)$. The condition $|r| < 1$ matters: substituting $r = 2$ into the infinite formula would produce a negative answer for a growing positive sum.

Example A.6 (Wait until the first success)**.** In independent trials with success probability $0 < p \leq 1$, let N count trials up to and including the first success. With $r = 1 - p$, $\mathbb{P}(N = k) = pr^{k-1}$ for $k \geq 1$, and the geometric series checks that these probabilities sum to one. For $0 < p < 1$, differentiating the geometric power series inside $|r| < 1$ gives

$$\sum_{k=1}^{\infty} kr^{k-1} = \frac{1}{(1 - r)^2}, \qquad \mathbb{E}[N] = \sum_{k=1}^{\infty} kpr^{k-1} = \frac{1}{p}.$$

Power series can be differentiated term by term strictly inside their interval of convergence. Alternatively, differentiate the finite geometric identity first and let its length tend to infinity; the extra terms involve mr^m and tend to zero. For $p = 1$, $N = 1$ with certainty, so the same mean formula holds directly.

Alternating sums come with a useful error bound

If b_k decreases to zero through positive values, the partial sums of $b_0 - b_1 + b_2 - \cdots$ alternate above and below its limit. The magnitude of the error is at most the first omitted term. To see why, group consecutive terms: the even partial sums decrease, the odd partial sums increase, and their difference tends to zero. The limit is trapped between the two adjacent partial sums. For a strictly decreasing positive tail, the limit is strictly between them. This is exactly the estimate needed for derangements.

A.9 Taylor polynomials and a reason to trust them

The degree-m Taylor polynomial at a matches the first m derivatives of a function there:

$$T_m(x) = \sum_{k=0}^{m} \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

The factorial appears because the kth derivative of $(x-a)^k$ at a is $k!$. The first two terms give the tangent line; the quadratic term also records curvature.

Theorem A.7 (Taylor's error estimate)**.** *Suppose f has continuous derivatives through order $m+1$ on the interval between a and x, and $|f^{(m+1)}(t)| \le M$ there. Then*

$$|f(x) - T_m(x)| \le \frac{M|x-a|^{m+1}}{(m+1)!}.$$

One way to understand the bound is to repeatedly integrate the derivative error using the fundamental theorem. The remainder has the form

$$R_m(x) = \frac{1}{m!}\int_a^x (x-t)^m f^{(m+1)}(t)\,dt.$$

Taking absolute values and integrating $M|x-t|^m/m!$ gives the estimate, with the same conclusion when $x < a$ after reversing the limits.

The exponential series

All derivatives of e^x are e^x. Expanding at zero gives

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^m}{m!} + R_m(x), \qquad |R_m(x)| \le \frac{e^{|x|}|x|^{m+1}}{(m+1)!}.$$

For fixed x, the successive factorial-denominator terms have ratio $|x|/(m+2)$, eventually less than $1/2$. They therefore tend to zero, proving

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} \quad \text{for every real } x.$$

For example, $e^{.1} \approx 1 + .1 + .1^2/2 = 1.105$; the error is at most $e^{.1}(.1)^3/6 < .000185$. The estimate is accurate because it includes a small remainder bound, not merely because a few terms were retained.

Logarithms from a geometric series

For $0 \le x < 1$, integrate the finite geometric identity for $1/(1-t)$ from 0 to x. For $m \ge 1$ this gives

$$-\log(1-x) = \sum_{k=1}^{m} \frac{x^k}{k} + \int_0^x \frac{t^m}{1-t}\,dt.$$

Since $1/(1-t) \le 1/(1-x)$ on this interval, the remainder is between zero and $x^{m+1}/[(m+1)(1-x)]$. It tends to zero for fixed $x < 1$, so

$$\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots.$$

At $x = .1$, the first two terms give $-.105$, with error less than $.1^3/[3(.9)] < .000371$. At $x = 1$, the logarithm is undefined and the corresponding harmonic series diverges. An expansion's domain is part of the formula.

Common trap

A function is not automatically equal to its infinite Taylor series just because many derivatives exist. For the exponential and logarithm used here, the explicit remainders above prove equality on the stated domains. For an approximation, keep enough terms to control the error at the inputs you actually use.

A.10 Birthdays: logarithms turn the product into a sum

In the independent uniform d-day birthday model, write $q_d(n)$ for the chance of a collision and $r_d(n) = 1 - q_d(n)$ for the chance of no collision. For $1 \le n \le d$,

$$r_d(n) = \prod_{j=0}^{n-1} (1 - j/d), \qquad \log r_d(n) = \sum_{j=0}^{n-1} \log(1 - j/d).$$

Keeping just the linear term $\log(1 - x) \approx -x$ yields

$$\log r_d(n) \approx -\frac{n(n-1)}{2d}, \qquad q_d(n) \approx 1 - \exp\left(-\frac{n(n-1)}{2d}\right).$$

There are $\binom{n}{2}$ pairs, so this exponent already suggests why a class much smaller than d can produce a match.

An error estimate, with a direction

For $0 \le x \le c < 1$,

$$-\log(1 - x) - x = \int_0^x \frac{t}{1 - t}\, dt.$$

Bounding the integrand between t and $t/(1 - c)$ gives

$$-x - \frac{x^2}{2(1 - c)} \le \log(1 - x) \le -x - \frac{x^2}{2} \le -x.$$

Set

$$c = \frac{n - 1}{d}, \qquad S_1 = \frac{n(n-1)}{2d}, \qquad S_2 = \frac{n(n-1)(2n-1)}{6d^2}.$$

Sum the inequalities over the birthday factors and exponentiate:

$$\boxed{1 - e^{-S_1 - S_2/2} \;\le\; q_d(n) \;\le\; 1 - e^{-S_1 - S_2/[2(1-c)]}.}$$

In particular, the simpler estimate $1 - e^{-S_1}$ is a lower bound. Its omitted correction in the logarithm is at most $S_2/[2(1 - c)]$. The approximation is useful when this correction is small; having each individual factor close to one is not by itself an error guarantee. For $n = 0$, the exact collision probability is zero; for $n > d$, it is one, and we use those exact cases instead of the logarithmic approximation.

n on 365 days	$1-e^{-S_1}$	$1-e^{-S_1-S_2/2}$	Exact $q_{365}(n)$
10	.115991	.116936	.116948
22	.468938	.475496	.475695
23	.500002	.507073	.507297
30	.696320	.705916	.706316
50	.965131	.970040	.970374

The entries are rounded, not exact inequalities between the printed decimals. At $n = 23$, the rigorous bounds give $.50707 < q_{365}(23) < .50753$; direct multiplication gives about .507297.

Where the number 23 comes from

Set the first approximation equal to $1/2$. Taking logarithms gives $n(n-1) = 2d \log 2$, hence the positive solution

$$n \approx \frac{1+\sqrt{1+8d\log 2}}{2}.$$

For $d = 365$, this is about 22.99994, suggesting a threshold of 23. The exact probabilities at 22 and 23, .475695 and .507297, verify the conclusion. This use of a smooth expression estimates an integer class size; the exact model still checks the adjacent integers.

Calculate with small tails intact

For numerical work, sum $\log(1-j/d)$ and then exponentiate instead of multiplying many factors. Software functions named `log1p(z)` compute $\log(1+z)$ accurately near zero, and `expm1(z)` computes $e^z - 1$. Thus $q_d(n) = -\texttt{expm1}(\sum_j \texttt{log1p}(-j/d))$ is a useful evaluation for $n \le d$. It evaluates the exact product formula with floating-point arithmetic; it is not the Taylor approximation.

A.11 Derangements: why the limiting probability is one over e

Section 2.7 obtained an exact finite expression:

$$\frac{D_n}{n!} = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!}.$$

Now set $x = -1$ in the exponential series. Its sum is e^{-1}. The positive terms in every omitted tail strictly decrease to zero, so the alternating-series estimate gives, for $n \ge 0$,

$$\left|\frac{D_n}{n!} - e^{-1}\right| < \frac{1}{(n+1)!}.$$

Even partial sums are above the limit and odd partial sums below it. For example,

$$\frac{44}{120} < e^{-1} < \frac{265}{720}, \qquad e^{-1} \approx .3678794412.$$

At $n = 10$, the exact probability is $1334961/3628800 \approx .3678794643$, within $1/11! < 2.51 \cdot 10^{-8}$ of the limit. This is why the approximation becomes excellent so quickly.

Proposition A.8 (Recover an integer count by rounding)**.** *For every $n \geq 1$, D_n is the nearest integer to $n!/e$.*

Proof. Multiply the strict error bound by $n!$ to obtain $|D_n - n!/e| < 1/(n+1) \leq 1/2$. Thus the integer D_n lies strictly within half a unit of $n!/e$. □

The condition $n \geq 1$ matters: $D_0 = 1$, whereas $1/e$ rounds to zero. For large n, the exact recurrence is preferable for computer calculations because ordinary floating-point arithmetic may not retain unit accuracy in $n!/e$.

A further connection: the number of fixed points

For a uniform permutation, the probability of exactly k fixed points is

$$\frac{\binom{n}{k} D_{n-k}}{n!} = \frac{1}{k!} \frac{D_{n-k}}{(n-k)!} \longrightarrow \frac{e^{-1}}{k!}$$

as $n \to \infty$ with k fixed. These limiting probabilities sum to one by the exponential series. They form the Poisson distribution with mean one, introduced in Chapter 5. Already, indicators tell us that the expected number of fixed points is exactly one for every $n \geq 1$.

Birthdays and derangements both lead to exponentials, through different calculations: birthdays use a logarithmic expansion of a product; derangements produce a finite alternating exponential sum by inclusion–exclusion. In each case, the remainder explains the accuracy.

A.12 Optional: a beta integral for Bayesian updating

For nonnegative integers s, f, consider

$$I_{s,f} = \int_0^1 p^s (1-p)^f \, dp.$$

This integral normalizes the likelihood from s successes and f failures when a uniform prior density is used. For $s \geq 1$, integration by parts with $u = p^s$ and $v' = (1-p)^f$ gives

$$I_{s,f} = \frac{s}{f+1} I_{s-1,f+1}.$$

The boundary term is zero at both ends. Starting from $I_{0,m} = 1/(m+1)$ and applying the recurrence s times gives

$$I_{s,f} = \frac{s!\,f!}{(s+f+1)!}.$$

For example, $I_{3,1} = 1/20$, so $20p^3(1-p)$ is a density on $[0,1]$. Its mean is $I_{4,1}/I_{3,1} = (1/30)/(1/20) = 2/3$. More generally the mean is $I_{s+1,f}/I_{s,f} = (s+1)/(s+f+2)$. Chapter 9 interprets this as a posterior mean and predictive probability.

A.13 Practice: rebuild the tools

Exercises

1. Simplify $\binom{n}{3}(n-3)!/n!$ for $n \geq 3$. Evaluate $\sum_{j=0}^{5} j$ and $\sum_{j=0}^{5} j^2$. [Fluency]

2. Find $\lim_{x\to 2}(x^2-4)/(x-2)$ and $\lim_{n\to\infty} n/(n^2+1)$. Explain why direct substitution is insufficient for the first. [Fluency]

3. Differentiate x^2e^{-x} and $\log(1-x^2)$, specifying the real domain of the second function. [Fluency]

4. Use a tangent line at zero to approximate $e^{.05}$. Use Taylor's estimate to bound its absolute error. [Reasoning]

5. Maximize $p^8(1-p)^2$ on $[0,1]$. Explain why maximizing p^{10} needs a boundary check instead. [Fluency]

6. Differentiate the binomial identity twice and set $x = 1$ to prove $\sum_{k=0}^{n} k(k-1)\binom{n}{k} = n(n-1)2^{n-2}$ for $n \geq 2$. [Reasoning]

7. Let a density be cx^2 on $[0,1]$, zero elsewhere. Find c, its CDF, $\mathbb{P}(X > 1/2)$, its mean, and its variance. [Fluency]

8. Evaluate $\int_0^2 2xe^{-x^2}\,dx$ using substitution. Show the new bounds. [Fluency]

9. Evaluate $\int_0^1 xe^{2x}\,dx$ by parts. [Fluency]

10. Normalize ce^{-3x} on $x \geq 0$. Find its mean, variance, and median. [Fluency]

11. For density $1/(1+x)^2$ on $x \geq 0$, find the survival probability $\mathbb{P}(X > t)$ for $t \geq 0$ and the median. Explain how a finite median is compatible with an infinite mean. [Reasoning]

12. Sum $\sum_{k=4}^{\infty}(1/2)^k$. If a geometric waiting time has success probability .2, find its mean and $\mathbb{P}(N > 5)$. [Fluency]

13. Explain why terms tending to zero do not guarantee that a series converges. Contrast $\sum 1/k$ with $\sum(1/2)^k$. [Reasoning]

14. Approximate $\log(.95)$ using $-x - x^2/2$ at $x = .05$. Bound the error using the logarithmic remainder. [Fluency]

15. For $n = 23, d = 365$, compute S_1, S_2, c and use the two logarithmic bounds to bracket the collision probability. Compare with the first-order estimate. [Reasoning]

16. Compute D_8 using the exact recurrence. Bound the error in $D_8/8! \approx e^{-1}$, determine which is larger, and check the nearest-integer conclusion. [Reasoning]

17. Evaluate $\int_0^1 p^2(1-p)^2\,dp$ both by expanding and by the beta formula. Normalize the integrand and find the resulting density's mean. [Fluency]

18. For a uniform permutation of four objects, find the probability of exactly two fixed points. Compare it with the limiting value $e^{-1}/2!$. Explain why a limiting formula need not be accurate for such a small permutation. [Reasoning]

A.14 Worked answers to the refresher

1. Cancel factorials to get $1/3! = 1/6$. The sums are $5 \cdot 6/2 = 15$ and $5 \cdot 6 \cdot 11/6 = 55$.

2. Away from $x = 2$, the fraction equals $x + 2$, so the first limit is 4. At 2 the original denominator is zero. For the sequence, $0 \le n/(n^2+1) \le 1/n \to 0$.

3. The product rule gives $(2x - x^2)e^{-x}$. The logarithm requires $1 - x^2 > 0$, so $-1 < x < 1$; its derivative is $-2x/(1-x^2)$ by the chain rule.

4. The approximation is $1 + .05 = 1.05$. On $[0, .05]$, the second derivative of e^x is at most $e^{.05}$, so the error is at most $e^{.05}(.05)^2/2 < .001315$. The positive quadratic and later terms show the tangent estimate is too small.

5. On $0 < p < 1$, the log derivative is $8/p - 2/(1-p)$, which vanishes at $p = .8$. Its derivative is negative, so this is the maximum; the original function is zero at both endpoints. The function p^{10} increases on $[0, 1]$ and is maximized at 1.

6. Twice differentiating gives $n(n-1)(1+x)^{n-2} = \sum_{k=2}^{n} k(k-1)\binom{n}{k}x^{k-2}$. Set $x = 1$. The terms $k = 0, 1$ contribute zero to the requested sum.

7. Normalization gives $c/3 = 1$, hence $c = 3$. The CDF is 0 below zero, x^3 on $[0, 1]$, and 1 above one. Thus $\mathbb{P}(X > 1/2) = 7/8$, $\mathbb{E}[X] = 3\int_0^1 x^3\,dx = 3/4$, and $\mathbb{E}[X^2] = 3/5$. The variance is $3/5 - 9/16 = 3/80$.

8. With $u = x^2$, $du = 2x\,dx$ and the bounds become $0, 4$. The integral is $\int_0^4 e^{-u}\,du = 1 - e^{-4}$.

9. Set $u = x$ and $v' = e^{2x}$, so $v = e^{2x}/2$. The result is $[xe^{2x}/2]_0^1 - \frac{1}{2}\int_0^1 e^{2x}\,dx = e^2/2 - (e^2-1)/4 = (e^2+1)/4$.

10. The total integral is $c/3$, so $c = 3$. This is an exponential density with rate 3: its mean is $1/3$, variance $1/9$, and median $(\log 2)/3$.

11. Integrating the tail gives $\mathbb{P}(X > t) = 1/(1+t)$. Setting this to $1/2$ gives median 1. A median divides probability in half; a mean weights outcomes by their magnitudes. The tail is heavy enough that those weighted contributions have infinite total.

12. The geometric tail is $(1/2)^4/(1-1/2) = 1/8$. The waiting-time mean is $1/.2 = 5$. The event $N > 5$ means five initial failures, with probability $.8^5 = .32768$.

13. Harmonic terms tend to zero, but grouping indices between consecutive powers of two supplies infinitely many contributions of at least $1/2$. The geometric partial sums instead approach a finite limit because the remainder is a vanishing power divided by $1 - r$.

14. The approximation is $-.05-.05^2/2 = -.05125$. The omitted magnitude is at most $.05^3/[3(.95)] < .00004386$. The exact logarithm is more negative, so $-.05129386 < \log(.95) < -.05125$.

15. Here $S_1 = 253/365$, $S_2 = 3795/133225$, and $c = 22/365$. The lower and upper expressions are approximately .507072664 and .507522764. Thus an outward-rounded bracket is $.50707 < q_{365}(23) < .50753$. The first-order estimate is .500001752, which is noticeably lower.

16. Continuing from $D_6 = 265$ gives $D_7 = 7 \cdot 265 - 1 = 1854$ and $D_8 = 8 \cdot 1854 + 1 = 14833$. Since 8 is even, $D_8/8! > e^{-1}$, with error less than $1/9! \approx .000002756$. Multiplying by $8!$ puts 14833 within $1/9$ of $8!/e$, so it is the nearest integer.

17. Expand $p^2(1-p)^2 = p^2 - 2p^3 + p^4$; its integral is $1/3 - 1/2 + 1/5 = 1/30$, also $2!2!/5!$. The density is $30p^2(1-p)^2$, with mean $30(1/4 - 2/5 + 1/6) = 1/2$.

18. Choose the fixed pair and swap the other two, giving $\binom{4}{2} D_2/4! = 6/24 = 1/4$. The limiting probability is about .18394. In the exact formula $\frac{1}{2!} D_2/2!$, only two objects remain to derange; their probability $1/2$ is still far from $1/e$.

Appendix B

Selected hints and solutions

Chapter 1

For the at-least-one-ace problem, count the complement: $\binom{52}{5} - \binom{48}{5}$. For divisibility by 2, 3, or 5, inclusion–exclusion gives

$$500 + 333 + 200 - 166 - 100 - 66 + 33 = 734.$$

The four-person selection counts from nine students are $\binom{9}{4} = 126$, $(9)_4 = 3024$, and $4\binom{9}{4} = 504$. The repeated-letter count for `LEVEL` is $5!/(2!2!) = 30$. The coefficient of x^2 in $(1+3x)^5$ is $\binom{5}{2}3^2 = 90$. The length-five strings using all three symbols number $3^5 - 3 \cdot 2^5 + 3 = 150$. For Exercise 3, choose the three letter positions, then fill all positions: $\binom{8}{3}26^3 10^5$. For Exercises 13–16:

13. A specified string with three successes has probability $(1/3)^3(2/3)^2 = 4/243$. There are $\binom{5}{3} = 10$ such strings, giving total probability $40/243$.

14. The ordered count is $3^4 = 81$. The unordered count is $\binom{4+3-1}{4} = 15$. A selection $AAAA$ has one ordering, whereas $AAAB$ has four; the multiplicity is not constant.

15. There are 11 letters, with four I's, four S's, two P's, and one M. The count is $11!/(4!4!2!) = 34650$.

16. Count by subset size to obtain $\sum_k k\binom{n}{k}$. Alternatively, choose the distinguished member in n ways and any subset of the remaining $n-1$ members in 2^{n-1} ways.

Chapter 2

For the maximum of two dice to equal 4, both dice must be at most 4 but not both at most 3, giving $(4^2 - 3^2)/36 = 7/36$. For two uniform points X, Y, the band $|X - Y| < .1$ has area $1 - 2(.9)^2/2 = .19$. For the weighted four-outcome space, the probability of an even outcome is $(2+4)/10 = 3/5$, not $2/4$. On a 30-day calendar, the birthday collision probability is about .41356 for six people and .53084 for seven; thus the first size above $1/2$ is seven. For four people on five days, there are $5 \cdot 4 \cdot 3 \cdot 2 = 120$ distinct-birthday lists among $5^4 = 625$ lists, so the collision probability is $1 - 120/625 = 101/125$. The quantity $\binom{n}{2}/d$ sums probabilities of overlapping events. It is an upper bound, not necessarily a probability. For Exercises 12–16:

12. There are $5 \cdot 4 = 20$ equally likely ordered pairs of distinct labeled balls. Exactly $3 \cdot 2 = 6$ have two reds, giving $3/10$. Unordered pairs give $\binom{3}{2}/\binom{5}{2} = 3/10$ as well.

13. Choose the fixed pair and derange the remaining three: $\binom{5}{2} D_3/5! = 10 \cdot 2/120 = 1/6$.

14. $\mathbb{P}(A_1) = 1/4$ and $\mathbb{P}(A_1 \cap A_2) = 2!/4! = 1/12$. The product of the single probabilities is $1/16$, so the events are dependent.

15. The preceding recurrence gives $(-1)^n = (n-1)D_{n-2} - D_{n-1}$. Substitute into $D_n = nD_{n-1} + (-1)^n$ to get the claimed formula. Then $D_7 = 6(265 + 44) = 1854$.

16. The probability is $\sum_{j=1}^{\infty} 4^{-j} = (1/4)/(1 - 1/4) = 1/3$.

Chapter 3

For the coin problem,

$$\mathbb{P}(D \mid HHH) = \frac{1 \cdot .2}{1 \cdot .2 + (1/2)^3 \cdot .8} = \frac{2}{3}.$$

A pairwise-not-mutual example: flip two fair coins; let A be first-heads, B second-heads, and C equal-results.

Chapter 4

Exercises 1–18 use discrete variables. Exercises 19–20 belong to the later continuous section; Exercises 21–25 give further finite linearity examples.

1. The masses at $0, 1, 2$ are $1/4, 1/2, 1/4$. The CDF is 0 for $x < 0$, $1/4$ for $0 \le x < 1$, $3/4$ for $1 \le x < 2$, and 1 for $x \ge 2$.

2. $\mathbb{E}[X] = (1 + 4 + 9 + 16)/10 = 3$ and $\mathbb{E}[X^2] = (1 + 8 + 27 + 64)/10 = 10$, so $\mathrm{Var}(X) = 1$. Also $\mathbb{P}(X \ge 3) = (3 + 4)/10 = 7/10$. The equally weighted average $5/2$ uses the wrong probabilities.

3. Write $m = \mathbb{E}[X]$ and expand: $\mathbb{E}[(X - m)^2] = \mathbb{E}[X^2] - 2m\mathbb{E}[X] + m^2 = \mathbb{E}[X^2] - m^2$.

4. Let I_i indicate that i is fixed and write $N = \sum I_i$. Since $\mathbb{P}(I_i = 1) = 1/n$, linearity gives $\mathbb{E}[N] = 1$ for $n \ge 1$. No independence is used.

5. One example pays 999 with probability $1/1000$ and -1 otherwise. Its mean is zero and its variance is $999^2/1000 + 999/1000 = 999$. A single play loses one unit with probability $999/1000$; a zero mean does not make the payoff certain or the chance of a loss small.

6. Chebyshev gives at most $1/9$. For a fair die the mean is $7/2$, the standard deviation is $\sqrt{35/12}$, and every value is within $5/2$ of the mean. Since $3\sqrt{35/12} > 5/2$, the exact probability is zero. The bound is valid but loose.

7. Equal masses at $-1, 1$ have mean zero and variance one. Another example has masses $1/4, 1/2, 1/4$ at $-\sqrt{2}, 0, \sqrt{2}$; its mean is zero and second moment is also one.

8. The event consists of $(4, 6), (5, 5), (6, 4), (5, 6), (6, 5), (6, 6)$, so its probability is $6/36 = 1/6$.

9. The masses at $0, 1, 2$ are $4/9, 4/9, 1/9$. The CDF is 0 below zero, $4/9$ on $[0, 1)$, $8/9$ on $[1, 2)$, and 1 on $[2, \infty)$. The first two moments are $2/3$ and $8/9$, giving variance $8/9 - 4/9 = 4/9$. Also $\mathbb{P}(X \ge 1) = 5/9$.

10. The values of Y at $X = 0, 1, 2, 3$ are $1, 0, 1, 4$. Its masses at $0, 1, 4$ are $3/8, 1/2, 1/8$, giving mean $0(3/8) + 1(1/2) + 4(1/8) = 1$.

11. The payoff probabilities are $1/2, 1/3, 1/6$. The mean is $-1+1/3+2/3 = 0$, and the second moment is $2 + 1/3 + 8/3 = 5$, so the variance is five. The positive-payoff probability is $1/3 + 1/6 = 1/2$. After a fee of one, the net values are $-3, 0, 3$: the mean is -1, variance remains five, and the probability of a positive net payoff becomes $1/6$.

12. At $r = 0, 1, 2, 3, 4$, the permutation counts are $9, 8, 6, 0, 1$, respectively, out of 24. Thus $\mathbb{E}[N] = (8 + 12 + 4)/24 = 1$ and $\mathbb{E}[N^2] = (8 + 24 + 16)/24 = 2$, giving variance one. Fixing three of four objects forces the fourth to be fixed as well.

13. There are $\binom{23}{2} = 253$ pairs, each matching with probability $1/365$, so the mean is $253/365 \approx .69315$. Multiple pairs can match in one outcome. Markov at threshold one gives $\mathbb{P}(C \geq 1) \leq 253/365$, whereas the exact probability is about .50730.

14. The new mean is $2(7/2) - 3 = 4$, variance is $4(35/12) = 35/3$, and standard deviation is $\sqrt{35/3}$.

15. Each of $I, J = 1 - I$ has variance $1/4$, but $I + J = 1$ has variance zero. With an independent second indicator, the sum instead has variance $1/4 + 1/4 = 1/2$.

16. The sum has mean seven and variance $35/6$. The difference has mean zero and variance $35/6$: negation changes the mean's sign but not the variance. The average has mean $7/2$ and variance $(35/6)/4 = 35/24$.

17. For $n \geq 2$, expand $N(N-1) = \sum_{i \neq j} I_i I_j$. Each of the $n(n-1)$ ordered terms has expectation $1/[n(n-1)]$, so $\mathbb{E}[N(N-1)] = 1$. Adding $\mathbb{E}[N] = 1$ gives $\mathbb{E}[N^2] = 2$ and variance one. For $n = 1$, $N = 1$ is constant and its variance is zero.

18. Write $X - a = (X - m) + (m - a)$, where $m = \mathbb{E}[X]$. The cross term has expectation $2(m - a)\mathbb{E}[X - m] = 0$. Thus $\mathbb{E}[(X - a)^2] = \mathrm{Var}(X) + (m - a)^2$, uniquely minimized by $a = m$.

19. Normalization gives $c \int_0^2 x \, dx = 2c = 1$, so $c = 1/2$. The mean is $\int_0^2 x^2/2 \, dx = 4/3$ and the second moment is $\int_0^2 x^3/2 \, dx = 2$. The variance is $2 - 16/9 = 2/9$.

20. The density is $3x^2$ on $[0, 1]$, zero elsewhere. The tail probability is $1 - (1/2)^3 = 7/8$. The first two moments are $3/4$ and $3/5$, hence the variance is $3/5 - 9/16 = 3/80$.

21. A position j is a record with probability $1/j$. Thus the total mean is $H_8 = 761/280$, and the mean for positions 5 through 8 is $H_8 - H_4 = 533/840$. Linearity sums the relevant position indicators without requiring them to be independent.

22. Each of the ten pairs is inverted with probability $1/2$, so the inversion mean is five. There are only four adjacent pairs, each descending with probability $1/2$, giving mean two. An inversion can involve nonadjacent positions, which a descent does not count.

23. Name the two strands' ends a, b and c, d. The matching $(ab)(cd)$ makes two loops; $(ac)(bd)$ and $(ad)(bc)$ each make one. All three are equally likely, so $\mathbb{P}(L_2 = 2) = 1/3$ and $\mathbb{P}(L_2 = 1) = 2/3$, giving mean $4/3$. Each specified original strand makes a one-strand loop precisely when its two ends are paired, with probability $1/(2n - 1)$. The expected number of such loops is $n/(2n - 1)$ by linearity. This does not count loops containing two or more original strands.

24. A head run starts either at a head on toss one or at a tail–head pair at positions $j - 1, j$, $j \geq 2$. Their indicators have means p and $(1 - p)p$, respectively. Summing gives $p + (n - 1)p(1 - p)$. At $p = 0$ there are no runs; at $p = 1$ there is one. Ten fair tosses give $1/2 + 9/4 = 11/4$ expected head runs.

25. Two of eight vertex colorings are monochromatic and give zero crossing edges; the other six give two. Thus the probabilities are $1/4, 3/4$, and the mean is $3/2$. The three edge indicators each have success probability $1/2$ but cannot all be one, so they are not mutually independent. For weights $w_e \geq 0$, the crossing weight is $\sum_e w_e I_e$, with mean $\frac{1}{2}\sum_e w_e$. Some coloring attains at least that mean.

Chapter 5

Exercises 1–14 concern discrete families and coupon collection; Exercises 15–17 use continuous distributions. Exercises 18–27 concern the additional mechanisms, unequal coupon collection, and Poissonization.

1. The complement gives $1 - .7^5 - 5(.3)(.7)^4 = .47178$. The mean is $5(.3) = 1.5$ and the variance is $5(.3)(.7) = 1.05$.

2. For $0 < p < 1$ and nonnegative integers s, t, divide $(1-p)^{s+t}$ by $(1-p)^s$ to get $(1-p)^t$. The denominator is positive. If $p = 1$, only $s = 0$ is admissible; conditioning on an event of probability zero is not defined.

3. Here $np = 2$. The exact probability is $.998^{1000} \approx .135064522$, compared with $e^{-2} \approx .135335283$ for a Poisson variable of mean two. The absolute error is about .000270761.

4. The mean is $9/2$, variance $(8^2 - 1)/12 = 21/4$, and the probability of exceeding $9/2$ is $4/8 = 1/2$.

5. The indicator has mean .2 and variance $.2(.8) = .16$. Ten independent repetitions have a count of mean two and variance 1.6. Recording one indicator ten times gives $10I$, still of mean two but variance $100(.16) = 16$. Its only possible values are zero and ten.

6. Without replacement, count each possible number of red balls: $\binom{2}{2}/\binom{5}{2} = 1/10$ for zero reds, $\binom{3}{1}\binom{2}{1}/\binom{5}{2} = 6/10$ for one, and $\binom{3}{2}/\binom{5}{2} = 3/10$ for two. The mean is $6/5$ and variance $9/25$. With replacement the masses are $4/25, 12/25, 9/25$, with mean $6/5$ and variance $12/25$.

7. Examples: a fair die fits discrete uniform, whereas a loaded die generally does not. A success indicator fits Bernoulli, whereas a count taking values $0, 1, 2$ does not. A count of independent fair tosses fits binomial; repeated copies of one toss do not. A uniform subset of a fixed urn fits hypergeometric; draws with replacement generally do not. Independent repeated trials with a fixed success chance fit geometric; changing that chance after each failure generally does not. For a Poisson example, take the limiting count of n independent opportunities each of probability λ/n. A binomial count with a fixed finite maximum and positive mean cannot be exactly Poisson, which has an unbounded set of possible counts.

8. The roll count is geometric with $p = 1/6$, so its mean is six, variance is thirty, and $\mathbb{P}(T > 5) = (5/6)^5 \approx .40188$. The failure count $T - 1$ has mean five.

9. $\mathbb{P}(X \geq 2) = 1 - e^{-3}(1+3) = 1 - 4e^{-3} \approx .80085$. Also $\mathbb{E}[X(X-1)] = 9$, hence $\mathbb{E}[X^2] = 9 + 3 = 12$ and variance $12 - 9 = 3$.

10. With replacement, the count is $\mathrm{Bin}(4, 3/10)$ with mean $6/5$ and variance $21/25$. Without replacement, the mean is still $6/5$ but the variance is $(21/25)(16/19) = 336/475$. The draws have the same marginal marked probability; without replacement, their dependence reduces the variance.

11. Decompose the collection time into independent geometric stages with success probabilities $1, 5/6, 4/6, 3/6, 2/6, 1/6$. The mean is $6H_6 = 147/10 = 14.7$ and the variance is $36\sum_{j=1}^{6} j^{-2} - 6H_6 = 3899/100 = 38.99$. When two faces remain, the expected further wait is $6/2 + 6/1 = 9$ rolls.

12. Stop the stage sum after W_{r-1}:

$$\mathbb{E}[T_{n,r}] = \sum_{k=0}^{r-1} \frac{n}{n-k} = n(H_n - H_{n-r}).$$

The remaining stages have expected total nH_{n-r}, even conditional on the particular history up to seeing r types. For $r = n$ this is zero, using $H_0 = 0$.

13. Require $100e^{-m/100} \le .05$, or $m \ge 100 \log 2000 \approx 760.09025$. Thus 761 draws suffice. The mean $100H_{100} \approx 518.74$ describes an average collection time; it does not specify a 95% completion deadline. The exponential bound is sufficient and may be conservative.

14. With a Poisson total, the six face counts are independent Poisson variables of mean two. Each is zero with probability e^{-2}, so the unseen-face count is $\mathrm{Bin}(6, e^{-2})$, with mean $6e^{-2}$. The completion probability is $(1 - e^{-2})^6 \approx .41791$. For exactly 12 rolls, inclusion–exclusion instead gives

$$\sum_{j=0}^{6} (-1)^j \binom{6}{j} (1 - j/6)^{12} \approx .43782.$$

Those face counts are dependent because they sum to 12. Randomizing the total changes the experiment and its completion probability.

15. The median solves $e^{-\lambda m} = 1/2$, giving $m = (\log 2)/\lambda$.

16. The standard deviation is three, so standardization gives $\mathbb{P}(8 < X < 13) = \Phi(1) - \Phi(-2/3)$.

17. For V uniform on $[0, 1]$, $\mathbb{E}[V] = 1/2$ and $\mathbb{E}[V^2] = 1/3$, so $\mathrm{Var}(V) = 1/12$. Write $X = a + (b-a)V$ to obtain mean $(a+b)/2$ and variance $(b-a)^2/12$.

18. The subset mean is $2 + 4 + 4 - 4/3 - 4/3 - 2 + 1 = 19/3$. Put $Q = \sum_{\varnothing \neq A} (-1)^{|A|+1}/p_A^2$. Here $Q = 4 + 16 + 16 - 16/9 - 16/9 - 4 + 1 = 265/9$. Hence $\mathbb{E}[T^2] = 2Q - \mathbb{E}[T] = 473/9$ and $\mathrm{Var}(T) = 473/9 - (19/3)^2 = 112/9$. With only types 2 and 3 missing, the original probabilities remain $1/4, 1/4$; the mean additional wait is $4 + 4 - 2 = 6$.

19. The exact mean is $10 + 10/9 - 1 = 91/9$ draws, lying between the bounds 10 and $100/9$. If a required type has probability zero, collection never completes and the mean is infinite; the positive-probability formulas no longer apply.

20. The counts are independent Poisson variables of means $6, 3, 3$. The mean missing count is $e^{-6} + 2e^{-3}$ and completion probability is $(1 - e^{-6})(1 - e^{-3})^2$. The missing count is a sum of three independent indicators with probabilities e^{-6}, e^{-3}, e^{-3}; it is not binomial. Conditional on $K = 12$, the first type's count is $\mathrm{Bin}(12, 1/2)$, and all three counts have a multinomial joint law.

21. The second success on trial four has probability $\binom{3}{1}(1/3)^2(2/3)^2 = 4/27$. The trial count has mean six and variance $2(2/3)/(1/3)^2 = 12$. Subtracting the two successes gives mean failure count four, with the same variance.

22. The wait is Gamma(2, rate 3), with density $9te^{-3t}$ for $t > 0$, mean $2/3$, variance $2/9$, and tail at one $e^{-3}(1 + 3) = 4e^{-3}$. The time unit is minutes.

23. This is Beta$(2,3)$ with density $12x(1-x)^2$ on $(0,1)$, mean $2/5$, and variance $1/25$. It exceeds $1/2$ exactly when zero or one of the four points is at or below $1/2$, giving $[\binom{4}{0}+\binom{4}{1}]/16 = 5/16$.

24. Each box count is independently Poisson with mean two. The probability is $[e^{-2}(1+2)]^2 = 9e^{-4}$. With exactly four requests, two boxes of capacity one cannot accommodate the total, so the probability is zero.

25. At rate one, the mean is $19/3$ and the variance is $112/9+19/3 = 169/9$. At rate two, divide times by two: the mean is $19/6$ and variance $169/36$. The discrete count itself retains the variance $112/9$; it is not multiplied by a random clock.

26. Here $\widetilde{F}(t) = (1-e^{-t/2})^2$. The lower error is $\mathbb{P}(\text{Pois}(1) > 2) = 1-(5/2)e^{-1}$, giving lower bound $(7/2)e^{-1} - 2e^{-1/2}$, approximately .0745. The upper error is $\mathbb{P}(\text{Pois}(4) < 2) = 5e^{-4}$, giving upper bound $1-2e^{-2}+6e^{-4}$, approximately .8392. Both bracket the exact fixed-draw probability $1/2$.

27. The exact expression is $2^{-25}\sum_{j=10}^{15}\binom{25}{j}$. The mean is $25/2$ and standard deviation $5/2$; replacing the integer range by the interval $(9.5, 15.5)$ gives the approximation $\Phi(6/5) - \Phi(-6/5)$. The independent fair-trial model makes the binomial expression exact, not the normal approximation.

Chapter 6

For X uniform on $\{-1,0,1\}$ and $Y = X^2$, symmetry gives $\mathbb{E}X = \mathbb{E}[XY] = 0$, hence covariance zero, but knowing X determines Y. For the tower property, take the weighted average of group means; each observation receives exactly its population weight.

Chapter 7

If $\sigma = 12, n = 64$, then $\text{SE}(\bar{X}) = 12/8 = 1.5$. The weak law follows from $\text{Var}(\bar{X}_n) = \sigma^2/n$ and Chebyshev.

Chapter 8

For 330 successes in 600 trials, $\hat{p} = .55$ and the plug-in standard error is

$$\sqrt{.55(.45)/600} \approx .0203.$$

The approximate 95% interval is $.55 \pm 1.96(.0203)$, or about $(.510, .590)$.

Chapter 9

Beta$(2,2)$ updated with 8 successes and 2 failures becomes Beta$(10,4)$, with posterior mean $10/14 \approx .714$. The posterior predictive success probability equals the posterior mean $\alpha'/(\alpha'+\beta')$.

Chapter 10

The numbering here matches Section 10.8.

1. The posterior probability of bias $3/4$ is $3^3/(3^3+3^1) = 9/10$. The predictive chance of heads is $(1/4)(1/10) + (3/4)(9/10) = 7/10$. Reordering the observed string does not change either calculation.

2. Let the newly chosen biases be independent variables $\Theta_1, \ldots, \Theta_n$. For any specified bits,

$$\mathbb{P}(X_1 = x_1, \ldots, X_n = x_n) = \mathbb{E}\prod_{i=1}^{n} \Theta_i^{x_i}(1-\Theta_i)^{1-x_i} = \prod_{i=1}^{n} \frac{1}{2} = 2^{-n}.$$

Here the tosses are generated independently given all the biases. The expectation factors because the biases are independent; it does not factor this way when a single shared Θ is reused.

3. The ordered path has probability $4!1!/6! = 1/30$. There are $\binom{5}{4} = 5$ such paths, so exactly four reds has probability $1/6$. After the path, the urn contains five red and two blue balls; the next-red probability is $5/7$.

4. Initially $\Theta \sim \text{Beta}(2,3)$. The posterior is $\text{Beta}(6,5)$ and the predictive probability is $6/11$. The updated urn also contains six red and five blue balls.

5. Write $Y = S_n/n$ and $m = \mathbb{E}\Theta$. Expand $Y - m = (Y - \Theta) + (\Theta - m)$. The cross term has expectation zero because $\mathbb{E}[Y - \Theta \mid \Theta] = 0$. Therefore

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y \mid \Theta)] + \text{Var}(\Theta) = \frac{\mathbb{E}[\Theta(1-\Theta)]}{n} + \text{Var}(\Theta).$$

Under uniform mixing, $\mathbb{E}\Theta = 1/2$, $\mathbb{E}\Theta^2 = 1/3$, so the two terms are $1/(6n)$ and $1/12$. This gives $\text{Var}(S_{10}/10) = 1/10$ and $\text{Var}(S_{100}/100) = 17/200$. The expression $1/(4n)$ wrongly treats the trials as independent Bernoulli$(1/2)$ variables; it misses the between-run variance.

6. Pairwise independence implies $\text{Cov}(X_1, X_2) = 0$, hence $\text{Var}(\Theta) = 0$ by the representation theorem. Thus $\Theta = p_0$ almost surely for some constant p_0, and (10.1) is the i.i.d. Bernoulli(p_0) product law. Infinite exchangeability is the additional hypothesis. An arbitrary finite collection of pairwise independent events need not have it.

7. Nonnegativity of variance gives

$$0 \le \text{Var}\left(\sum_{i=1}^{N} X_i\right) = Nv + N(N-1)c.$$

Divide by $N(N-1)$ to obtain the bound. The always-disagreeing pair has $v = 1/4, c = -1/4$, so an exchangeable extension would require $-1/4 \ge -1/[4(N-1)]$, possible only for $N \le 2$.

8. Fair-toss frequencies cluster near $1/2$; shared-coin frequencies cluster near $1/4$ and $3/4$; urn frequencies spread across $[0,1]$. Finite simulations fluctuate, and 100 runs will not give a perfectly flat histogram. The theorem identifies the limiting laws as $\delta_{1/2}$, $\frac{1}{2}\delta_{1/4} + \frac{1}{2}\delta_{3/4}$, and the uniform law, respectively.

9. Conditional on the total number of ones, compare sampling without replacement with independent sampling at that population's success proportion; then average over the total. For any event, the absolute probability error is at most $\binom{5}{2}/1000 = .01$. With 100 draws the union bound is $\binom{100}{2}/1000 = 4.95$; capped at one, it is only the trivial bound. This does not say the approximation error actually equals one. The collision argument is useful when n^2/N is small.

Appendix C

Reference sheets

Counting and overlap corrections

For $0 \le k \le n$, ordered selections without replacement number $(n)_k = n!/(n-k)!$; unordered selections number $\binom{n}{k} = n!/[k!(n-k)!]$. Ordered k-term lists with replacement number n^k. Here $0! = 1$.

$$\begin{aligned}
\binom{n}{k} &= \binom{n}{n-k}, \\
\binom{n}{k} &= \binom{n-1}{k-1} + \binom{n-1}{k}, \\
(x+y)^n &= \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k, \\
|A \cup B \cup C| &= |A| + |B| + |C| - |A \cap B| - |A \cap C| \\
&\quad - |B \cap C| + |A \cap B \cap C|.
\end{aligned}$$

The same alternating formula holds for event probabilities, without an independence assumption. Pairwise intersections include the triple region.

Finite probability spaces and birthdays

For finite Ω, assign weights $w_\omega \ge 0$ summing to one; then $\mathbb{P}(A) = \sum_{\omega \in A} w_\omega$. Only in the uniform case does this simplify to $|A|/|\Omega|$.

For n independent uniform birthdays on d days,

$$\mathbb{P}(\text{some birthday match}) = \begin{cases} 1 - \displaystyle\prod_{j=0}^{n-1} (1 - j/d), & 0 \le n \le d, \\ 1, & n > d. \end{cases}$$

For a designated person among $n \ge 1$, the probability of a match with someone else is $1-(1-1/d)^{n-1}$, taking it to be zero when $n = 1$. For 365 days, the any-pair probability first exceeds $1/2$ at 23 people.

Derangements and fixed points

For a uniform permutation of n objects,

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}, \qquad \mathbb{P}(\text{exactly } r \text{ fixed points}) = \frac{\binom{n}{r} D_{n-r}}{n!}.$$

Here $D_0 = 1$, $D_1 = 0$, and $D_n = nD_{n-1} + (-1)^n$. The error in $D_n/n! \approx e^{-1}$ is less than $1/(n+1)!$; for $n \geq 1$, D_n is the nearest integer to $n!/e$.

Random variables and threshold events

The event $\{X \geq a\}$ means $\{\omega : X(\omega) \geq a\}$. For a PMF, group outcome probabilities with the same value of X. For a density, integrate over the requested range:

$$F_X(x) = \mathbb{P}(X \leq x), \qquad \mathbb{P}(a < X \leq b) = F_X(b) - F_X(a), \qquad \mathbb{P}(X > a) = 1 - F_X(a).$$

For integer-valued X and integer k, $\mathbb{P}(X \geq k) = 1 - F_X(k-1)$. For a variable with a density, individual values have probability zero.

Discrete expectation and variance

For a finite probability table,

$$\mathbb{E}[X] = \sum_x x p_X(x), \qquad \mathbb{E}[g(X)] = \sum_x g(x) p_X(x),$$

$$\operatorname{Var}(X) = \sum_x (x - \mathbb{E}[X])^2 p_X(x) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2, \qquad \sigma_X = \sqrt{\operatorname{Var}(X)},$$

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b, \qquad \operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X).$$

Linearity of expectation needs no independence. Variances of independent sums add; dependent sums need the covariance terms below. An event indicator has mean p and variance $p(1-p)$. For independent copies with mean m and variance v, their r-term average has mean m and variance v/r. For the correct-hat count, the mean is one for $n \geq 1$ and the variance is one for $n \geq 2$ (zero for $n = 1$). For a uniform permutation, the record count has mean H_n and the inversion count has mean $n(n-1)/4$. Uniform matching of the $2n$ free ends of n spaghetti strands gives mean loop count $\sum_{j=1}^{n} (2j-1)^{-1} = H_{2n} - H_n/2$. Independent fair vertex colors give expected cut size half the edge count; some cut therefore attains at least that value.

Coupon collection

For independent uniform draws from $n \geq 1$ types, let T_n be the number of draws to see them all. With $H_n = \sum_{j=1}^{n} 1/j$,

$$\mathbb{E}[T_n] = nH_n, \qquad \operatorname{Var}(T_n) = n^2 \sum_{j=1}^{n} \frac{1}{j^2} - nH_n.$$

For integers $m \geq 1$, the number M_m of missing types has mean $n(1-1/n)^m$, and $\mathbb{P}(T_n > m) \leq \min\{1, ne^{-m/n}\}$. For an independent $K \sim \text{Pois}(t)$, $t > 0$, instead $M_K \sim \text{Bin}(n, e^{-t/n})$ and $\mathbb{P}(T_n \leq K) = (1-e^{-t/n})^n$. This product uses a random number of draws; see Section 5.7 for the fixed-draw formula.

For positive unequal probabilities p_i summing to one, put $p_A = \sum_{i \in A} p_i$. Then

$$\begin{aligned}
\mathbb{E}[T] &= \sum_{\varnothing \neq A} \frac{(-1)^{|A|+1}}{p_A}, \\
\mathbb{E}[T^2] &= \sum_{\varnothing \neq A} (-1)^{|A|+1} \frac{2 - p_A}{p_A^2}, \\
\mathbb{E}[M_m] &= \sum_i (1-p_i)^m, \qquad \mathbb{P}(T > m) \leq \min\{1, \sum_i (1-p_i)^m\}.
\end{aligned}$$

All subsets are of the required type set. A zero-probability required type makes completion impossible. For the same number of types, uniform probabilities minimize the mean, giving $\mathbb{E}[T] \geq nH_n$.

Poissonization and clocks

An independent total $K \sim \text{Pois}(t)$ and i.i.d. labels of probabilities p_i give independent counts $C_i \sim \text{Pois}(tp_i)$. Therefore

$$\mathbb{P}(T \leq K) = \prod_i (1 - e^{-tp_i}).$$

Conditioning on $K = m$ returns the original fixed-total model; its multinomial counts are dependent when $n \geq 2, m \geq 1$. Independent rate-one exponential gaps give a continuous collection time τ equal to the maximum of independent exponentials of rates p_i, with

$$\mathbb{E}[\tau] = \mathbb{E}[T], \qquad \text{Var}(\tau) = \text{Var}(T) + \mathbb{E}[T].$$

The product formula and the fixed-total CDF are different. The de-Poissonization bounds in Section 5.12.4 compare them with explicit Poisson-tail errors.

Calculus used in the course

$$\begin{aligned}
\int_a^b f(x)\,dx &= G(b) - G(a) \quad (G' = f), \\
e^x &= \sum_{k=0}^{\infty} \frac{x^k}{k!}, \\
\log(1-x) &= -\sum_{k=1}^{\infty} \frac{x^k}{k} \quad (0 \leq x < 1), \\
\sum_{k=0}^{\infty} r^k &= \frac{1}{1-r} \quad (|r| < 1).
\end{aligned}$$

The birthday estimate is $q_d(n) \approx 1 - e^{-n(n-1)/(2d)}$; it is a lower bound when $0 \leq n \leq d$. See Appendix A for the derivations, hypotheses, worked integrations, and quantitative error estimates.

Core probability identities

$$\begin{aligned}
\mathbb{P}(A^c) &= 1 - \mathbb{P}(A), \\
\mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B), \\
\mathbb{P}(A \mid B) &= \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \\
\mathbb{P}(A) &= \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i), \\
\mathbb{P}(B_j \mid A) &= \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.
\end{aligned}$$

Moments and dependence

$$\begin{aligned}
\mathbb{E}[aX + bY] &= a\mathbb{E}X + b\mathbb{E}Y, \\
\operatorname{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}X)^2, \\
\operatorname{Cov}(X, Y) &= \mathbb{E}[XY] - \mathbb{E}X\mathbb{E}Y, \\
\operatorname{Var}(X + Y) &= \operatorname{Var} X + \operatorname{Var} Y + 2\operatorname{Cov}(X, Y), \\
\rho_{X,Y} &= \operatorname{Cov}(X, Y)/(\sigma_X \sigma_Y).
\end{aligned}$$

Common distributions

The hypergeometric row uses $N \geq 2$ population objects, K marked, m sampled without replacement, and $p = K/N$. Its feasible counts are given in Section 5.4. The negative-binomial row counts trials through success r, and the gamma row uses an integer shape $r \geq 1$ and a rate $\lambda > 0$.

Distribution	Support	Mean	Variance
Discrete uniform	$1, \ldots, m$	$(m+1)/2$	$(m^2-1)/12$
Bernoulli(p)	$\{0, 1\}$	p	$p(1-p)$
Binomial(n, p)	$0, \ldots, n$	np	$np(1-p)$
Hypergeometric	feasible k	mp	$mp(1-p)(N-m)/(N-1)$
Geometric(p)	$1, 2, \ldots$	$1/p$	$(1-p)/p^2$
Neg. binomial (trials)	$r, r+1, \ldots$	r/p	$r(1-p)/p^2$
Poisson(λ)	$0, 1, \ldots$	λ	λ
Uniform(a, b)	$[a, b]$	$(a+b)/2$	$(b-a)^2/12$
Exponential(λ)	$[0, \infty)$	$1/\lambda$	$1/\lambda^2$
Gamma(r, rate λ)	$[0, \infty)$	r/λ	r/λ^2
Normal(μ, σ^2)	$\mathbb{R}$	μ	σ^2
Beta(α, β)	$[0, 1]$	$\alpha/(\alpha+\beta)$	$\alpha\beta/[(\alpha+\beta)^2(\alpha+\beta+1)]$

Distribution mechanisms: fixed trials give binomial counts; a fixed success target gives negative-binomial waits; fixed time in a Poisson stream gives Poisson counts; exponential gaps give gamma arrival times. The kth smallest of m independent uniform points has Beta($k, m+1-k$) law. Normal limits of sums require the central limit theorem's hypotheses; they are approximations unless normality is separately part of the model.

Inference checklist

1. What is the target population and parameter?
2. How were units sampled or assigned?
3. What model connects observations to the parameter?
4. What is the estimator and its sampling uncertainty?
5. Which assumptions are needed for the interval or test?
6. Is the effect practically meaningful?
7. What diagnostic or predictive check could reveal model failure?

Exchangeability and a shared random bias

For an *infinite* exchangeable zero–one sequence, de Finetti's theorem gives a unique mixing law μ on $[0,1]$. Choose $\Theta \sim \mu$ once; conditional on Θ, the trials are i.i.d. Bernoulli(Θ). For a specified string with s ones,

$$\begin{aligned}
\mathbb{P}(X_1 = x_1, \ldots, X_n = x_n) &= \int p^s (1-p)^{n-s}\, \mu(dp), \\
\mathbb{P}(S_n = s) &= \binom{n}{s} \int p^s (1-p)^{n-s}\, \mu(dp), \\
\mathrm{Cov}(X_i, X_j) &= \mathrm{Var}(\Theta) \quad (i \neq j), \\
\mathrm{Var}(S_n/n) &= \frac{\mathbb{E}[\Theta(1-\Theta)]}{n} + \mathrm{Var}(\Theta), \\
S_n/n &\longrightarrow \Theta \quad \text{with probability one.}
\end{aligned}$$

The predictive success probability is the posterior mean of Θ. With a Beta(a,b) mixing law and s successes in n trials,

$$\Theta \mid \text{data} \sim \mathrm{Beta}(a+s, b+n-s), \qquad \mathbb{P}(X_{n+1} = 1 \mid \text{data}) = \frac{a+s}{a+b+n}.$$

Pólya's urn, with one-ball reinforcement and a, b initial balls of the two colors, realizes this model for positive integer a, b. For $a = b = 1$, the mixing law is uniform and S_n is uniform on $\{0, \ldots, n\}$.

Common trap

Finite exchangeability alone does not imply a mixture of i.i.d. trials. Sampling without replacement can have negative covariance. Exchangeability also does not choose a particular prior or identify a unique physical mechanism.

For a prefix of length $n \leq N$ from an exchangeable zero–one vector of length N, the finite-mixture approximation using bias S_N/N has absolute probability error at most $\min\{1, \binom{n}{2}/N\}$ for any event on that prefix. See Chapter 10 for the coupling proof.

Further reading

- D. Freedman, R. Pisani, and R. Purves, *Statistics*.
- J. Blitzstein and J. Hwang, *Introduction to Probability* (named laws through their generating experiments).
- Stanford EE 178/278A, *Lecture Notes 4: Expectation* (indicator arguments, including spaghetti loops).
- Stanford CS265, *Balls and bins and Poissonization* (random totals and independent box counts).
- K. Siegrist, *Probability, Mathematical Statistics, and Stochastic Processes*: Poisson thinning, gamma waiting times, and order statistics.
- L. Wasserman, *All of Statistics* (for a concise next course).
- A. Gelman et al., *Bayesian Data Analysis* (for a more advanced Bayesian treatment).
- W. Kirsch, *An elementary proof of de Finetti's Theorem*, arXiv:1809.00882 (2018).
- P. Diaconis and D. Freedman, *Finite Exchangeable Sequences*, Annals of Probability **8**(4) (1980), 745–764.

Course packet designed as an adaptable teaching draft. Classical results and examples are attributed where appropriate; exposition, exercises, and diagrams are prepared for this packet.